

**COURSE STRUCTURE
AND
DETAILED SYLLABUS**

R22

**ELECTRONICS AND
COMMUNICATION
ENGINEERING**

For

B. Tech 4-Year Degree Course

(Applicable for the students admitted into E1 from the Academic Year 2022-23)

(I – IV Years Syllabus)



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

Basar, Nirmal, Telangana – 504107

Course Structure of Credits

S.No	Course Category	Course Type	Standard Credits as per AICTE	Department Credits
1	BSC	Basic Science Course	25	24
2	ESC	Engineering Science Course	24	28
3	HSMC	Humanities and Social Sciences including Management Course	12	11
4	PCC	Professional Core Course	48	58
5	PEC	Professional Elective Course	18	15
6	OEC	Open Elective Course	18	09
7	MC	Mandatory Course	0	0
8	SIP	Seminar + Internship + Project Work + Comprehensive Viva	15	15
Total			160	160

S. No.	Course Category	Credit Breakup for ECE students	Percentage of the Credit
1	BSC	24	15%
2	ESC	28	17.50%
3	HSMC	11	6.875%
4	PCC	58	36.25%
5	PEC	15	9.375%
6	OEC	09	5.625%
7	MC	0	0
8	SIP	15	9.375%
Total		160	

Definition of Credit

1	1Hr Lecture(L) per week	1credit
2	1 Hr Tutorial(T) per week	1credit
3	1 Hr Practical(P)per week	0.5credit
4	2 Hours Practical (Lab)/week	1credit

FIRST YEAR (E1) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Linear Algebra & Calculus	BSC	3	1	0	4	4
2		Engineering workshop	ESC	0	1	2	3	2
3		Programming for Problem Solving	ESC	3	0	0	3	3
4		Programming for Problem Solving Lab	ESC	0	0	3	3	1.5
5		Element of Electronics and Communications Engineering	PCC	0	0	2	2	1
6		Digital Electronic Circuits	PCC	3	0	0	3	3
7		Digital Electronic Circuits Lab	PCC	0	0	3	3	1.5
8		Constitution of India	MC	2	0	0	2	0
Total				11	2	10	23	16

FIRST YEAR (E1) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Differential Equations and Vector Calculus	BSC	3	1	0	4	4
2		Physics	BSC	3	0	0	3	3
3		Physics Lab	BSC	0	0	3	3	1.5
4		Engineering Graphics	ESC	0	1	4	5	3
5		Network Analysis	PCC	3	1	0	4	4
6		Electronic Devices and Circuits	PCC	3	1	0	4	4
7		Electronic Devices and Circuits lab	PCC	0	0	3	3	1.5
8		English	HSMC	2	0	0	2	2
9		English Language Lab	HSMC	0	0	2	2	1
Total				14	4	12	30	24

SECOND YEAR (E2) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Chemistry	BSC	3	0	0	3	3
2		Chemistry Lab	BSC	0	0	3	3	1.5
3		Transform Calculus and Complex Analysis	BSC	3	1	0	4	4
4		Electrical Technology	ESC	3	0	0	3	3
5		Electrical Technology lab	ESC	0	0	3	3	1.5
6		Signals and Systems	PCC	3	1	0	4	4
7		Probability Theory and Stochastic Processes	PCC	3	1	0	4	4
8		Essence of Indian Traditional Knowledge	MC	2	0	0	2	0
9		Electronic Mini Project	SIP	0	0	2	2	1
Total				17	3	8	28	22

SECOND YEAR (E2) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Numerical Methods	BSC	3	0	0	3	3
2		Object Oriented Programming	ESC	3	0	0	3	3
3		Object Oriented Programming Lab	ESC	0	0	2	2	1
4		Control systems	ESC	3	1	0	4	4
5		Analog Circuits	PCC	3	1	0	4	4
6		Analog Circuits Lab	PCC	0	0	2	2	1
7		Electromagnetic Waves	PCC	3	0	0	3	3
8		Digital System Design	PCC	3	0	0	3	3
9		Digital Systems Design and VLSI Lab	PCC	0	0	2	2	1
10		Environmental Science	MC	2	0	0	2	0
Total				20	2	6	28	23

THIRD YEAR (E3) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Analog and Digital Communications	PCC	3	1	0	4	4
2		Analog and Digital Communications lab	PCC	0	0	2	2	1
3		Computer Architecture	PCC	3	0	0	3	3
4		Digital Signal Processing	PCC	3	0	0	3	3
5		Digital Signal Processing Lab	PCC	0	0	2	2	1
6		VLSI Engineering	PCC	3	0	0	3	3
7		Managerial Economics and Financial analysis	HSMC	3	0	0	3	3
8		Soft Skills	HSMC	0	0	2	2	1
9		Open Elective -I	OEC	3	0	0	3	3
Total				18	1	6	25	22

THIRD YEAR (E3) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Computer Networks	ESC	3	0	0	3	3
2		Operating Systems	ESC	3	0	0	3	3
3		Microprocessor and Microcontrollers	PCC	3	0	0	3	3
4		Microprocessor lab	PCC	0	0	2	2	1
5		RF and Microwave Engineering	PCC	3	0	0	3	3
6		RF and Microwave Eng. Lab	PCC	0	0	2	2	1
7		Pattern recognition	PEC	3	0	0	3	3
8		Communication Skills	HSMC	0	0	2	2	1
9		Professional Law and Ethics	HSMC	3	0	0	3	3
10		Mini Project-I	SIP	0	0	4	4	2
Total				18	0	10	28	23

FOURTH YEAR (E4) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Signal Processing stream elective	PEC	3	0	0	3	3
2		Communications stream elective	PEC	3	0	0	3	3
5		Program Elective (Preferably course on Antennas)	PEC	3	0	0	3	3
7		Embedded systems stream elective	PEC	3	0	0	3	3
4		Major Project -I	SIP	0	0	8	8	4
6		Summer Internship	SIP					1
7		Comprehensive viva	SIP					1
Total				12	0	8	20	18

FOURTH YEAR (E4) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1		Open Elective-II	OEC	3	0	0	3	3
2		Open Elective-III	OEC	3	0	0	3	3
3		Major Project-II	SIP	0	0	12	12	6
Total				6	0	12	18	12

Program Elective Courses (PEC)

Sl. No.	Course Code	Course Title	Course Category
1	ECPE1_	Digital Image Processing	PEC-I
2	ECPE1_	Adaptive Signal Processing	PEC-I
4	ECPE2_	Wireless Communications	PEC-II
5	ECPE2_	Information Theory and Coding	PEC-II
6	ECPE2_	Satellite Communication	PEC-II
7	ECPE3_	IoT and Applications	PEC-III
8	ECPE3_	Mixed Signal Design	PEC-III
9	ECPE3_	CMOS Design	PEC-III
10	ECPE4_	Antennas and Propagation	PEC-IV
11	ECPE4_	Microwave Theory and Techniques	PEC-IV
12	ECPE4_	Fiber Optic Communication	PEC-IV

Open Elective Courses (OEC)

Sl. No.	Course Code	Course Title	Course Category
1		Robotics	OEC-I
2		Data Structures	OEC-I
3		Material Science	OEC-II
4		Artificial intelligence	OEC-II
5		Block chain	OEC-II
6		Cryptography	OEC-III
7		IPR	OEC- III
8		Operation research	OEC- III
9		Game theory	OEC- III

Humanities and Social Sciences including Management Courses [HSMC]

Sl. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1		Communication Skills	0	0	2	2	1	
2		English	2	0	0	2	2	
3		English Language Lab	0	0	2	2	1	
4		Managerial Economics & Financial Analysis	3	0	0	3	3	
5		Professional Law and Ethics	3	0	0	3	3	
6		Soft skills	0	0	2	2	1	
Total:			8	0	6	14	11	N/A

Basic Science Courses [BSC]

Sl. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1		Linear Algebra and Calculus	3	1	0	4	4	
2		Differential equations and Vector Calculus	3	1	0	4	4	
3		Physics	3	0	0	3	3	
4		Physics Lab	0	0	3	3	1.5	
5		Chemistry	3	0	0	3	3	
6		Chemistry Lab	0	0	3	3	1.5	
7		Transform Calculus and Complex Analysis	3	1	0	4	4	
8		Numerical Methods	3	0	0	3	3	
Total:			18	3	6	27	24	

Engineering Science Courses [ESC]

SI. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1		Engineering Workshop	0	1	2	3	2	
2		Programming for Problem Solving	3	0	0	3	3	
3		Programming for Problem Solving Lab	0	0	3	3	1.5	
4		Electrical Technology	3	0	0	3	3	
5		Electrical Technology lab	0	0	3	3	1.5	
6		Computer Aided Graphics	0	1	4	5	3	
7		Object Oriented Programming	3	0	0	3	3	
8		Object Oriented Programming lab	0	0	2	2	1	
9		Operating Systems	3	0	0	3	3	
10		Computer Networks	3	0	0	3	3	
11		Control systems	3	1	0	4	4	
Total:			18	3	14	35	28	

Professional Core Courses [PCC]

Sl. No.	Code No.	Course Title	Hours per Week				Total Credits	Semester
			L	T	P	Total		
1		Digital Electronic Circuits	3	0	0	3	3	
2		Digital Electronic Circuits Lab	0	0	3	3	1.5	
3		Electronic Devices and Circuits	3	1	0	4	4	
4		Electronic Devices and Circuits lab	0	0	3	3	1.5	
5		Network Analysis	3	1	0	4	4	
6		Signals and Systems	3	1	0	4	4	
7		Analog Circuits	3	1	0	4	4	
8		Analog Circuits lab	0	0	2	2	1	
10		Electromagnetic Waves	3	0	0	3	3	
11		Probability Theory and Stochastic Processes	3	1	0	4	4	
12		Analog and Digital Communications	3	1	0	4	4	
13		Analog and Digital Communications lab	0	0	2	2	1	
14		Computer Architecture	3	0	0	3	3	
15		Digital Signal Processing	3	0	0	3	3	
16		Digital Signal Processing lab	0	0	2	2	1	
17		RF and Microwave Engineering	3	0	0	3	3	
18		RF and Microwave Eng .Lab	0	0	2	2	1	
19		VLSI Engineering	3	0	0	3	3	
20		Microprocessor and Microcontroller	3	0	0	3	3	
21		Microprocessor lab	0	0	2	2	1	
22		Digital System Design	3	0	0	3	3	
23		Digital Systems Design and VLSI Lab	0	0	2	2	1	
24		Element of Electronics and Communications Engineering	0	0	2	2	1	
		TOTAL:	42	6	20	68	58	N/A

Professional Elective Courses [PEC]

SI. No.	CodeNo.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1	PEC	Signal Processing stream elective (DIP)	3	0	0	3	3	
2	PEC	Communications stream elective (WC)	3	0	0	3	3	
3	PEC	Embedded systems stream elective (IoT)	3	0	0	3	3	
4	PEC	Program Elective-4 (AWP)	3	0	0	3	3	
5	PEC	Pattern recognition	3	0	0	3	3	
Total:			15	0	0	15	15	N/A

Open Elective Courses [OEC]

SI. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1	OEC	Open Elective-I	3	0	0	3	3	
2	OEC	Open Elective-II	3	0	0	3	3	
3	OEC	Open Elective -III	3	0	0	3	3	
Total:			9	0	0	9	9	

Mandatory Courses (MC)

SI. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1		Essence of Indian Traditional Knowledge	2	0	0	2	0	
2		Environmental Sciences	2	0	0	2	0	
3		Constitution of Indian	2	0	0	2	0	
Total:			6	0	0	6	0	

SIP

SI. No.	Code No.	Course Title	Hours per week				Total Credits	Semester
			L	T	P	Total		
1		Electronic Mini Project	0	0	2	2	1	
2		Mini Project -I	0	0	4	4	2	
3		Major Project-I	0	0	8	8	4	
4		Summer Internship					1	
5		Comprehensive viva					1	
6		Major Project-II	0	0	12	12	6	
Total:			0	0	26	26	15	

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
	Linear Algebra and Calculus				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4

Course Objectives

- To introduce students to the fundamental concepts and operations of matrix theory and their applications in linear algebra and other fields.
- To enable students to understand the properties of vectors, vector spaces, and linear transformations.
- To familiarize students with the methods of solving systems of linear equations and finding the inverse of a matrix.
- To provide students with an understanding of sequences, series, and their convergence

Course Outcomes

1. Students will be able to understand and apply the fundamental concepts and operations of matrix theory, such as types of matrices, elementary operations, and matrix inverse.
2. Students will be able to analyze the properties of vectors, vector spaces, and linear transformations, and apply them to real-world problems.
3. Students will be able to solve systems of linear equations using elementary operations and Gauss elimination method, and find the inverse of a matrix using Gauss-Jordan method.
4. Students will be able to determine the convergence or divergence of sequences and series using various tests, such as comparison test, ratio test, and root test.

Detailed Contents

UNIT-I: Matrix Theory

Types of Matrices, Symmetric, Hermitian, Skew-Symmetry, Skew-Hermitian, Orthogonal matrices, Unitary matrices; Elementary row and column operations on a matrix, Rank of a matrix by Echelon form and Normal form, Inverse of a Non-singular matrix by Gauss-Jordan method; Consistency and solutions of system of linear equations using elementary operations, Gauss elimination method.

UNIT-II: Vector spaces:

Definition of vector spaces and subspaces with example; Linear dependence of vectors, basis and dimension, Linear Transformation, range and kernel of linear map, rank and nullity theorem, Characteristic roots and vectors of a matrix; Cayley-Hamilton theorem (without proof); finding inverse and power of a matrix by Cayley-Hamilton Theorem; Quadratic forms and Nature of the Quadratic forms.

UNIT-III: Sequences & Series

Definition of a sequence, limit; Convergent, Divergent and Oscillatory sequences. Convergent, Divergent and Oscillatory Series; Series of positive terms; Comparison test, p-test, D'Alembert's ratio test; Raabe's test; Cauchy's Integral test; Cauchy's root test; Logarithmic test. Alternating series; Leibnitz test; Alternating Convergent series; Absolute and conditionally convergence.

UNIT-IV: Calculus

Mean value theorems: Rolle's theorem, Lagrange's Mean value theorem, Cauchy's Mean value theorem; Taylor's and Maclaurin's series with remainders, Expansions; Definition of Improper Integrals and Beta and Gamma functions and their applications.

UNIT-V: Multivariable Calculus (Partial Differentiation and applications)

Functions of two variables, Definitions of Limits and continuity, Partial Differentiation; Euler's theorem; Total Derivative; Jacobian; Functional dependence and independence; Maxima and minima of functions of several variables (two and three variables) using Lagrange Multipliers.

Text Books

1. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8th Edition,
2. R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House, 2008

References

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
3. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
4. N.P. Bahl and Manish Goyal, A textbook of Engineering Mathematics, Laxmi Publications, Reprint, 2008.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
	Engineering Workshop				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	1	2	40	60	2

Course Objectives

- To understand the basic manufacturing process of producing a component by casting, forming plastic molding, joining processes, machining of a component either by conventional or by unconventional processes.
- To understand the advanced manufacturing process of additive manufacturing process.

Course Outcomes

1. Students will gain knowledge of the different manufacturing processes which are commonly employed in the industry, to fabricate components using different materials

Detailed Contents

Module – 1: Metal Casting: Introduction, Tools, Types of Patterns, Pattern Materials, Types of casting – Sand, Die and other casting processes and Applications

Module – 2: Metal Forming: Introduction, Classification, Types of Bulk and sheet metal forming and Applications.

Module – 3: Powder Metallurgy: Introduction, Powder production methods, Compaction, Sintering, Secondary operations and Applications.

Module – 4: Joining: Types of Joining, Introduction to Welding, Brazing and soldering, Arc, Solid state welding processes.

Module – 5: Conventional Machining Processes: Introduction to machining operations; Lathe operations, Drilling, Milling and Grinding.

Module – 6: Unconventional Machining Processes.

Module – 7: CNC Machining and Additive manufacturing.

List of Experiments

1. **Fitting** – Step and V Fit.
2. **Carpentry** – Half lap joint and Dove tail joint.
3. **House Wiring** – Series, Parallel, Staircase and Godown wiring.
4. **Tin Smithy** – Tray and Cylinder.
5. **Welding** – Bead formation, Butt and Lap joint welding.
6. **Foundry** – Mold preparation with Single piece and Split piece pattern.
7. **Machining** – Plain turning, Facing, Step and Taper turning.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
	Programming for Problem Solving				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3

Course Objectives

- To learn the fundamentals of computers.
- To understand the various steps in program development.
- To learn the syntax and semantics of the C programming language.
- To learn the usage of structured programming approaches in solving problems.

Course Outcomes

1. To write algorithms and to draw flowcharts for solving problems.
2. To convert the algorithms/flowcharts to C programs.
3. To code and test a given logic in the C programming language.
4. To decompose a problem into functions and to develop modular reusable code.
5. To use arrays, pointers, strings and structures to write C programs.
Searching and sorting problems.

Detailed Contents

UNIT-I: Introduction to Programming

Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.)

Representation of Algorithm - Algorithms for finding roots of a quadratic equations, finding minimum and maximum numbers of a given set, finding if a number is prime number Flowchart/Pseudo code with examples, Program design and structured programming

Introduction to C Programming Language: variables (with data types and space requirements), Syntax and Logical Errors in compilation, object and executable code, Operators, expressions and precedence, Expression evaluation, Storage classes(auto, extern, static and register), Type conversion, The main method and command line arguments Bitwise operations: Bitwise AND, OR, XOR and NOT operators

UNIT-II: Conditional Branching Loops Arrays, Strings:

Conditional Branching and Loops: Writing and evaluation of conditionals and consequent branching with if, if-else, switch-case, ternary operator, goto, Iteration with for, while, do-while loops I/O: Simple input and output with scanf and printf, formatted I/O, Introduction to stdin, stdout and stderr. Commandline arguments Arrays: one and two dimensional arrays, creating, accessing and manipulating elements of arrays.

UNIT-III: Structures, Pointers, Preprocessor and File handling in C:

Strings: Introduction to strings, handling strings as array of characters, basic string functions available in C (strlen, strcat, strcpy, strstr etc.), arrays of strings

Structures: Defining structures, initializing Structures, unions, Array of structures

Pointers: Idea of pointers, Defining pointers, Pointers to Arrays and Structures.

UNIT-IV: Preprocessor, Function and Dynamic Memory Allocation:

Preprocessor: Commonly used Preprocessor commands like include, define, undef, if, ifdef, ifndef Files: Text and Binary files, Creating and Reading and writing text and binary files, Appending data to existing files, Writing and reading structures using binary files, Random access using f seek, ftell and rewind functions. Functions: Designing structured programs, Declaring a function, Signature of a function, Parameters and return type of a function, passing parameters to functions, call by value, Passing arrays to functions, passing pointers to functions, idea of call by reference.

UNIT-V: Searching and Sorting:

Recursion: Simple programs, such as Finding Factorial, Fibonacci series etc., Limitations of Recursive functions. Dynamic memory allocation: Allocating and freeing memory, Allocating memory for arrays of different data types. Basic searching in an array of elements (linear and binary search techniques), Basic algorithms to sort array of elements (Bubble and Insertion sort).

Text Books

1. Jeri R. Hanly and Elliot B. Koffman, Problem solving and Program Design in C 7th Edition, Pearson
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition).

References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India.
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill.
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB.
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression).
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
	Programming for Problem Solving Lab				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives ● To work with an IDE to create, edit, compile, run and debug programs. ● To analyze the various steps in program development. ● To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc. ● To develop modular, reusable and readable C Programs using the concepts like functions, arrays etc. ● To Write programs using the Dynamic Memory Allocation concept. ● To create, read from and write to text and binary files.						
Course Outcomes 1. Formulate the algorithms for simple problems. 2. Translate given algorithms to a working and correct program. 3. Correct syntax errors as reported by the compilers and identify and correct logical errors encountered during execution. 4. Represent and manipulate data with arrays, strings and structures and use pointers of different types. 5. Create, read and write to and from simple text and binary files. and modularize the code with functions so that they can be reused.						

Lab Experiments

Practice sessions:

- A. Write a simple program that prints the results of all the operators available in C (including pre/ post increment , bitwise and/or/not , etc.). Read required operand values from standard input.
- B. Write a simple program that converts one given data type to another using auto conversion and casting. Take the values from standard input.

Simple numeric problems:

- A. Write a program for finding the max and min from the three numbers.
- B. Write the program for the simple, compound interest.
- C. Write a program that declares Class awarded for a given percentage of marks, where mark $<40\%$ = Failed, 40% to $<60\%$ = Second class, 60% to $<70\%$ = First class, $\geq 70\%$ = Distinction. Read percentage from standard input.
- D. Write a program that prints a multiplication table for a given number and the number of rows in the table. For example, for a number 5 and rows = 3, the output should be:
$$\begin{array}{l} 5 \times 1 = 5 \\ 5 \times 2 = 10 \\ 5 \times 3 = 15 \end{array}$$
- E. Write a program that shows the binary equivalent of a given positive number between 0 to 255.

Expression Evaluation:

- A. A building has 10 floors with a floor height of 3 meters each. A ball is dropped from the top of the building. Find the time taken by the ball to reach each floor. (Use the formula $s = ut + \frac{1}{2}at^2$ where u and a are the initial velocity in m/sec ($= 0$) and acceleration in m/sec^2 ($= 9.8 \text{ m/s}^2$)).
- B. Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators $+$, $-$, $*$, $/$, $\%$ and use Switch Statement).
- C. Write a program that finds if a given number is a prime number
- D. Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome.
- E. A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.
- F. Write a C program to generate all the prime numbers between 1 and n , where n is a value supplied by the user.
- G. Write a C program to find the roots of a Quadratic equation.
- H. Write a C program to calculate the following, where x is a fractional value. i. $1 - \frac{x}{2} + \frac{x^2}{4} - \frac{x^3}{6}$
- I. Write a C program to read in two numbers, x and n , and then compute the sum of this geometric progression: $1 + x + x^2 + x^3 + \dots + x^n$. For example: if n is 3 and x is 5, then the program computes $1 + 5 + 25 + 125$.

Arrays, Pointers and Functions:

- A. Write a C program to find the minimum, maximum and average in an array of integers.
- B. Write a function to compute mean, variance, Standard Deviation, sorting of n elements in a single dimension array.
- C. Write a C program that uses functions to perform the following:
 - a. Addition of Two Matrices

- b. Multiplication of Two Matrices
- c. Transpose of a matrix with memory dynamically allocated for the new matrix as row and column counts may not be the same.
- d. To find the GCD (greatest common divisor) of two given integers.
- e. To find x^n .
- D. Write C programs that use both recursive and non-recursive functions
To find the factorial of a given integer.
- E. Write a program for reading elements using a pointer into an array and display the values using the array.
- F. Write a program for display values reverse order from an array using a pointer.
- G. Write a program through a pointer variable to sum of n elements from an array.

Files:

- A. Write a C program to display the contents of a file to standard output device.
- B. Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.
- C. Write a C program to count the number of times a character occurs in a text file. The file name and the character are supplied as command line arguments.
- D. Write a C program that does the following:
 - a. It should first create a binary file and store 10 integers, where the file name and 10 values are given in the command line. (hint: convert the strings using atoi function).
 - b. Now the program asks for an index and a value from the user and the value at that index should be changed to the new value in the file. (hint: use fseek function) .
 - c. The program should then read all 10 values and print them back.
- E. Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Strings:

- A. Write a C program to convert a Roman numeral ranging from I to L to its decimal equivalent.
- B. Write a C program that converts a number ranging from 1 to 50 to Roman equivalent.
- C. Write a C program that uses functions to perform the following operations:
 - a. To insert a sub-string into a given main string from a given position.
 - b. To delete n Characters from a given position in a given string.
- D. Write a C program to determine if the given string is a palindrome or not (Spelled same in both directions with or without a meaning like madam, civic, noon, abcba, etc.)
- E. Write a C program that displays the position of a character ch in the string S or – 1 if S doesn't contain ch.
- F. Write a C program to count the lines, words and characters in a given text.

Miscellaneous:

- A. a. Write a menu driven C program that allows a user to enter n numbers and then choose between finding the smallest, largest, sum, or average. The menu and all the choices are to be functions. Use a switch statement to determine what action to take. Display an error message if an invalid choice is entered.

- B. b. Write a C program to construct a pyramid of numbers as follows:

```
1      *      1      1      *
1 2    * *    2 3    2 2    * *
1 2 3  * * *  4 5 6  3 3 3  * * *
                        4 4 4 4 * *
                          *
```

Sorting and Searching:

- A. Write a C program that uses a non recursive function to search for a Key value in a given list of integers using linear search method.
- B. c. Write a C program that uses non recursive function to search for a Key value in a given sorted list of integers using binary search method.
- C. e. Write a C program that implements the Bubble sort method to sort a given list of integers in ascending order.
- D. Write a C program that sorts the given array of integers using selection sort in descending order.
- E. Write a C program that sorts the given array of integers using insertion sort in ascending order.
- F. Write a C program that sorts a given array of names.

Text Books

1. Jeri R. Hanly and Elliot B. Koffman, Problem solving and Program Design in C 7th Edition, Pearson
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)

References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, PHI.
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill.
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB.
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression).
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
6. Herbert Schildt, C: The Complete Reference, Mc Graw Hill, 4th Edition.
7. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title					Course Type
	Elements of Electronics and Communication Engineering (EECE)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	2	40	60	1
Course Objectives • To introduce and motivate students to learn design and decision processes in ECE domains • To understand the purpose of studying B.Tech in E.C.E with practical demonstrations						
Course Outcomes 1. Understand, apply and design electronic systems 2. Gain practical skills in lab work and team projects 3. Prepare for ECE career through certification and skills 4. Implement signal processing and applications 5. Analyze and design communication and networking systems						

Course syllabus:

Block diagram of a typical Electronic system with essential components like I/O Devices (Sensors, actuators), Communication devices, controllers, software, etc, Familiarization of electronic components, Typical design flow of an electronic product manufacturing, Two examples of Electronic systems from daily life, Block diagram of a typical Communication system with essential components, Overview of 1G to 5G communications, Two examples of Electronic systems from daily life, Analog and Digital Signal processing and applications in various domains, Block diagram of a typical Networking (Wired and Wireless technologies) and introduction to communication protocols, two examples of systems using networking from daily life, Inter disciplinary: AI and ML, Data Science, IoT, Robotics, Drones, Bio medical Engineering, Instrumentation, Imaging, Solar devices, Electric vehicles, Ethics, and Morals in professional environment, Career planning in ECE (Career opportunities, Certification courses, skill development, tools).

Exposure to the labs in ECE and demonstration, Team projects/proposals designed to replicate the decision process in real-world applications of the EC engineering process.

Scheme of Evaluation:

Internal: Participation of students (20 marks),

Internal exam with objective questions (20 Marks)

External: Evaluation of project/proposal and viva (60 marks)

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
EC1101	Digital Electronic Circuits (DEC)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • This course aims at understanding various number systems, numbers representation and number base conversions. • Learn Boolean Algebra, Logic gates, and Simplification techniques. • Design combinational logic circuits using MSI devices. • Design Sequential circuits using FFs, Shift Registers and counters. • Design Finite State Machines and Synchronous circuits.						
Course Outcomes Upon completing this course, the student will be able to 1. Understand the numerical information in different forms and Have a thorough understanding of the fundamental concepts and techniques used in digital electronics. 2. To understand Postulates of Boolean algebra and to minimize combinational functions and its application in digital design. 3. To understand, analyze and design various combinational and sequential circuits. 4. Ability to identify basic requirements for a design application and propose a cost effective solution. 5. To Represent a sequential circuit using Finite State machine and apply state minimization techniques to design a FSM						

Detailed Contents

UNIT-I: Introduction

Digital & analog signals, Number System, BCD & its arithmetic, Binary, Decimal, Octal, Hexadecimal, Negative numbers & its arithmetic, Number base conversions,

Unit II: Logic Realization & Simplification

Boolean Algebra and De Morgan's Theorem, SOP & POS forms, Canonical forms, Karnaugh maps up to 6 variables, Binary codes, Code Conversion. Logic Gates and its

realization.

Unit III: Combinational Logic Design

MSI devices like Comparators, Multiplexers, Encoder, Decoder, Driver & Multiplexed Display, Half and Full Adders, Subtractors, Serial and Parallel Adders, BCD Adder, Barrel shifter and ALU, parity generator, checker

Unit IV: Sequential Logic Design

Building blocks like S-R, JK and Master-Slave JK FF, T-FF, D-FF and Flip-Flop conversions. Shift Registers (SISO, SIPO, PISO, PIPO), universal shift register. Synchronous and Asynchronous counters and its realization. Programmable logic Families: PAL, PLA, PROM.

Unit V: Finite State Machines:

Design of synchronous FSM, Mealy model, Moore model, state diagrams and state reduction method, overlapping & Non-overlapping models. Designing synchronous circuits like Pulse train generator, Pseudo Random Binary Sequence generator.

Text Books

1. M. M. Mano, "Digital Design - Third Edition," Pearson Education/PHI, 2002.
2. A. P. Malvino and D. P. Leach, "Digital Principles and Applications," TATA McGraw Hill Edition, 2007

References

1. C. H. Roth, Jr., "Fundamentals of Logic Design," 5th edition, Thomson, 2004.
2. Kohavi, Z. "Switching and finite automata theory". Tata McGraw Hill 1978.
3. Rao, C. V. S. "Switching and logic design". Pearson Education 2002.
4. Givone, D. "Digital principles and design". Tata McGraw Hill Education 2003.
5. Rafiquzzaman, M. "Fundamentals of digital logic and microcomputer design" 5th edition John Wiley & Sons 2005.
6. A. Anand Kumar, "Switching Theory and Logic Design" 3rd edition, PHI 2008.

Video links:

1. <https://youtu.be/fwTRzApMCgs>.
2. <https://youtu.be/CeD2L6KbtVM>.
3. <https://youtu.be/o56BWcHZteQ>.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
EC1701	Digital Electronic Circuits Lab				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives • To introduce the basics of digital electronics and the concepts of gates and logic circuits. • To provide hands-on experience in designing and testing digital circuits. • To impart the knowledge of digital components such as flip flops, registers, and counters. • To teach the design and implementation of complex digital circuits such as encoders, decoders, and multiplexers.						
Course Outcomes At the end of this course Students will be able to: 1. Understand the working of basic digital components such as gates and their application in logic circuits. 2. Design and test simple digital circuits such as full adders, subtractors, and comparators. 3. Design and test more complex digital circuits such as shift registers, flip flops, and counters. 4. Design and implement digital circuits using multiplexers, decoders, and encoders. 5. Design, Analyze, Implementation of application-level projects.						

LIST OF EXPERIMENTS

1. I/O characteristics of Universal, Basic, Arithmetic gates.
2. Design of a digital comparator.
3. Check the functionality of a 1 bit full adder circuit and subtractor.
4. Develop 4 bit RCA .
5. Realize the functionalities of encoders and decoders.
6. Design sr-latch and flip flop.
7. Design jk-latch and flip flop.
8. Functioning of shift register, master slave flip flop, ALU.
9. Design of asynchronous and synchronous counters.
10. Verify the functionality of a n*1 multiplexer and 1*n demultiplexer.
11. Design of a 7-segment LED display.

FIRST YEAR (E1) – SEMESTER – I

Course Code	Course Title				Course Type	
BM1105	Constitution of India				MC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	2	0	0	40	60	0
Course Objectives ● To provide basic information about the Indian constitution. ● To identify individual roles and ethical responsibility towards society. ● To understand human rights and its implications.						
Course Outcomes 1. Understand the formation and principles of Indian Constitution. 2. Understand Fundamental Rights and its implications in life. 3. Understand Fundamental Duties of Individual toward country and society 4. Understand Directive principles to govern the policy formation 5. Understand the Way of running the Government and basic Governance						

Detailed Contents

UNIT-I

Introduction to Indian Constitution:

- Meaning of the term Constitution
- Historical background of Indian constitution
- Making of Indian constitution
- Constituent Assembly

UNIT-II

Features of Indian Constitution

- Preamble of the Constitution , Importance, Scope, Relevance
- The Salient Features of Indian Constitution , Importance, Scope, Relevance

UNIT-III

Fundamental Rights

- Fundamental Rights
- Importance and scope of fundamental rights
- Categorization of Fundamental Rights

UNIT-IV

Fundamental Duties & The Directive Principles of State Policy:

- Fundamental Duties
- Importance and scope of fundamental Duties
- The Directive Principles of State Policy - Importance, Scope, Relevance

UNIT-V

Union/Central Government:

- Union Government
- Union Legislature (Parliament)
- Lok Sabha and Rajya Sabha (with Powers and Functions)
- Union Executive
- President of India (with Powers and Functions)
- Prime Minister of India (with Powers and Functions)

TextBooks:

1. 'Indian Polity' by Laxmikanth
2. 'Indian Administration' by Subhash Kashyap

References:

1. 'Indian Constitution' by D.D. Basu
2. 'Indian Administration' by Avasti and Avasti

Course Code	Course Title				Course Type	
MA1202	Differential Equations and Vector Calculus				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4

Course Objectives

- Methods of solving the differential equations of first and higher order.
- To study the methods of solving improper integrals and the concepts of multiple integrals.
- The basic properties of vector valued functions and their applications to line, surface and volume integrals.
- To study numerical methods to analyze experimental data.

Course Outcomes

1. Solve first order linear differential equations and special non linear first order equations like Bernouli , Riccati & Clairaut's equations and compute double integrals over rectangles and type I and II" regions in the plane.
2. Explain the concept of a vector field and make sketches of simple vector fields in the plane
3. Explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics
4. Apply gradient to solve problems involving normal vectors to level surfaces.
5. Recognize the statements of Stokes' Theorem and the Divergence Theorem and understand how they are generalizations of the Fundamental Theorem of Calculus.
6. Able to solve the problems in diverse fields in engineering science using numerical methods.

Detailed Contents

UNIT-I: Ordinary Differential Equations of first order

Exact first order differential equation, finding integrating factors, linear differential equations, Bernoulli's ,Riccati , Clairaut's differential equations, finding orthogonal trajectory of family of curves, Newton's Law of Cooling, Law of Natural growth or decay.

UNIT-II: Ordinary Differential Equations of higher order

Second order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax} , $\sin(ax)$, $\cos(ax)$, polynomials in x , $e^{ax}V(x)$, $xV(x)$; method of variation of parameters; Equations reducible to linear ODE with constant coefficients: Legendre's equation, Cauchy-Euler equation.

UNIT-III: Integral Calculus

Evaluation of the double integrals (Cartesian and Polar), change of order of integration (only Cartesian form), Evaluation of Triple integrals. Change of variables (Cartesian to polar) in case of double integrals (Cartesian to spherical and cylindrical) in case of Triple Integrals-Jacobians of transformations. Differentiation of integrals with variable limits - Leibnitz rule.

Applications: Finding Areas (using double integrals) and volumes (using double and Triple Integrals), Centre of mass, Centre of gravity for constant and variable densities by double and triple integrals (applications involving cubes, Sphere and rectangular parallelepiped)

UNIT-IV: Vector Differentiation

Vector point functions and scalar point functions. Gradient, Divergence and Curl. Directional derivatives, Tangent plane and normal line. Vector Identities. Scalar potential functions. Solenoidal and Irrotational vectors.

UNIT-V: Vector Integration

Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

Numerical Methods: Introduction and motivation about numerical methods, True value, approximate value, error, error percentage, algebraic equations, transcendental equations, Newton-Raphson method, Bisection method.

Text Books

1. Advanced Engineering Mathematics (3rd Edition) by R. K. Jain and S. R. K. Iyengar, Narosa Publishing House, New Delhi
2. Advanced Engineering Mathematics (8th Edition) by Erwin Kreyszig, Wiley-India.

References

1. Dr. M.D. Raisinghania, Ordinary and Partial differential equations, S. CHAND, 17th Edition 2014.

Course Code	Course Title				Course Type	
	Engineering Physics				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3

Course Objectives

- To acquire skills allowing the student to identify and apply formulas of physics using course literature.
- To be able to identify and illustrate physical concepts and terminology used and to be able to explain them in appropriate detail.
- To be able to make approximate judgments of physical aspects and other phenomena when necessary.
- To acquire skills allowing the student to organize and plan simpler laboratory course experiments and to prepare an associated oral and written report.

Course Outcomes

1. The basic laws of physics, their corollaries, and comprehension of how they can be applied to explain specific natural phenomena within the five key topic-areas* as described in the mission statement of the physics undergraduate program.
2. Use of critical thinking, hypothesis building, and application of the scientific method to physics concepts, theoretical models and calculations, and laboratory experimentation.
3. Problem solving skills and relevant mathematical methods to approach, conceptualize, and achieve analytical or numerical solutions to physics problems within important sub-categories of the five topic-areas*.
4. Laboratory skills and exposure to a variety of important experiments at appropriate levels that illustrate phenomena discussed in the lecture classes. Instrumentation and experimental techniques; methods for quantitative analysis of data and measurement uncertainty.
5. General knowledge of the development of physics and the nature of scientific inquiry, particularly the progression from classical physics to the modern physics ideas of quantum mechanics, statistical mechanics, and relativity.
6. Contemporary areas of physics inquiry as introduced in upper-level physics and interdisciplinary elective courses, as well as in faculty-mentored undergraduate research available to all majors who seek this experience.
7. Written and oral communication skills for dissemination of scientific results in report, article, or oral presentation formats; standard citation methods; ethics in science and

scholarship and its importance to scientific inquiry and professionalism.
(*key-topic areas as given in the Mission Statement of the physics undergraduate program are:

- i) Electrodynamics; ii) Quantum Mechanics; iii) Electron Structure of Solids;
- iv) Semiconductor Physics; v) experimental methods.

Detailed Contents

Unit I: Vectors and Mathematical Physics

Gradient, Divergence, Curl and its applications, Line, surface and volume integrals, Stokes and Gauss theorem: Applications, Curvilinear Coordinates: Polar, Cylindrical and spherical co-ordinates, Problems. Electrodynamics before Maxwell, Fixing of Ampere's Law, Maxwell Equations in Matter, Boundary Conditions, Continuity Equation, Poynting Theorem, Wave equation for E and B, Monochromatic Plane Waves, Energy and Momentum in EM Waves. Propagation in Linear Media, Reflection and Transmission at Normal Incidence. EM Waves in Conductors, Reflection at Conducting Surface.

Unit II: Quantum Mechanics

Introduction to Quantum Mechanics, De-Broglie's waves and uncertainty principle, Wave Function and its Significance, Time dependant and time independent Schrodinger wave equations, Particle in a box - Problems.

Unit III: Electron Structure of Solids

Introduction to Crystallography, Bravais Lattices and crystal systems, Atomic Packing, Atomic Radii, Crystal Structures (SC, BCC and FCC), Miller Indices, Classical Free electron Theory, Kronig Penny model (E vs K), Band theory of solids.

Unit IV: Semiconductor Physics

Intrinsic and extrinsic semiconductors, Fermi level and carrier- concentration, Effect of temperature on Fermi level. Mobility of charge carriers and effect of temperature on mobility,

Unit V: Optics

Interference- Introduction and examples, Young's; double slit experiment, Diffraction – Types, Single Slit, Double Slit, Diffraction Grating.

Text Books

1. Engineering Physics by H.K. Malik and Singh
2. Electrodynamics by David.J.Griffiths

References

1. Quantum Mechanics by Aruldas.
2. Solid State Physics by C. Kittel

Course Code	Course Title				Course Type	
PH1801	Engineering Physics Lab				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives - To gain practical knowledge by applying experimental methods to correlate with the theory. - Apply the analytical techniques and graphical analysis to the experimental data. - To develop intellectual communication skills and discuss the basic understanding of various experimental principles involved.						
Course Outcomes - Prepare and perform individually a wide spectrum of experiments. - Present experimental data in various appropriate forms like tabulation and plots. - Analyze, interpret and summarize the experimental results. - Communicate clearly the understanding of various experimental principles, instruments/setup, and procedure.						

List of experiments:

1. Photoelectric effect
2. Hall effect
3. Ultrasonic Interferometer
4. Melde's Experiment
5. Four probe Method
6. Frank hertz Experiment
7. Seebeck and Peltier effect
8. Solar cell
9. Couple pendulum

Course Code	Course Title				Course Type	
CY2101	Engineering Chemistry				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To provide an understanding of electrochemistry and its practical applications. • To introduce the concepts of corrosion and water treatment. • To study different energy sources and their characteristics. • To understand the basics of chemical kinetics and reaction rates. • To introduce the properties and synthesis of nanomaterials.						
Course Outcomes 1. Understand the fundamentals of electrochemistry and its applications in various fields. 2. Identify different types of corrosion and apply corrosion prevention methods. 3. Analyze different types of fuels and their characteristics. 4. Understand the basics of chemical kinetics and solve problems related to reaction rates. 5. Recognize different types of nanomaterials and their properties, and apply the synthesis techniques to produce them.						

Detailed Contents

Unit I: Electrochemistry

Introduction to electrochemistry: Galvanic cell (Daniel cell), Nernst equation. Types of electrodes: metal-metal ion electrodes, metal-insoluble salt-anion electrodes, calomel electrode, gas-ion electrodes, hydrogen and chlorine electrodes, oxidation-reduction electrodes (quinhydrone electrode), amalgam electrodes and ion exchange electrode (glass electrode). EMF and applications of EMF: determination of pH of the solution, potentiometric titrations, Classification of commercial cells - primary cells (dry cell) and secondary cells (Lithium ion battery, Pb-acid storage battery). Fuel cells: H₂-O₂ fuel cell.

UNIT - II: Corrosion and water treatment

Dry and wet corrosion and their mechanisms. Pilling - Bedworth Rule. Types of Corrosion: galvanic corrosion, concentration cell corrosion, pitting corrosion and stress corrosion. Factors influencing the rate of corrosion: Temperature, pH and dissolved oxygen. Corrosion Prevention methods: Cathodic protection – Sacrificial Anodic method and Impressed current method. Metallic coatings: galvanization and tinning methods. Water:

Hardness of water, Degrees of hardness. Calculation of hardness by EDTA method. Disadvantages of hard water in boilers: priming, foaming, scales, sludges and caustic embrittlement. Treatment of boiler feed water: Zeolite process, Ion exchange process.

UNIT - III: Energy sources

Introduction. Definition and classification of fuels. Calorific value of a fuel, Characteristics of a good fuel. Coal, types of Coal. Analysis of Coal: Proximate and Ultimate analysis. Bomb Calorimeter and Junker's gas Calorimeter. Problems on calculation of calorific value. Liquid fuels. Introduction. Synthetic Petrol: Fisher Tropsch process. Introduction to Bio-fuels: Bio-diesel, Bio-gas

Unit IV: Chemical kinetics

Introduction to rate of reaction and rate constant determination. Factors influencing rate of reaction. Complex reactions: definition and classification of complex reactions, definition of reversible reactions with examples, rate law derivation for reversible reactions. Consecutive reactions: definition, rate law derivation and examples of consecutive reactions. Parallel reactions: definition, rate law derivation and examples of parallel reactions. Steady-state approximation: introduction, kinetic rate law derivation by applying steady state approximation in case of the oxidation of NO and pyrolysis of methane.

UNIT - V: Nanochemistry

Introduction to nanomaterials, classification: Carbon based nanomaterials, metallic nanoparticles, metal oxide nanoparticles. Properties at nanoscale. Synthetic approaches: Top-Down (Photolithography, ball milling) and Bottom-Up (Sol-gel, Hydrothermal). Brief overview on characterization of nanomaterials: X-ray, SEM and TEM. Applications of nanomaterials.

Text Books

1. S. Agarwal, Engineering Chemistry. New Delhi, India: Dhanpat Rai & Co., 2017.
2. S. Chawla, A Textbook of Engineering Chemistry. New Delhi, India: Dhanpat Rai & Co., 2013

References.

1. S. Chawla, Essentials of Experimental Engineering Chemistry. New Delhi, India: Dhanpat Rai & Co., 2009.
2. O.P. Pandey, Practical Chemistry. New Delhi, India: S. Chand and Company Ltd., 2011.
3. V.K. Ahluwalia, College Practical Chemistry. New Delhi, India: New Age International Publishers, 2015.
4. K. Mukkanti, Practical Engineering Chemistry. New Delhi, India: BS Publications, 2011.
5. R. Kulakarni and Adil, Laboratory Manual. New Delhi, India: Prentice-Hall of India Pvt. Ltd., 2010

Course Code	Course Title				Course Type	
CY2701	Engineering Chemistry Lab				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives - To learn the preparation of organic compounds in the laboratory. - To estimate the hardness and alkalinity of the given sample of water. - To understand the Job's method for determining the composition. - Learn how to use the pH meter and polarimeter. - Synthesis of a pharmaceutically active drug.						
Course Outcomes Minimum knowledge on basic synthesis, quantitative and qualitative analysis is being imparted.						

List of Experiments

- Determination of the strength of weak acid (CH_3COOH) by pH metry.
- Conductometric titration (strong acid (HCl) vs strong base (NaOH)).
- Throwing power of Copper.
- Estimation of alkalinity of water.
- Determination of total hardness of water by complexometric method using EDTA.
- Determination of the calorific value of fuel sample by using bomb calorimeter.
- Preparation of bio-diesel from palm oil by trans esterification method.
- The rate constant and order of the reaction of the hydrolysis of an ester catalyzed by an acid (dil.HCl).
- Preparation of Nano particle (ZnO).

References

- V.K. Ahluwalia, S. Dhingra, and A. Gulati, "College Practical Chemistry," New Delhi, India: S. Chand Publishing, 2016.
- K. Mukkanti, "Practical Engineering Chemistry," Hyderabad, India: Universities Press, 2010.
- S. Chawla, "A Textbook of Engineering Chemistry," 16th ed., New Delhi, India: Dhanpat Rai & Co., 2017.
- S. Chawla, "Essentials of Experimental Engineering Chemistry," 2nd ed., New Delhi, India: Dhanpat Rai & Co., 2015.
- V.K. Ahluwalia and R. Aggarwal, "Comprehensive Practical Organic Chemistry – Preparation and Quantitative Analysis," New Delhi, India: S. Chand Publishing, 2010

Course Code	Course Title					Course Type
HS1201	English					HSMC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	2	0	0	40	60	2
Course Objectives: • Improve the language proficiency of students in English with an emphasis on Vocabulary, Grammar, Reading and Writing skills. • Equip students to study academic subjects more effectively and critically using The theoretical and practical components of English syllabus. • Develop study skills and communication skills in formal and informal situations.						
Course Outcomes 1. Use English Language effectively in spoken and written forms. 2. Comprehend the given texts and respond appropriately. 3. Communicate confidently in various contexts and different cultures. 4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.						

Introduction

In view of the growing importance of English as a tool for global communication and the consequent emphasis on training students to acquire language skills, the syllabus of English has been designed to develop linguistic, communicative and critical thinking competencies of Engineering students. In English classes, the focus should be on the skills development in the areas of vocabulary, grammar, reading and writing. For this, the teachers should use the prescribed text for detailed study. The students should be encouraged to read the texts leading to reading comprehension and different passages may be given for practice in the class. The time should be utilized for working out the exercises given after each excerpt, and also for supplementing the exercises with authentic materials of a similar kind, for example, newspaper articles, advertisements, promotional material etc.

The focus in this syllabus is on skill development, fostering ideas and practice of language skills in various contexts and cultures.

Detailed Contents

UNIT-I

‘The Raman Effect’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary Building: The Concept of Word Formation --The Use of Prefixes and Suffixes.

Grammar: Identifying Common Errors in Writing with Reference to Articles and Prepositions.

Reading: Reading and Its Importance- Techniques for Effective Reading.

Basic Writing Skills: Sentence Structures -Use of Phrases and Clauses in Sentences

Importance of Proper Punctuation- Techniques for writing precisely – Paragraph writing – Types, Structures and Features of a Paragraph - Creating Coherence-Organizing Principles of Paragraphs in Documents.

UNIT-II

‘Ancient Architecture in India’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Synonyms and Antonyms.

Grammar: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement.

Reading: Improving Comprehension Skills – Techniques for Good Comprehension

Writing: Format of a Formal Letter-Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Job Application with Resume.

UNIT-III

‘Blue Jeans’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Acquaintance with Prefixes and Suffixes from Foreign Languages in English to form Derivatives-Words from Foreign Languages and their Use in English.

Grammar: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.

Reading: Sub-skills of Reading- Skimming and Scanning.

Writing: Nature and Style of Sensible Writing- Defining- Describing Objects, Places and Events – Classifying- Providing Examples or Evidence.

UNIT-IV

‘What Should You Be Eating’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Standard Abbreviations in English

Grammar: Redundancies and Clichés in Oral and Written Communication.

Reading: Comprehension- Intensive Reading and Extensive Reading.

Writing: Writing Practices--Writing Introduction and Conclusion - Essay Writing-Précis Writing.

UNIT-V

‘How a Chinese Billionaire Built Her Fortune’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Technical Vocabulary and their usage

Grammar: Common Errors in English

Reading: Reading Comprehension-Exercises for Practice

Writing: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report.

Text Books

1. Sudarshana, N.P. and Savitha, C. (2018). English for Engineers. Cambridge University Press.
2. Swan, M. (2016). Practical English Usage. Oxford University Press

References

1. Kumar, S and Lata, P.(2018). Communication Skills. Oxford University Press.
2. Wood, F.T. (2007).Remedial English Grammar. Macmillan.
3. Zinsser, William. (2001). On Writing Well. Harper Resource Book.
4. Hamp-Lyons, L. (2006).Study Writing. Cambridge University Press.
5. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.

Course Code	Course Title				Course Type	
CE1801	Computer Aided Engineering Graphics				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	1	0	4	40	60	3
Course Objectives • To introduce the students to the “Universal Language of Engineers” for effective communication through drawing. • To understand the basic concepts of drawing through modern techniques. • To impart knowledge about standard principles of projection of objects. • To provide the visual aspects of Engineering drawing using Auto-CAD.						
Course Outcomes 1. Use Engineering principles and techniques to understand and interpret engineering drawings. 2. Understand the concepts of Auto-CAD. 3. Draw orthographic projections of lines, planes and solids using Auto-CAD. 4. Use the techniques, skills and modern engineering tools necessary for engineering practices.						

UNIT-I: Introduction to Engineering Drawing

Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, types of lines and Dimensioning.

Over view of Auto-CAD: Theory of CAD software (The Menu System, Tool Bars, drawing area, Dialogue boxes, Shortcut Menu, the command lines, Select and erase objects, Introduction to layers etc.), Drawing simple figures- lines, planes, solids.

UNIT-II: Geometrical constructions

Construction of regular polygons.

Conic sections: Construction of Ellipse, Parabola, Hyperbola (General method only), Cycloid, Epicycloid, Hypocycloid and Involute.

Scales: Construction of Plain, Diagonal and Vernier scales.

UNIT-III:

Orthographic projections: Principles of Orthographic Projections

Projections of Points: Projections of Points placed in different quadrants

Projection of lines: lines parallel and inclined to both the planes (Determination of true lengths and true inclinations and traces)

Projection of planes: Planes inclined to both the reference planes

UNIT-IV:

Projection of Solids: Projection of solids whose axis is parallel to one of the reference planes and inclined to the other plane, axis inclined to both the planes

Projection of sectioned solids: Sectioning of simple solids like prism, pyramid, cylinder and cone in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section.

UNIT-V:

Development of surfaces: Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone

Isometric Projections: Principles of Isometric projection – Isometric Scale, Isometric Views of planes and simple solids

Perspective projections: Basic concepts of perspective views.

Text Books

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House
2. Gopalakrishna K.R., “Engineering Drawing” (Vol. I&II combined), Subhas Stores, Bangalore, 2007

References

1. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
2. Venugopal K. and Prabhu Raja V., “Engineering Graphics”, New Age publications
3. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication
4. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers (Corresponding set of) CAD Software Theory and User Manuals

Course Code	Course Title					Course Type
	Soft Skills					HSMC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	2	40-	60	1
Course Objectives • To enable students to speak effectively in formal and informal situations. • To equip the students with necessary writing skills in order to face the corporate world. • To strengthen the writing skills of the students and help them in documentation. • To enable students to sharpen their communication skills towards writing a persuasive resume and effective job application letters. • To equip students with pre-presentation steps, to understand the structure of a good presentation, and devise various techniques for delivering a successful presentation. • To make students understand the importance of teamwork and group presentations and group discussions.						
Course Outcomes 1. Communicate effectively in formal and informal situations. 2. Understand the structure and mechanics of writing resumes, reports, documents and E-mails. 3. Present effectively in academic and professional contexts. 4. Develop communication in writing for a variety of purposes. 5. Identify areas of evaluation in Group Discussions conducted by organizations as part of the selection procedure. 6. Overcome stage fear and tackle questions.						

Detailed Contents

UNIT-I: Activities on Fundamentals of Inter-personal Communication

Activities on Fundamentals of Interpersonal Communication Starting a conversation - responding appropriately and relevantly - using the right body language-Role Play in different situations & Discourse Skills using visuals.

UNIT-II: Activities on Reading Comprehension

General Vs Local comprehension- reading for facts- guessing meanings from context- scanning- skimming- inferring meaning- critical reading – surfing Internet

UNIT-III: Activities on Writing Skills

Structure and presentation of different types of writing- Resume writing/ e-correspondence/ Technical report writing- planning for writing - improving one's writing.

UNIT-IV: Activity on Presentation Skills

Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations

UNIT-V: Activities on Group Discussion, Debate and Interview Skills

Dynamics of group discussion- intervention- summarizing-modulation of voice-body language-relevance-fluency and organization of ideas and rubrics for evaluation- Concept and process-pre-interview planning-opening strategies-answering strategies- interview through tele-conference & video-conferencing - Mock Interviews

Text Books

1. Soft Skills Training: A workbook to Develop Skills for Employment – By Frederick H.
2. Everyone Communicates, Few People Connect: What the Most Effective People do Differently – By John C. Maxwell

References

1. Soft Skills Training: A workbook to Develop Skills for Employment – By Frederick H. Wentz
2. How to Talk to Anyone: 92 Little Tricks to Have Big success in Relationships – By Leil Lowndes
3. Teamwork101: What Every Leader Needs to Know – By John C. Maxwell
4. AdaptAbility: How to Survive Change You Didn't Ask For- By M.J. Ryan
5. Conflict Communication: A New Paradigm in Conscious Communication – By Rory Miller

Course Code	Course Title				Course Type	
	English Language Lab				HSMC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	2	40	60	1

Course Objectives

- To facilitate computer-assisted multimedia instruction enabling individualized and independent language learning.
- To sensitize students to the nuances of English speech sounds, word accent, intonation and rhythm.
- To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking.
- To improve the fluency of students in spoken English and neutralize their mother tongue influence.
- To train students to use language appropriately for public speaking and interview.
- To enable students to develop their listening skills so that they may appreciate its role in the LSRW skills approach to language and improve their pronunciation.
- To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.
 - Listening for general content.
 - Listening to fill up information.
 - Intensive listening.
 - Listening for specific information.

Speaking Skills

- To involve students in speaking activities in various contexts.
- To enable students to express themselves fluently and appropriately in social and professional contexts.
- Oral practice: Just A Minute (JAM) Sessions.
- Describing objects/situations/people.
- Role play – Individual/Group activities.

Course Outcomes

1. Better understanding of nuances of English language through audio- visual experience and group activities.
2. Neutralization of accent for intelligibility.
3. Speaking skills with clarity and confidence which in turn enhances their employability.

Detailed Contents**UNIT – I**

Understand Listening Skill- Its importance – Purpose- Process- Types- Barriers of Listening - Communication at Work Place- Spoken vs Written language.

Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants -Ice-Breaking Activity and Just a Minute (JAM) Session- Situational Dialogues – Greetings – Taking Leave Introducing Oneself and Others.

UNIT – II

Understand Structure of Syllables – Word Stress and Rhythm– Weak Forms and Strong Forms in Context- Features of Good Conversation – Non-verbal Communication.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms in Context-Situational Dialogues – Role-Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

UNIT - III

Understand Intonation-Errors in Pronunciation- Mother Tongue Influence (MTI)- How to make

Formal Presentations.

Practice: Common Indian Variants in Pronunciation – Differences in British and American Pronunciation- Formal Presentations.

UNIT – IV

Understand Listening for General Details-Public Speaking – Exposure to Structured Talks.

Practice: Listening Comprehension Tests- Making a Short Speech – Extempore

UNIT – V

Understand Listening for Specific Details- Interview Skills.

Practice: Listening Comprehension Tests- Mock Interviews.

References

1. Clarity English Success - Software.
2. Connected Speech- Software.
3. Issues in English 2- Software.
4. <http://www.clarityenglish.com/program/practicalwriting/>
5. <http://www.clarityenglish.com/program/roadtoielts/>
6. <http://www.clarityenglish.com/program/clearpronunciation1/>
7. <http://www.clarityenglish.com/program/resultsmanager>

Course Code	Course Title				Course Type	
	ELECTRONIC DEVICES AND CIRCUITS (EDC)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4

Course Objectives

- To introduce the fundamental concepts of semiconductor materials and its characteristics.
- To study about Rectifier and Filter circuits.
- To understand the applications of Bipolar Junction Transistors ..
- To understand the applications of Field effect Transistors .
- To understand the construction and working of Tunnel Diode and Varactor Diode.

Course Outcomes

1. Students will be good at fundamental concepts of semiconductor materials and its characteristics.
2. Students will be able to understand about the Rectifiers and Filter Circuits.
3. Students will be good at the basic structure of BJT and Its Applications with its different modes of operation.
4. Students will know the applications of Field Effect Transistors and its different modes of operation.
5. Students will be good at the basic structure of Tunnel diode , Varactor diode with its working.

Detailed Contents

UNIT-1: Semiconductor Physics

Review of semiconductor physics. Energy bands in intrinsic and extrinsic silicon; Carrier transport: diffusion current, drift current, mobility and resistivity; Generation and recombination of carriers; Poisson and continuity equation, Hall effect.

Diodes: Qualitative Theory of P-N Junction, P-N Junction as a Diode, Diode Equation, Volt- Ampere Characteristics, Temperature dependence of V-I characteristic, Ideal versus Practical – Resistance levels (Static and Dynamic), Transition and Diffusion Capacitances, Diode Equivalent Circuits, Load Line Analysis, Breakdown Mechanisms in Semiconductor Diodes, Zener Diode Characteristics and its applications.

UNIT-II: Rectifiers and Filters

The P-N junction as a Rectifier, Half wave Rectifier, Full wave Rectifier, Bridge Rectifier, Harmonic components in a Rectifier Circuit, Inductor Filters, Capacitor Filters, L-Section Filters, π -Section Filters, Comparison of Filters.

Opto-Electronics: Optical sources: LED, Direct & Indirect band gap semiconductors. Optical detectors: Photo diode, Pin diode, Solar cell.

UNIT-III: Bipolar Junction Transistor

The Junction Transistor, Transistor Current Components, Transistor as an Amplifier, Transistor Construction, non-ideal effects of BJT: Base width modulation, Emitter band gap narrowing, non-uniform base doping, breakdown voltage. BJT Operation, Common Base, Common Emitter and Common Collector Configurations, Limits of Operation, BJT Specifications.

Transistor Biasing and Stabilization: Operating Point, The DC and AC Load lines, Need for Biasing, Fixed Bias, Collector Feedback Bias, Emitter Feedback Bias, Collector - Emitter Feedback Bias, Voltage Divider Bias, Bias Stability, Stabilization Factors, Stabilization against variations in V_{BE} and β , Bias Compensation using Diodes and Transistors, Thermal Runaway, Thermal Stability.

UNIT-4: Field Effect Transistor

The Junction Field Effect Transistor (Construction, principle of operation, symbol) – Pinch-off Voltage - Volt-Ampere characteristics, The JFET Small Signal Model, MOSFET (Construction, principle of operation, symbol), MOSFET Characteristics in Enhancement and Depletion modes.

FET Amplifiers: FET Common Source Amplifier, Common Drain Amplifier, Generalized FET Amplifier, Biasing FET, FET as Voltage Variable Resistor, Comparison of BJT and FET.

UNIT-V: Special Purpose Electronic Devices

Principle of Operation and Characteristics of Tunnel Diode (with the help of Energy Band Diagram), Varactor Diode.

Text Books

1. G. Streetman, and S. K. Banerjee, “Solid State Electronic Devices,” 7th edition, Pearson, 2014.
2. D. Neamen , D. Biswas "Semiconductor Physics and Devices," McGraw-Hill Education.

References

1. Jacob Millman, Christos C Halkias and Satyabrata JIT, “ Electronics Devices and Circuits”, 3rd Edition.
2. S. M. Sze and K. N. Kwok, “Physics of Semiconductor Devices,” 3rd edition, John Wiley&Sons, 2006.
3. C.T. Sah, “Fundamentals of solid state electronics,” World Scientific Publishing Co. Inc, 1991.
4. Y. Tsividis and M. Colin, “Operation and Modeling of the MOS Transistor,” Oxford Univ.Press, 2011.

Online Resources:

1. <https://www.youtube.com/@esegatepreparationforecast7171>
2. <https://www.youtube.com/watch?v=xhn188JafbM&list=PL350612601E2DBFDE>

Course Code	Course Title					Course Type
	Signals and Systems (SS)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	4	40	60	4
Course Objectives • To provide basics of signals and systems required for analyzing various areas related to signal processing. • To analyze spectral behavior of signals and systems using different transform techniques. • To develop mathematical skills to solve problems involving convolution. • To understand signals and systems in terms of both the time and frequency domains. • To understand the importance of sampling theorem						
Course Outcomes 1. To define the basic concepts of signals and describe the classification of continuous signals and systems 2. To outline and express the spectral characteristics of continuous time signals using Fourier series representation and Fourier transform techniques. 3. To apply the contents of convolution in time domain, frequency domain and construct the graphical representation of convolution of two different signals to interpret the response of an LTI system. 4. To establish sampling theorem for band limited signals, distinguishing between different types of samplings and examine the signal reconstruction. 5. To explain the need for Laplace Transformation, evaluate the Laplace Transform and Inverse Laplace Transform of different types of signals assessing Region of Convergence and apply the Laplace Transform techniques to engineering problems.						

Detailed Contents

UNIT I: Introduction to Signals and Systems

Signals and systems as seen in everyday life, and in various branches of engineering and science. Continuous and discrete time signals, Analog and Digital signals and some special signals of importance: the unit step, the unit impulse, the sinusoid, the complex exponential, some special time-limited signals; Signal properties: periodicity, absolute integrability, determinism and stochastic character, energy and power, odd and even, Operations on

independent variables of the signal: Time shifting, reversal, scaling; System properties: linearity: additivity and homogeneity, shift-invariance, causality, stability, realizability with examples.

Unit II: Behavior of continuous and discrete-time LSI systems

Impulse response and step response, convolution with examples, cascade interconnections. Characterization of causality and stability of LSI systems. System representation through differential equations and difference equations. Periodic inputs and semi periodic inputs to an LSI system, the notion of a frequency response and its relation to the impulse response. notion of eigen functions of LSI systems, a basis of eigen functions.

Unit III: Fourier Transform

The idea of signal space and orthogonal bases, Fourier series representation of periodic signals with examples, Waveform Symmetries. Continuous Time Fourier Transform and it's properties including Parseval's Theorem and Duality, magnitude and phase response, The DiscreteTime Fourier Transform (DTFT). Ideal Filters, R-C first order LPF, HPF, BPF circuits and its frequency domain analysis using transfer function.

Unit IV: Laplace Transform

Review of the Laplace Transform for continuous time signals and systems, Region of Convergence, system functions, poles and zeros of system functions and signals, Laplace domain analysis, solution to differential equations and system behavior. Butterworth LPF, HPF and BPF and its realization.

Unit V: Sampling and Reconstruction

The Sampling Theorem and its implications. Spectra of sampled signals. Reconstruction: ideal interpolator, zero-order hold, first-order hold. Aliasing and its effects. Relation between continuous and discrete time systems.

Text books

1. Simon Haykin and Van Veen, Signals & Systems, 2nd edition, Wiley, 2003
2. A. V. Oppenheim, A.S. Willsky and S.H. Nawab, Signals and Systems, 2nd edition, 2013, PHI.

References

1. A. V. Oppenheim, A.S. Willsky and S.H. Nawab, Signals and Systems, 2nd edition, 2013, PHI.
2. R.F. Ziemer, W.H. Tranter and D.R. Fannin, "Signals and Systems - Continuous and Discrete", 4th edition, Prentice Hall, 1998.
3. Papoulis, "Circuits and Systems: A Modern Approach", HRW, 1980.
4. B.P. Lathi, "Signal Processing and Linear Systems", Oxford University Press, c1998.

5. Douglas K. Lindner, "Introduction to Signals and Systems", McGraw Hill International Edition: c1999.
6. Robert A. Gabel, Richard A. Roberts, "Signals and Linear Systems", John Wiley and Sons, 1995.
7. M. J. Roberts, "Signals and Systems - Analysis using Transform methods and MATLAB", TMH, 2003.
8. J. Nagrath, S. N. Sharan, R. Ranjan, S. Kumar, "Signals and Systems", TMH New Delhi, 2001.
9. Ashok Ambardar, "Analog and Digital Signal Processing", 2nd Edition, Brooks/ Cole Publishing Company (An international Thomson Publishing Company), 1999.

Online Resources

1. Signals and Systems, Prof. Dennis Freeman, Massachusetts Institute of Technology:
https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/video_galleries/lecture-re-videos/
2. Signals and Systems, Sai Radha Krishna G, Assistant Professor, Rajiv Gandhi University of Knowledge Technologies, Basar
https://www.youtube.com/watch?v=jt1s_zRg9u4&list=PL7Y2n876EILCJSRTVEJNPAVaCUe9_8dKP

EC2901

ELECTRONICS MINI PROJECT-I

L-T-P-C

0-0-2-1

Topics:

Analog Filter Design:

1. Butterworth, Chebyshev, Elliptic filter design for LPF, HPF, BPF, BSF (Need to verify with MultiSim software and realize the same using passive components)
2. Concepts related to Instrumentation and Measurements

Course Code	Course Title					Course Type
	Transform Calculus and Complex Analysis (TCCA)					BSC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	4	40	60	4
Course Objectives • Teach linear algebra and complex variables for engineering problem solving • Emphasize application of SVD in image processing • Teach calculus of complex variables including differentiation, integration, and residue calculus • Equip students with skills for definite integration and problem solving • To Solve the Differential & integral equations using Laplace Transform. • To know the Applications of Laplace Transforms						
Course Outcomes 1. Use shifting theorems to compute the Laplace transform and inverse Laplace transform 2. Solve Differential equations and Partial differential equations using Laplace Transforms. 3. Write the LU, QR, SVD decompositions for given matrices. 4. Finding the Orthogonal basis for a given Inner Product space. 5. Evaluate improper real integrals using Residue theorem. 6. Find the Fourier series for a given function. 7. Apply Fourier Transform to solve the differential and Partial Differential Equations.						

UNIT-I: Complex Variable – Differentiation

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding Harmonic conjugate; Elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties

UNIT-II: Complex Variable-Integration

Contour integrals, Cauchy-Goursat theorem (without proof), and Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, Zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (Without proof), Evaluation of definite integral involving sine and cosine functions. Evaluation of certain improper integrals using the Bromwich contour

Unit-III: Laplace Transform

Definition of Laplace Transform, linearity property, conditions for existence of Laplace Transform. First and second shifting properties, Laplace Transform of derivatives and integrals, unit step functions, Dirac delta-function, error function. Differentiation and

integration of transforms, convolution theorem. Inverse Laplace Transform, periodic functions. Evaluation of integrals by Laplace Transform. Solution of initial and boundary value problems and solving Differential Equations & Integral Equations.

UNIT-IV: Fourier Series

Periodic functions, Fourier series representation of a function, half range series, sine and cosine series, Fourier integral formula, Parseval's identity

UNIT-V: Fourier Transform

Fourier Transform, Fourier sine and cosine transforms. Linearity, scaling, frequency shifting and time shifting properties. Self-reciprocity of Fourier Transform, convolution theorem. Applications to boundary value problems

Text Books

1. Introduction to Linear Algebra ,Gilbert Strang fourth edition
2. Advanced Engineering Mathematics (3rd Edition) by R. K. Jain and S. R. K. Iyengar, Narosa Publishing House, New Delhi

References

1. B.S. Grewal and J.S. Grewal, “Higher Engineering Mathematics”,(40th Edition), Khanna Publishers,2007
2. Advanced Engineering Mathematics (8th Edition) by Erwin Kreyszig, Wiley-India.
3. Dr .M .D. Raisinghania,Ordinary and Partial differential equations,S. CHAND,17th Edition2014.
4. R.V.Churchill, “ Complex Variables & its applications”,Mc Graw-Hill Company,INC.

Course Code	Course Title					Course Type
	Essence of Indian Traditional Knowledge(EITK)					MC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	2	0	0	40	60	0
Course Objectives - To provide an overview of the Indian knowledge system and its various components - To explore the relationship between modern science and the Indian knowledge system - To introduce the schools of Indian philosophy - To cover the Indian linguistic tradition and the artistic traditions of India						
Course Outcomes - Understand the basic structure of the Indian knowledge system - Identify the different types of Vedas, Upavedas, and Vedangas - Recognize the major philosophical traditions of India, including orthodox and heterodox schools - Gain knowledge about the Indian linguistic and artistic traditions - Appreciate the relationship between modern science and the Indian knowledge system - Develop a broader understanding of India's cultural heritage and traditions.						

Unit – I: Basic structure of Indian knowledge system

Veda (Ayurveda, Dhanurveda, Gandharva Veda, Sthapatya aati (Shilpa Veda), Artha Veda, Veedanga (Shiksha, Kalpa, Chhanda, Niruktha, Vyakarana, Jyotishya) Dharma Shastra, Mimasha, Purana, Tarka Shastra

Unit – II

Modern science and Indian knowledge system Yoga holistic health care

Unit – III

Indian philosophical tradition:

- Orthodox (Hindu) school: Samkhya, Yoga, Nyaya, Vaisheshika, Purva Mimamsa, Vedanta,
- Heterodox (non-Hindu) schools: Carvaka, Jain, Buddha

Unit-IV

Indian linguistic tradition: phonology, morphology, syntax and semantics

Unit –V

Indian artistic tradition: Chitra kala, Mantra kala, Vaastu kala, Sangeetha kala, Nruthyu evam sahityam

Text Books

1. "Indian Knowledge Systems" by Dr. K. Ramakrishna Rao
2. "Indian Philosophy: A Very Short Introduction" by Sue Hamilton

Course Code	Course Title				Course Type	
	Environmental Science (ES)				MC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	2	0	0	40	60	0

Course Objectives

- To introduce students to the multidisciplinary nature of environmental studies and its scope and importance.
- To create awareness among students about the need for conservation and sustainable use of natural resources.
- To educate students about the structure, function, and value of ecosystems and biodiversity, as well as the threats they face and methods for their conservation.
- To familiarize students with the causes, effects, and control measures of different types of pollution and the legal framework for environmental protection.
- To sensitize students to the social issues related to the environment and their role in promoting sustainable development.

Course Outcomes

1. Students will be able to understand the complex interrelationships between different components of the environment and their impact on human well-being.
2. Students will develop an appreciation for the importance of conservation and sustainable use of natural resources and become aware of their individual and collective responsibilities towards the environment.
3. Students will be able to understand the functioning of different ecosystems and their role in maintaining the ecological balance.
4. Students will be able to identify different types of pollution and their causes, effects, and control measures, and develop an understanding of the legal framework for environmental protection.
5. Students will become aware of the social issues related to the environment and the role of education, technology, and individual actions in promoting sustainable development.

Detailed Contents

UNIT 1: MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL STUDIES

Definition, scope and importance, need for public awareness.

UNIT 2: NATURAL RESOURCES

Renewable and non-renewable resources: Natural resources and associated problems.

- a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.
- b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.
- c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.
- d) Food resources: World food problems, changes caused by agriculture and over-grazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.
- e) Energy resources: Growing energy needs, renewable and non renewable energy sources, use of alternate energy sources.
- f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification.
- .Role of an individual in conservation of natural resources.
 - Equitable use of resources for sustainable lifestyles.

UNIT 3: ECOSYSTEMS & BIODIVERSITY

Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs and ecological pyramids.

Introduction, types, characteristic features, structure and function of the following ecosystems:-

- a. Forest ecosystem, b. Grassland ecosystem, c. Desert ecosystem, d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).
- b. Biodiversity- Definition : genetic, species and ecosystem diversity. Biogeographical classification of India Value of biodiversity : consumptive use, productive use, social, ethical, aesthetic and option values.
- c. Biodiversity at global, National and local levels. India as a mega-diversity nation Hot-spots of biodiversity.
- d. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT 4: ENVIRONMENTAL POLLUTION

Definition, Cause, effects and control measures of: - Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards

- Solid waste Management: Causes, effects and control measures of urban and industrial wastes.
- Role of an individual in prevention of pollution
- Disaster management: floods, earthquake, cyclone and landslides.
- Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies.
- Environment Protection Act., Air (Prevention and Control of Pollution) Act. Water Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act.

UNIT 5 : SOCIAL ISSUES & THE ENVIRONMENT

Human Rights. Value Education. HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.

Field work : Visit to a local area to document environmental assets river/forest/grassland/hill/mountain Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, birds. Study of simple ecosystems-pond, river, hill slopes, etc.

References

1. Agarwal, K.C. 2001 Environmental Biology, Nidi Publ. Ltd. Bikaner.
2. Bharucha Erach, The Biodiversity of India, Mapin Publishing Pvt. Ltd., Ahmedabad-380013, India, Email:mapin@icenet.net (R)
3. Brunner R.C., 1989, Hazardous Waste Incineration, McGraw Hill Inc. 480p
4. Clark R.S., Marine Pollution, Clanderson Press Oxford (TB)
5. Cunningham, W.P. Cooper, T.H. Gorhan i, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumabai, 1196p
6. De A.K., Environmental Chemistry, Wiley Eastern Ltd.
7. Down to Earth, Centre for Science and Environment (R)
8. Gleick, H.P. 1993. Water in Crisis, Pacific Institute for Studies in Dev., Environment & Security. Stockholm Env. Institute Oxford Univ. Press. 473p
9. Hawkins R.E., Encyclopedia of Indian Natural History, Bombay Natural History Society, Bombay (R) j) Heywood, V.H &Waston, R.T. 1995. Global Biodiversity Assessment. Cambridge Univ. Press 1140p.
10. Jadhav, H & Bhosale, V.M. 1995. Environmental Protection and Laws. Himalaya Pub. House, Delhi 284 p.
11. Mckinney, M.L. & School, R.M. 1996. Environmental Science Systems & Solutions, Web enhanced edition. 639p.
12. Mhaskar A.K., Matter Hazardous, Techno-Science Publication (TB)
13. Miller T.G. Jr. Environmental Science, Wadsworth Publishing Co. (TB)
14. Odum, E.P. 1971. Fundamentals of Ecology. W.B. Saunders Co. USA, 574p

15. Rao M N. & Datta, A.K. 1987. Waste Water treatment. Oxford & IBH Publ. Co. Pvt. Ltd. 345p.
16. Sharma B.K., 2001. Environmental Chemistry. Geol Publ. House, Meerut
17. Survey of the Environment, The Hindu (M)
18. Townsend C., Harper J, and Michael Begon, Essentials of Ecology, Blackwell Science.
19. Trivedi R.K., Handbook of Environmental Laws, Rules Guidelines, Compliances and Standards, Vol I and II, Enviro Media (R).
20. Trivedi R. K. and P.K. Goel, Introduction to air pollution, Techno-Science Publication (TB).
21. Wanger K.D., 1998 Environmental Management. W.B. Saunders Co. Philadelphia, USA 499p (M) Magazine (R) Reference (TB) Textbook Members of the Expert Committee on Environmental Studies .

- ★ Prof. ErachBharucha, Director Bharati Vidyapeeth, Institute of Environment Education & Research, Pune
- ★ Prof. C. Manoharachary Department of Botany Osmania University Hyderabad
- ★ Prof. S. Thayumanavan Director Centre for Environmental Studies Anna University, Chennai
- ★ Prof. D.C. Goswami Head, Dept. Of Environment Science Gauhati University Guwahati-781014
- ★ Shri R. Mehta Director EE Division Ministry of Environment & Forest Prayavaran Bhawan, CGO Complex Lodhi Road, New Delhi-110 003 UGC OFFICIALS
- ★ Dr. N. K. Jain Joint Secretary UGC, New Delhi

Course Code	Course Title				Course Type	
	ELECTRONIC DEVICES AND CIRCUITS LAB				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives • To understand usage of basic electronic equipments like Oscilloscope, Function generator, Multimeter ..etc. • To understand the basic electronic components like passive components, active components, bread board, etc. • To design the basic circuits by using diodes, zener diode etc. • To understand the characteristics of the diodes, Transistors.						
Course Outcomes 1. Students get the ability for usage of basic electronic equipment like Oscilloscope, Function generator, Multimeter ..etc. 2. Students will be having knowledge on the basic electronic components like passive components, active components, bread board, etc. 3. Students can design the basic circuits by using diodes, zener diode etc. 4. Students having the knowledge on characteristics of the diodes, Transistors.						

Section 1: Basics of Electronic Circuits

Section 2: Electronic Device Characteristics

Lab 1: Basics of Electronic Circuits

1. Familiarization of electric components and usage of multimeter.
2. Familiarization with Oscilloscope and function generator.
3. Frequency response and square wave rectifying of RC, CR and RL networks.
4. Half wave and full wave rectifiers, Rectification with capacitance filters, Zener diode and IC regulation.
5. Study of CE, CB, CC amplifier on kit.

Lab 2: Device Characteristics

1. Characteristics of Diodes
 - a. PN Junction Diode
 - b. Zener Diode
 - c. Light Emitting Diode
 - d. Tunnel Diode
 - e. Varactor Diode
2. Characteristics of Transistor
 - a. Bipolar Junction Transistor (Common Base, Common Collector, Common Emitter)
 - b. Unipolar Junction Transistor (p-channel JFET, n-channel JFET)
 - c. MOSFET (p-channel enhancement mode MOSFET, n-channel enhancement mode MOSFET, p-channel depletion mode MOSFET, n-channel depletion mode MOSFET)

Text /Reference Books

1. Electronic Devices and Circuit Theory – Robert L.Boylestad, Louis Nashelsky, 9th edition, 2008 PE.
2. Electronic Devices and Circuits- David A. Bell- 5th Edition, Oxford University Press.
3. Jacob Millman, Christos C Halkias and Satyabrata JIT, “ Electronics Devices and Circuits”, 3rd Edition.
4. G. Streetman, and S. K. Banerjee, “Solid State Electronic Devices,” 7th edition, Pearson,2014.
5. D. Neamen , D. Biswas "Semiconductor Physics and Devices," McGraw-Hill Education.
6. S. M. Sze and K. N. Kwok, “Physics of Semiconductor Devices,” 3rd edition, John Wiley&Sons, 2006.
7. C.T. Sah, “Fundamentals of solid state electronics,” World Scientific Publishing Co. Inc, 1991.
8. Y. Tsividis and M. Colin, “Operation and Modeling of the MOS Transistor,” Oxford Univ.Press, 2011.

Course Code	Course Title				Course Type	
	Analog Circuits (AC)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4
Course Objectives • The concepts of small signal equivalent circuits of BJT, FET and it's frequency response. • The concept of multistage amplifiers, differential amplifiers and current mirrors for high input impedance. • The fundamental concepts of positive and negative feedback and their applications. • The performance analysis of Operational amplifiers and it's applications. • The concept of large signal amplifiers and radio frequency amplifiers.						
Course Outcomes 1. An ability to design and analyze the BJT & FET amplifiers at low frequency, mid frequency and high frequency regions. 2. An ability to analyze a given differential amplifier or design a differential amplifier to meet the given specifications with constant current bias circuit. 3. An ability to design and analyze the positive feedback and negative feedback amplifiers for a given specifications. 4. An ability to design and perform op-amp based circuits and it's applications for a given specifications. 5. An ability to understand the large signal amplifiers (i.e. power amplifiers) and its efficiency calculations. 6. An ability to understand the waveform generators, timers and analog to digital converters & digital to analog convertors						

UNIT-I

Small Signal Analysis: Amplifier models: Voltage amplifier, current amplifier, trans conductance amplifier and trans-resistance amplifier. Biasing schemes for BJT and FET amplifiers, various configurations (such as CE/CS, CB/CG, CC/CD) and their features, small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance etc., design procedure for particular specifications

High Frequency Analysis: High frequency transistor models, frequency response of single stage and multistage amplifiers, cascode amplifier.

UNIT-II:

Power Amplifiers: Various classes of operation (Class A, B, AB, C etc.), their power efficiency and linearity issues.

Feedback topologies: Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits.

UNIT-III:

Oscillators: Review of the basic concepts, Barkhausen criterion, RC oscillators (phase shift, Wien bridge etc.), LC oscillators (Hartley, Colpitt, Clapp etc.), non-sinusoidal oscillators.

Current mirror: Basic topology and its variants, V-I characteristics, output resistance and minimum sustainable voltage (V_{ON}), maximum usable load. **Differential amplifier:** Basic structure and principle of operation, calculation of differential gain, common mode gain, CMRR.

UNIT-IV:

OP-AMP: Introduction of op-amp and its internal circuit diagram. Ideal and practical op-amp with transfer characteristics.

OP-AMP applications: Review of inverting and non-inverting amplifiers, virtual ground concept. Linear op-amps (Adders, Subtractors, V-V, V-I, I-V, I-I amplifiers, Instrumentation amplifier); Non-linear op-amps (Rectifiers, Peak detector, Clipper, Clamper, Logarithmic amplifier) and multipliers; Open loop op-amps (Comparator, Detector); Positive Feedback op-amps (Schmitt trigger, Multivibrators)

Active filters Design: Low pass, high pass, band pass and band stop, design guidelines.

UNIT-V:

Digital-to-analog converters (DAC): Weighted resistor, R-2R ladder, resistor string etc.

Analog-to- digital converters (ADC): Single slope, dual slope, successive approximation, flash etc

Text Books :

1. Electronics Devices and Circuit Theory Boylestad, Robert & Louis, Nashelsky Pearson, 10th Edition
2. Microelectronic Circuits-Theory and applications by Adel S. Sedra and Kenneth C.Smith, Fifth Edition , (Oxford International Student Edition)

References

1. Electronic Devices and Circuits- Millman and Halkias, TMH
2. Op-Amps and Linear Integrated Circuits Gayakwad , Ramakant A PHI, Learning, 4th Edition Electronic Devices and Circuits Dr. Sharma, Sanjay KATSON, 2012
3. Fundamentals of Electronic Devices and Circuits David, A Bell Oxford Press, 5th Edition, 2008
4. Electronic Principles - with simulation CD Malvino, A.P. Tata McGraw- Hill , Education, 7th Edition
5. Basic Electronics and Linear Circuits Bhargava, N., Kulshreshtha D., S. Gupta Tata McGraw- Hill Education, 2011
6. Electronics Devices and Circuits Mottershead, Allen PHI Learning, 2011
7. Electronic Devices and Circuits- David A Bell - PHI 4th edition

Online Resources

1. <https://www.mooc-list.com/course/electronic-systems-and-digital-electronics-uninet-tun-o?static=true>
2. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-012-microelectronic-devices-and-circuits-spring-2009>
3. Introductory Analog Electronics Laboratory (Spring 2007) by MIT Open Courseware | Reviews and Ratings

Course Code	Course Title				Course Type	
	Control Systems (CS)				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4

Course Objectives

- To familiarize the students with the need for modeling of systems and to represent the system in various ways mathematically.
- To teach them the various well-established techniques to analyze the stability of systems and related issues.
- Ability to find time response of given control system model & plot Root Locus and Bode
- plots for given control system model
- Ability to design Lead, Lag, Lead-Lag systems in control system & Ability to design PID controllers for given control system model.
- Ability to learn state space analysis and optimal control system.

Course Outcomes

1. To narrate the basic features of Open Loop and Closed Loop control systems, illustrate the physical and electrical systems and formulate their transfer function.
2. To categorize first order and second order systems and examine their time domain responses for different inputs and analyze the transfer function of proportional (P), proportional plus derivative (PD), proportional plus integral (PI), proportional plus derivative plus integral controllers (PID).
3. To define the concept of stability and analyze the system stability using RH criteria and Root Locus technique.
4. To assess the stability in frequency domain using Bode, Nyquist plots and develop compensators to meet given frequency domain specifications.
5. To define the basic concepts of state variable analysis, derive state equations and establish transfer functions for a given system and discuss their controllability and observability. To define the basics of Digital control system and its Transfer function.

UNIT-I: INTRODUCTION

Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Examples of control systems- Classification of control systems, Feed-Back Characteristics, Effects of Feedback, Mathematical modeling of physical systems: Differential equation and Transfer functions, Examples of modeling different types (e.g. electrical, mechanical, chemical, biological, social etc.) of systems, Equivalence between the elements of different types of systems. Block diagram algebra –Signal flow graph -Reduction using Mason 's gain formula. Translational and rotational mechanical systems.

UNIT-II: TIME DOMAIN ANALYSIS

Standard test signals - Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems - Time domain specifications. Steady state response - Steady state errors and error constants, Frequency domain response – Transfer function and its interpretation in terms of frequency responses peak and peaking frequency, bandwidth and cut-off rate; Link between time and frequency domain response features. Advantages of closed loop operation: Sensitivity and complementary sensitivity, Disturbance and noise reduction.

UNIT-III: STABILITY ANALYSIS IN S-DOMAIN

The concept of stability – Routh ‘s stability criterion – qualitative stability and conditional stability – limitations of Routh ‘s stability. The root locus concept - construction of root loci- and relative stability using root-locus approach, effects of adding poles and zeros to $G(s)H(s)$ on the root loci.

UNIT-IV: FREQUENCY DOMAIN ANALYSIS

Polar Plots-Nyquist Plots-Stability Analysis. Bode diagrams- Determination of Frequency domain specifications and transfer function from the Bode Diagram-Phase margin and Gain Margin-Stability Analysis from Bode Plots. P, PD, PI, PID Controllers and Compensation techniques – Lag, Lead, Lead-Lag Controllers design in frequency Domain.

UNIT-V: STATE VARIABLE ANALYSIS

State variable Analysis- Concepts of state, state variable, state model, state models for linear continuous time functions, diagonalization of transfer function, solution of state equations, concept of controllability & observability. Introduction to Optimal control & Nonlinear control, Optimal Control problem, Regulator problem, Output regulator, tracking problem. Nonlinear system – Basic concept & analysis.

Text Books

1. Automatic Control Systems– by B. C. Kuo and Farid Golnaraghi – John Wiley and Sons, 8th edition, 2003.
2. Control Systems Engineering – by I. J. Nagrath and M. Gopal, New Age International (P) Limited, Publishers, 5th edition, 2007.

References

1. Modern Control Engineering – by Katsuhiko Ogata – Prentice Hall of India Pvt. Ltd., 5th edition, 2010.
2. Control Systems Engineering - by NISE 5th Edition – John Wiley.
3. M. Gopal, Control Systems Principles and Design, 4th edition, Tata McGraw-Hill, 2012.
4. A. Nagoorkani, Control Systems, 3rd Edition, RBA Publications, 2017.
5. A. K. Jairath, Solutions and Problems of Control Systems, 7th edition, CBS publications and distributors, 2021.

Online Resources

1. Control Engineering, Dr. Rama Krishna Pasumarthy, Associate Professor, IIT Madras.
<https://nptel.ac.in/courses/108106098/>
2. Control Engineering, Prof. S. D. Agashe, Professor, IIT Bombay.
<https://www.digimat.in/nptel/courses/video/108101037/L01.html>

Course Code	Course Title				Course Type	
	Electromagnetic Waves (EMW)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To familiarize the students with the importance of transmission lines • To teach them the various engineering graphical tools to design transmission lines • To understand the importance of maxwell's equations in electromagnetic fields • To understand the importance of electromagnetic waves in wireless communications • Ability to learn the wave guides and antenna basics						
Course Outcomes 1. Understand characteristics and wave propagation on high frequency transmission lines 2. Carryout impedance transformation on TL 3. Use sections of transmission line sections for realizing circuit elements 4. Characterize uniform plane wave 5. Calculate reflection and transmission of waves at media interface 6. Analyze wave propagation on metallic waveguides in modal form 7. Understand principle of radiation and radiation characteristics of an antenna						

Unit I: INTRODUCTION TO TRANSMISSION LINES:

Concept of distributed elements, equations of voltage and current, standing waves and impedance transformation, lossless and low loss transmission lines, power transfer on a transmission line, short circuit and open circuit lines, parameters of transmission line.

SMITH CHART: applications, applications of transmission line, impedance matching using transmission lines.

Unit II: BASIC LAWS OF ELECTROMAGNETICS:

Gauss's law, Ampere's circuital law, Faraday's law of electromagnetic induction.

Maxwell's equations: Surface charge and Surface current, Displacement current and continuity equation, Boundary conditions at media interface.

Unit III: UNIFORM PLANE WAVES I:

Wave equation for time harmonic fields, Solution of the wave equation, Uniform plane wave, Wave polarization, Wave propagation in conducting medium, Phase velocity and Poynting vector

Uniform plane waves II : Plane wave in arbitrary direction, Reflection and refraction of waves at dielectric and conducting interface, Total internal reflection, Brewster angle.

Unit IV: WAVE GUIDES:

Parallel plane wave guide, TE mode, TM mode, TEM mode,

Rectangular wave guides: Group velocity and dispersion, Analysis of rectangular wave guides.

Unit V: ANTENNAS:

Introduction, Radiation parameters of antenna, potential functions and their solutions.

FIELDS: Near and far fields, Radiation resistance and radiation pattern of Hertz dipole, total power radiated by a dipole.

Text Books:

1. M.N.O. Sadiku, "Principles of Electromagnetics", Oxford University press, 2009.
2. R.K. Shevgaonkar, "Electromagnetic Waves", Tata McGraw Hill, 2005.

References

1. C.A. Balanis, "Advanced Engineering Electromagnetics" , John Wiley and sons, 2012.
2. D. K. Cheng, "Field and Wave Electromagnetics", Addison-Wesley, 1989.
3. C. A. Balanis, "Antenna Theory: Analysis and Design" , John Wiley and sons, 2005

Online resources

1. <https://youtube.com/playlist?list=PLQzh033j0lVKTC5vOQ0E0njh6UV8M1UR4>
2. <https://youtube.com/playlist?list=PLuv3GM6-gsE3-hVNaw-YEb7EeY5XVPZdz>
3. <https://youtube.com/playlist?list=PL0925FD10648D664E>

Course Code	Course Title					Course Type
	Probability Theory and Stochastic Processes (PTSP)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4
Course Objectives • Understand the fundamental concepts of probability theory and their applications in modeling and analyzing real-world phenomena. • Learn the properties and characteristics of various discrete and continuous random variables and distributions. • Develop skills in computing and analyzing joint distributions, conditional distributions, and moments of random variables. • Learn the basics of random sequences and modes of convergence, and apply them to limit theorems. • Acquire knowledge of stationary processes and their properties, and analyze the transmission of random processes through LTI systems.						
Course Outcomes 1. Ability to model and analyze real-world phenomena using probability theory and random variables. 2. Proficiency in computing and analyzing the properties and characteristics of various random variables and distributions. 3. Capability to apply joint distributions, conditional distributions, and moments of random variables in real-world scenarios. 4. Mastery in analyzing and applying limit theorems to random sequences and modes of convergence. 5. Expertise in understanding stationary processes and their properties, and analyzing their transmission through LTI systems.						

Unit I

Sets and set operations; Probability space; Conditional probability and Bayes theorem; Independent events, Combinatorial probability and sampling models.

Unit II

Discrete random variables, probability mass function, probability distribution function, example random variables and distributions; Uniform, Geometric and its memoryless property, Bernoulli, Binomial, Poisson distributions.

Continuous random variables, probability density function, probability distribution function, example distributions; Uniform, Exponential and its memoryless property, Gaussian distribution, Standard Normal distribution, Q(.) function, Heavy tailed Pareto distribution.

Unit III

Joint distributions, Jointly Gaussian random variables, Marginal distributions, Independent random variable, functions of one and two random variables, Sum of random variables, minimum, maximum of random variables, log normal distribution, Rayleigh distribution, Chi-square distribution, Square of Rayleigh random variable, moments of random variables; Conditional distribution, densities and moments; Mean, Variance, Covariance, Correlation, Correlation coefficient, Covariance Matrix. Uncorrelated random variables. Conditional expectation. Characteristic functions of a random variable; Markov, Chebyshev and Chernoff bounds; Cauchy Schwarz inequality. Use of MATLAB for generating: random Gaussian samples, Rayleigh distributed samples generation using two IID Gaussian samples. Transformation of random variables: Generating Exponential distributed samples using Uniform distributed samples.

Unit IV

Random sequences and modes of convergence (everywhere, almost everywhere, probability, distribution and mean square); Limit theorems; Strong and weak laws of large numbers, central limit theorem.

Unit V

Random process, Stationary processes: Strict sense stationary, Wide sense stationary processes. Gaussian Random process, Mean and covariance functions. Ergodicity. Transmission of random process through LTI. Power spectral density.

Text Books

1. A.Papoulis and S. Unnikrishnan Pillai, "Probability, Random Variables and Stochastic Processes," Fourth Edition, McGraw Hill
2. Introduction to Probability 2nd edition Dimitri P. Bertsekas and John N. Tsitsiklis, Massachusetts Institute of Technology

References

1. H. Stark and J. Woods, "Probability and Random Processes with Applications to Signal Processing," Third Edition, Pearson Education
2. K. L. Chung, Introduction to Probability Theory with Stochastic Processes, Springer International
3. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability, UBS Publishers,
4. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Stochastic Processes, UBS Publishers
5. S. Ross, Introduction to Stochastic Models, Harcourt Asia, Academic Press.

Course Code	Course Title				Course Type	
CS2102	Object Oriented Programming				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • The course will introduce standard tools and techniques for software development, using object oriented approach, use of a version control system, an automated build process, an appropriate framework for automated unit and integration tests. • To understand Object oriented programming concepts, and apply them in Problem solving. • To learn the basics of Java Console and GUI based programming						
Course Outcomes 1. Specify simple abstract data types and design implementations, using abstraction functions to document them. 2. Recognise features of object-oriented design such as encapsulation, polymorphism, inheritance, and composition of systems based on object identity 3. Name and apply some common object-oriented design patterns and give examples of their use Design applications with an event-driven graphical user interface.						

Detailed Contents:

UNIT-1:

Introduction to OOPS: Paradigms of Programming Languages, Basic concepts of Object Oriented Programming, Differences between Procedure Oriented Programming and Object Oriented Programming, Objects and Classes, Data abstraction and Encapsulation, Inheritance, Polymorphism, benefits of OOP , application of OOPs.

Java :History, Java features, Java Environment, JDK, API.

Introduction to Java :Types of java program, Creating and Executing a Java program, Java Tokens, Keywords, Character set, Identifiers, Literals, Separator, Java Virtual Machine (JVM), Command Line Arguments, Comments in Java program.

UNIT -2:

Elements: Constants, Variables, Data types, Scope of variables, Type casting, Operators: Arithmetic, Logical, Bit wise operator, Increment and Decrement, Relational, Assignment, Conditional, Special operator, Expressions – Evaluation of Expressions

Decision making and Branching: Simple if statement, if, else statement, Nesting if, else, else if Ladder, switch statement, Decision making and Looping: While loop, do-While loop, for loop, break, labelled loop, continue Statement, Simple programs

Arrays: One Dimensional Array, Creating an array, Array processing, Multidimensional Array, Vectors, Wrapper classes, Simple programs

UNIT-3:

Strings: Exploring String class, String Class Methods, String Buffer Class, Simple programs

Class and objects: Defining a class, Methods, Creating objects, Accessing class members, Constructors, Static members, Nesting of Methods, this keyword, Command line input.

Polymorphism – Static Polymorphism, Dynamic Polymorphism, Method overloading, Polymorphism with Static Methods, Private Methods and Final Methods.

Inheritance: Defining a sub class, Deriving a sub class, Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Overriding methods, Final variables and methods, Final classes, Finalizer methods, Abstract methods and classes, Visibility Control: Public access, Private access, default and protected. Abstract classes.

Interfaces - Interfaces vs Abstract classes, defining an interface, implementing interfaces, accessing implementations through interface references, extending interfaces. Inner classes - uses of inner classes, local inner classes, anonymous inner classes, static inner classes, examples.

UNIT- 4:

Packages: Java API Packages, System Packages, Naming Conventions, Creating & Accessing a Package, Adding Class to a Package, Hiding Classes, Programs

Exception Handling: Limitations of Error handling, Advantages of Exception Handling, Types of Errors, Basics of Exception Handling, try blocks, throwing an exception, catching an exception, finally statement

Multi threading: Creating Threads, Life of a Thread, Defining & Running Thread, Thread Methods, Thread Priority, Synchronization, Implementing runnable interface, Thread scheduling.

I/O Streams: File, Streams, Advantages, The stream classes, Byte streams, Character streams.

JDBC, ODBC Drivers, JDBC ODBC Bridges, Seven Steps to JDBC, Importing java SQL Packages, Loading & Registering the drivers, Establishing connection. Creating & Executing the statement.

UNIT-5:

AWT Components and Event Handlers: Abstract window tool kit, Event Handlers, Event Listeners, AWT Controls and Event Handling: Labels, TextComponent, ActionEvent, Buttons, CheckBoxes, ItemEvent, Choice, Scrollbars, Layout Managers- Input Events, Menus, Programs

Design patterns - Introduction to Creational design patterns, Structural design patterns and Behavioral design patterns.

GUI Programming with Java - Introduction to Swing, limitations of AWT, Swing vs AWT, MVC architecture, Hierarchy for Swing components, Containers - JFrame, JApplet, JDialog, Jpanel. Overview of some swing components JButton, JLabel, JTextField, JTextArea, simple swing applications.

TEXT BOOKS

1. Java the complete reference, 7 th edition, Herbert Schildt, TMH.
2. Understanding OOP with Java, updated edition, T. Budd, Pearson Education.

REFERENCES

1. An Introduction to programming and OO Design using Java, J.Nino and F.A. Hosch, John wiley& Sons.
2. Introduction to Java Programming, Y. Daniel Liang, Pearson Education
3. An Introduction to Java programming and Object Oriented Application Development, R.A. Johnson-Thompson
4. Programming with Java - E. Balagurusamy
5. Object oriented Programming in Java - Dr. G.Thampi
6. Let us Java – YashavantKanetkar - BPB Publications, New Delhi - First Edition 2012
7. Core Java, An Integrated Approach, Dr. R. Nageswara Rao
8. An Introduction to OOPS with Java - C Thomas WU - TataMc-Graw Hill, New Delhi - 4th Edition
9. Object oriented Programming through Java - ISRD Group - TataMc-Graw Hill, New Delhi - Eight Reprint 2011

Course Code	Course Title				Course Type	
CS2102	Network Analysis (NA)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4
Course Objectives • To provide the knowledge of basic components of circuit and their relations in various forms of connection. • To provide the knowledge of D.C analysis of circuits • To learn the basics of Java Console and GUI based programming • To understand concept and various parameters of single-phase AC circuits. • To understand concept of resonance and various theorems. • To give the knowledge of two port network and Graphs in circuit theory.						
Course Outcomes 1. Will be able to articulate in working of various components of a circuit. 2. Will be familiar with solving DC circuits. 3. Will have ability to analyze AC circuit and measure related parameters. 4. To solve the given circuit with various theorems and methods. 5. Ability to Express given Electrical Circuit in terms of A, B, C, D and Z, Y Parameter Model and Solve the circuits 6. To distinguish between tie set and cut set methods for solving various circuits						

UNIT- I: DC CIRCUIT ANALYSIS R-L-C Parameters, Voltage and Current Independent and Dependent Sources, Source Transformation – V–I relationship for Passive elements, Kirchhoff's Laws, Network reduction techniques – series, parallel, series parallel, star-to-delta, delta-to-star transformation, Nodal Analysis

UNIT- II: DC TRANSIENTS DC Transients: RL, RC and RLC, Laplace transforms and their adaptation to networks

UNIT- III: SINGLE PHASE AC CIRCUITS Single Phase AC Circuits - R.M.S. and Average values, Form Factor, steady state analysis of series, Parallel and Series parallel Combinations of R, L and C with Sinusoidal excitation, concept of reactance, Impedance, Susceptance and Admittance – phase and phase difference, Concept of Power Factor, j-notation, complex and Polar forms of representation.

UNIT- IV: RESONANCE and NETWORK THEOREMS Resonance – Series resonance and Parallel resonance circuits, concept of bandwidth and Q factor, Locus Diagrams for RL, RC

and RLC Combinations for Various Parameters. Network Theorems - Thevenin's, Norton's, Maximum Power Transfer, Superposition, Reciprocity, Tellegen's, Millman's and Compensation theorems for DC and AC excitations.

UNIT- V: TWO PORT NETWORKS and GRAPH THEORY Two port networks: Z, Y, h and ABCD parameters Graphs: Paths, connectedness, circuits, cutsets, trees; Matrix representation of directed graphs: incidence, cutset and circuit matrices; Methods of analysis of linear networks: nodal-cutsetmesh- and loop-analysis. Transfer functions: poles and zeros; Elements of Filter Theory.

TEXT BOOKS

1. Network Theory by N.Sreenivasulu, REEM Publications
2. Circuits & Networks by A. Sudhakar and Shyammoohan S Palli, Tata McGraw- Hill 3. Electric Circuits- Schuam Series

REFERENCES

1. Network Analysis by M.E Van Valkenberg, Prantice Hall India, 3rd Edition.
2. Electric circuit Analysis by C.L. Wadhwa, New Age international
3. Electric circuits by David A. Bell, Oxford University press

BM0007 MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Course Objective:

- Enable the students to learn managerial economics principles applied in industries and equip them to handle the tasks in their career by making a real sense of what is happening economically in the organization.
- The course describes the Nature and Scope of Managerial Economics. It gives complete study on the demand and elasticity of demand and methods of demand forecasting.
- It provides a detailed structure on the pricing strategies and shows clear picture methods and sources of raising finance.
- It gives clear cut information of preparing final accounts and capital Budgeting techniques.

Course Outcome:

After the successful completion of this course, the learner will be able to know:

1. The dynamic game of demand and supply, and how the trinity of Economics i.e. Demand, Supply and Scarcity make the things move around the globe.
2. Principles of Microeconomics applied to industries.
3. Concept of forecasting and applying forecasting techniques to address the challenges and opportunities in the organization they work.
4. Cost and Production analysis, Break-Even analysis, Opportunity Cost, how to optimize organizational resources and how to minimize cost and maximize production, revenue and profit
5. Different pricing structure and discount mechanism suitable for business firms.
6. Market structure and how to exploit market structure for optimizing the benefits of organization.
7. Capital requirements and sources of capital.

UNIT I: Introduction to Managerial Economics:

Definition, Nature and Scope of Managerial Economics-Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT II: Theory of Production and Cost Analysis:

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs. Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale.

Cost Analysis: Cost concepts, Opportunity cost. Fixed vs. Variable costs, Explicit costs Vs. Implicit costs. Out of pocket costs vs. Imputed costs. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems)- Managerial Significance and limitations of BEA.

UNIT III: Markets & Pricing Policies:

Market structures: Types of competition, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Objectives and Policies of Pricing- Methods of Pricing: Cost Plus Pricing. Marginal Cost Pricing, Sealed Bid Pricing, Going Rate Pricing, Limit Pricing, Market Skimming Pricing, Penetration Pricing.

UNIT IV: Introduction to Financial Accounting: Introduction to Financial Accounting: Double entry Book Keeping, Journal, Ledger, Trail Balance and Final Accounts (Trading account, Profit and Loss Account and Balance sheet with simple adjustments).

UNIT V: Capital and Capital Budgeting:

Capital and Capital Budgeting: Capital and its significance. Types of Capital. Estimation of Fixed and Working capital requirements. Methods and sources of raising finance. Nature and scope of capital budgeting, features of capital budgeting proposals. Methods of Capital Budgeting: Payback Method. Accounting Rate of Return (ARR) and Net Present Value Method, Internal Rate of Return (IRR).

Reference Books:

1. Aryasri: Managerial Economics and Financial Analysis, TMH,2009.
2. Varshney &Maheswari : Managerial Economics, Sulthan Chand,2009.
3. Raghunatha Reddy &Narasimhachary: Managerial Economics& Financial Analysis, Scitech. 2009.
4. V.Rajasekarn&R.Lalitha. Financial Accounting, Pearson Education. New Delhi. 2010
5. Suma Damodaran, Managerial Economics, Oxford University Press. 2009.

Course Code	Course Title					Course Type
	Electrical Technology					ESC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives ● To acquire the basic concepts of AC and DC Machines ● To learn the basics of Alternators, Transformers. ● To learn the measurement characteristics of Voltmeter, Ammeter, Wattmeter and Energy meter.						
Course Outcomes 1.						

UNIT- I: THREE PHASE AC CIRCUITS Three phase EMF generation, delta and Y connections, line and phase quantities, solution of three phase circuits, balanced supply voltage and balanced load, phasor diagram, measurement of power in three phase circuits, Three phase four wire circuits.

UNIT -II: TRANSFORMERS Magnetic Circuits : Ampere's circuital law, B- H curve, solution of magnetic circuits, hysteresis and eddy current losses, relays, an application of magnetic force, basic principles of stepper motor. Transformers : Construction, EMF equation, ratings, phasor diagram on no load and full load, equivalent circuit, regulation and efficiency calculations, open and short circuit tests, auto-transformers.

UNIT -III: INDUCTION MOTORS The revolving magnetic field, principle of orientation, ratings, equivalent circuit, Torque-speed characteristics, starters for cage and wound rotor type induction motors.

UNIT -IV: DC MACHINES Construction, EMF and Torque equations, Characteristics of DC generators and motors, speed control of DC motors and DC motor starters.

UNIT- V: ELECTRICAL MEASURING INSTRUMENTS DC PMMC instruments, shunt and multipliers, multimeters, Moving iron ammeters and voltmeters, dynamometer, wattmeter, AC watthour meter, extension of instrument ranges.

TEXT BOOKS

1. Electric machinery - A.E. Fitzgerald, C.Kingsley and S.Umans, Mc Graw Hill Companies, 5th edition
2. Electrical machines-PS Bhimbra, Khanna Publishers.
2. Electromechanics – II , by Kamakshaiah

REFERENCES:

1. Performance and Design of AC Machines by MG.Say, BPB Publishers
2. Theory of Alternating Current Machinery- by Langsdorf, Tata McGraw-Hill Companies, 2nd edition.
3. Electric Machines –by I.J.Nagrath & D.P.Kothari,Tata Mc Graw Hill, 7th Edition.2005

4. Fundamentals of Electric Machines by B. R. Gupta, Vandana singhal, 3rd Edition, New age international Publishers
5. Electrical Measurements and measuring Instruments – by E.W. Golding and F.C. Widdis, 5th Edition, Reem Publications.
6. Electrical & Electronic Measurement & Instruments by A.K.Sawhney Dhanpat Rai & Co. Publications

Electrical Technology Lab

List of Experiments:

1. Characteristics of Fluorescent and Incandescent Lamp
2. Verification of Network Theorems
3. R-L-C Series Circuit
4. Three phase power measurement by two Wattmeter methods
5. Single Phase Energy Meter
6. OC and SC Test of Single Phase Transformer
7. OCC of separately excited DC Shunt Generator
8. Load test of Three Phase Induction Motor

Course Objectives

1. To design and Characterize of small signal equivalent circuits of BJT, FET and its frequency response.
2. To design differential amplifiers & differential amplifier with active load and its frequency response.
3. To design of simple current mirror circuit using BJT and MOSFET.
4. To design of cascode current mirror circuit using BJT and MOSFET.
5. To design the positive feedback amplifiers for a given specifications and tuned amplifiers & timers.

List of Experiments:**SECTION-A**

1. Clipping and Clamping circuits.
2. LC, CLC filters
3. Voltage Regulators.
4. RC-coupled amplifier (single stage & two-stage).
5. Darlington Emitter follower & Tuned voltage amplifier.
6. Power amplifiers (Class-B push pull power amplifier).
7. Feedback amplifiers:
 - (i) Voltage series feedback amplifier
 - (ii) Voltage shunt feedback amplifier
 - (iii) Current shunt feedback amplifier
 - (iv) Current series feedback amplifier
8. Oscillators:
 - (i) RC-phase shift oscillator
 - (ii) Wein-bridge oscillator
 - (iii) Hartley oscillator
 - (iv) Colpits oscillator
 - (v)

SECTION-B**Operational Amplifiers**

1. Parameters of Operational Amplifiers.
 - (i) Input bias current, Input Offset current, Input Offset voltage
 - (ii) Common Mode Rejection Ratio (CMRR)
2. Applications of Operational Amplifiers.
 - (i) Inverting op-amp & Non-Inverting op-amp
 - (ii) Voltage follower, Summing amplifier
 - (iii) ZCD, Schmitt trigger
 - (iv) Full wave precision rectifier etc.

3. Wave form generators by using op-amp
 - (i) Monostable Multi vibrator
 - (ii) Astable Multi vibrator

Data Converters

4. Digital to Analog converters.
 - (i) Weighted Resistor Type D/A converter
 - (ii) R-2R ladder Type D/A converter
5. Analog to Digital converter.
 - (i) Single slope & Dual slope A/D converters
 - (ii) Flash type & Successive Approximation Type A/D converters
6. Switched capacitor circuit.

Course Outcomes:

- ☐ An ability to design and Characterize of small signal equivalent circuits of BJT, FET and its frequency response.
- ☐ An ability to design differential amplifiers & differential amplifier with active load and its frequency response.
- ☐ An ability to design of simple current mirror circuit using BJT and MOSFET.
- ☐ An ability to design of cascode current mirror circuit using BJT and MOSFET.
- ☐ An ability to design the positive and negative feedback amplifiers for a given specifications and tuned amplifiers & timers.

Text Books

1. Electronics Devices and Circuit Theory Boylestad, Robert & Louis, Nashelsky Pearson, 10th Edition
2. Microelectronic Circuits-Theory and applications by Adel S. Sedra and Kenneth C.Smith, Fifth Edition , (Oxford International Student Edition)
3. Electronic Devices and Circuits- Millman and Halkias, TMH
4. Op-Amps and Linear Integrated Circuits Gayakwad , Ramakant A PHI, Learning,4 th Edition
Electronic Devices and Circuits Dr. Sharma, Sanjay KATSON,2012

References

1. Fundamentals of Electronic Devices and Circuits David, A Bell Oxford Press, 5thEdition, 2008
2. Electronic Principles - with simulation CD Malvino, A.P. Tata McGraw- Hill , Education,7 thEdition
3. Basic Electronics and Linear Circuits Bhargava, N.,Kulshreshtha D., S.Gupta Tata McGraw- Hill Education, 2011
4. Electronics Devices and Circuits Mottershead, Allen PHI Learning,2011
5. Electronic Devices and Circuits- David A Bell - PHI 4th edition

MOOCs

1. <https://www.mooc-list.com/course/electronic-systems-and-digitalelectronics-uninettuno?static=true>
2. <http://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-012-microelectronic-devices-and-circuits-spring-2009/>
3. Introductory Analog Electronics Laboratory (Spring 2007) by MIT Open Courseware | Reviews and Ratings.

Course Code	Course Title				Course Type	
CS 2702	Object Oriented Programming Lab				ESC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	0	0	3	40	60	1.5
Course Objectives 1. To understand basic Java programming concepts and syntax. 2. To develop Java applications using control structures, arrays, and object-oriented programming concepts. 3. To implement Java applications using Strings, Wrapper classes, and file I/O. 4. To apply inheritance, polymorphism, and exception handling in Java programming. 5. To develop multi-threaded Java applications and understand the concept of threading.						
Course Outcomes 1. Understand basic Java programming concepts and syntax, Develop simple Java applications. 2. Use control structures and arrays in Java programs. 3. Implement object-oriented programming concepts in Java. 4. Develop Java applications using Strings and Wrapper classes. 5. Implement various types of inheritance, polymorphism and Use exception handling in Java. 6. Develop Java applications using file I/O. 7. Understand the concept of threading and implement multi-threaded Java applications.						

Detailed Contents

Week-I

1. Write a Java program print "Hello World"
2. Write a Java program that prints all real and imaginary solutions to the quadratic equation $ax^2 + bx + c = 0$. Read in a, b, c and use the quadratic formula
3. Write a Java program to implement calculator operations
4. Write a java program to find prime factors of given number
5. Write a java program to find whether given number is Palindrome or not
6. Write an application that declares 5 integers, determines and prints the largest and smallest in the group.

Week-II

1. Write a Java program to sort a given list of numbers.
2. Write a Java program to implement linear search.
3. Write a Java program to implement binary search.
4. Write a java program to add two given matrices.

5. Write a java program to multiply two given matrices.
6. Write a java program for sorting a given list of names.
7. Write a Java program to give an example for command line arguments

Week-III

1. Write a program to display details of the required employee based on his Id. The details of employees includes, Emp_name, Emp_age, Emp_gender, Emp_designation, Emp_salary, Emp_Address etc.
2. A mail-order house sells five products whose retail prices are as follows : Product 1 : Rs. 99.90 , Product 2 : Rs. 20.20 , Product 3 : Rs. 6.87 , Product 4 : Rs. 45.50 and Product 5:Rs. 40.49 . Each product has Prdouct_Id, Product_Name,Product_Quantity, Product_Price. Write an application that reads a series of pairs of numbers as follows :
 - a. product Id
 - b. quantity sold. your program should calculate and display the total retail value of all products sold
3. Write a java program that inputs 5 numbers, each between 10 and 100 inclusive. As each number is read display it only if it's not a duplicate of any number already read display the complete set of unique values input after the user enters each new value
4. Write a java program : rolling a pair of dice 10 times [each attempt should be delayed by 10000 ms] and count the number Successful attempts. successful attempt : If the pair of Dice results in the same values.
5. Implement the following case study using OOP concepts in Java.E-Book stall : Every book has Properties i.e. Book_Name, Book_Author, Book_Count ; Every Customer has properties as : Customer_Id, Customer_Name,Customer_Address and he can buy Books from the E-Book stall by giving book name, author name and number of books he/she want to buy .Write a Program which will display the list books bought by the customer and remaining text books in the E-book stall with the the count.

Week -IV:

1. Write an application that uses the String method "*compareTo*" to compare two strings defined by the user.
2. Write an application that uses the String method equals and equalsIgnoreCase to test any two string objects for equality.
3. Write an application that uses String method indexOf to determine the total number of occurrences of any given alphabet in a defined text.
4. Write an application that uses String method concat to concatenate two defined strings.

5. Write a Java program to print all vowels in given string and count number of vowels and consonants present in given string
6. Write an application that finds the length of a given string.
7. Write an application that uses the String method charAt to reverse the string.
8. Write an application that finds the substring from any given string using substring method and startsWith & endsWith methods.
9. Write an application that changes any given string with uppercase letters, displays it, changes it back to lowercase letters and displays it.

Week-V

1. Write a Java Program to implement Wrapper classes and their methods.
2. Write an application that prompts the user for the radius of a circle and uses a method called circleArea to calculate the area of the circle and uses a method circlePerimeter to calculate the perimeter of the circle.
3. Write a JAVA program for the following.
 - a. Call by value
 - b. Call by object
4. Create a class Account with an instance variable balance (double). It should contain a constructor that initializes the balance, ensuring that the initial balance is greater than 0.0. Acct details: Acct_Name, Acct_acctno, Acct_Bal, Acct_Address. Create two methods namely credit and debit, getBalance. The Credit adds the amount (passed as parameter) to balance and does not return any data. Debit method withdraws money from an Account. GetBalance displays the amount. Ensure that the debit amount does not exceed the Account's balance. In that case the balance should be left unchanged and the method should print a message indicating "Debit amount exceeded account balance".
5. Write Java program for the following
 - a. Example for this operator and the use of this keyword.
 - b. Example for super keywords.
 - c. Example for static variables and methods.

Week-VI

1. Write a Java program to find Area and Circle of different shapes using polymorphism concept
2. Write a Java program which can give example of Method overloading and overriding

3. Write an application to create a super class Employee with information first name & last name and methods getFirstName(), getLastName() derive the subclasses ContractEmployee and RegularEmployee with the information about department, designation & method displayFullName() , getDepartment(), getDesig() to print the salary and to set department name & designation of the corresponding sub-class objects respectively.
4. Derive sub-classes of ContractEmployee namely HourlyEmployee & WeeklyEmployee with information number of hours & wages per hour, number of weeks & wages per week respectively & method calculateWages() to calculate their monthly salary. Also override getDesig () method depending on the type of contract employee.
5. Write an application to create a superclass Vehicle with information vehicle number,insurance number,color and methods getConsumption() displayConsumption(). Derive the sub-classes TwoWheeler and FourWheeler with method maintenance() and average() to print the maintenance And average of the vehicle.
6. Extend the above Two Wheeler class with methods getType() and getName() which gives the information about the type and the name of the company.Crea sub-classes Geared and Non Geared with method average() to print the average of a geared and non-geared two wheeler.

Week-VII

1. Create an abstract class Shape which calculates the area and volume of 2-d and 3-d shapes with methods getArea() and getVolume(). Reuse this class to calculate the area and volume of square ,circle ,cube and sphere.
2. Create an abstract class Employee with methods getAmount() which displays the amount paid to employees. Reuse this class to calculate the amount to be paid to WeeklyEmployee and HourlyEmployee according to no. of hours for HourlyEmployee and no. of weeks for WeeklyEmployee.
3. Create an Interface payable with method getAmount ().Calculate the amount to be paid to Invoice and Employee by implementing Interface.
4. Create an Interface Vehicle with methods getColor(),getNumber(), getConsumption() calculate the fuel consumed, name and color for TwoWheeler and Four Wheeler By implementing interface Vehicle.
5. Create an Interface Fare with method getAmount() to get the amount paid for fare of traveling. Calculate the fare paid by bus and train implementing interface Fare.
6. Create an Interface StudentFee with methods getAmount(), getFirstName(), getLastName() , getAddress(), getContact(). Calculate the amount paid by the Hostler and Non Hostler student by implementing interface Student Fee

Week-VIII

1. Write a Program to create your own package. Package should have more than two classes.write a Program that uses the classes from the package.

2. Create a package named org.shapes. Create some classes in the package representing some common geometric shapes like Square, Triangle, Circle and so on. write a Program that uses the classes from the package.
3. Write a Java program to create a package called dept. Create four classes as CSE, ECE, ME and CE adds methods in each class which can display subject names of your respective year. access this package classes from main class
4. Write a Calculator program : Include all calculator operations as classes in a Package “Calculator” and import it to the main class.
5. 5. Write a program for the following
 - a. Example to use interfaces in Packages.
 - b. Example to create a sub package in a package

Week-IX

1. Write a program to create MyThread class with run() method and then attach a thread to this MyThread class object.
2. Write a program where the consumer thread checks the data production status [is over or not] for every 10 ms.
3. Write a Program using Threads to simulate a traffic light. The Signal lights should glow after each 10 second, one by one. For example: Firstly Red, then after 10 seconds, red will be put to off and yellow will start glowing and then accordingly green.
4. Write a Program using Threads for the following case study: Movie Theatre To watch a movie the following process is to be followed, at first get the ticket then show the ticket. Assume that N persons are trying to enter the Theatre hall all at once, display their sequence of entry into theater. Note: The person should enter only after getting a ticket and showing it to the boy.
5. Write a Program using Threads for the following case study: Train Reservation system To reserve a berth the following process need to be followed, at first check the number of available berths with the requested berths, if the number of requested berths are less than or equal to available berths then allot berth and print ticket or else display no berths are available. Assume that N persons are trying to reserve the berth, display their sequence of reservation status along with the number of available berths. Note : The person can print ticket only if berth is confirmed.

Week-X

1. Write a program for the following
 - a. display a frame with title MyFrame

- b. draw a horizontal line.
 - c. Draw one line perpendicular to other. One line parallel to other.
- 2. Create an application to display a circle within rectangle and fill different colors in the circle & rectangle
- 3. Write an application that displays any string. Choose color from combo box to change the color of this displayed string and choose its size & type respectively from another two combo boxes.
- 4. Create a GUI with title STUDENT which has labels roll no., name, course, gender, class, address with textboxes for taking input from the user(without any functionality) and checkboxes for selecting the course, radio buttons for selecting gender with appropriate background color.

Week-XI

- 1. Write a program to create a frame by creating an object to JFrame class and include close button to terminate the application of the frame.
- 2. Write a program to create a push button, when the button is clicked an image is displayed in the frame.
- 3. Write a program to create a menu with several menu items.
- 4. Create an application Form for University Enrollment with the following Fields.
 - a. Check box
 - b. Text area
 - c. List box
 - d. Display text
 - e. Push buttons
 - f. Combo box.
 - g. Radio buttons.
 - h. Background color

Week-XII

- 1. Write a program to insert data into Student Table.
- 2. Write a program to retrieve the data from the table Student.

Course Code	Course Title					Course Type
	Analog and Digital Communications					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	1	0	40	60	4
Course Objectives • To Understand principles of analog modulation and demodulation • To Analyze noise characteristics in communication systems and calculate SNR • Comprehend pulse modulation techniques and digital modulation schemes. • Study techniques for optimal signal detection in the presence of noise • To Learn about information theory and coding techniques.						
Course Outcomes 1. Analyze and compare different analog modulation schemes for their efficiency and bandwidth 2. Analyze the behavior of a communication system in presence of noise 3. Investigate pulsed modulation system and analyze their system performance 4. Analyze different digital modulation schemes and can compute the bit error performance 5. Apply coding techniques for efficient data transmission.						

UNIT-I: Analog Communications:

Amplitude Modulation schemes: AM, DSBSC, SSBSC and VSB modulation and demodulation. Angle Modulation schemes: FM and PM, Spectral characteristics of angle modulated signals. Super hetrodyne receivers, Frequency Division Multiplexing.

UNIT-II: Noise in analog communication systems:

Gaussian and White noise characteristics, Noise in Amplitude modulation and Angle modulation systems, and SNR calculations. Pre-emphasis and De-emphasis.

UNIT-III: Pulse modulation:

Sampling process and Quantization, SQNR, A-law and μ -law companding. PAM, PCM, DPCM, DM, ADM, Time Division Multiplexing.

Digital modulation schemes:

ASK, PSK, FSK, QAM and their constellations.

UNIT-IV: Detection Theory

Gram-Schmidt Orthogonalization, Optimal detection of signals in the presence of noise: MAP rule, ML rule. Matched filter receiver. BER calculations for PCM, ASK, PSK, FSK, QAM in AWGN channel.

UNIT-V: Information theory and coding:

Entropy, Source coding theorem, Channel coding theorem, Huffman code, Repetition code, Hamming code.

Text Books

1. Simon Haykin, —Communication Systems, Wiley-India edition, 3rd edition, 2010
2. B.P. Lathi, & Zhi Ding, —Modern Digital & Analog Communication Systems , Oxford University Press, International 4th edition, 2010.

References

1. John G. Proakis & M. Salehi --- Digital Communications, 5th edition, Mc Graw Hill education, 2014.
2. A. Bruce Carlson, & Paul B. Crilly, —Communication Systems – An Introduction to Signals & Noise in Electrical Communication , McGraw-Hill International Edition, 5th Edition, 2010.
3. Herbert Taub & Donald L. Schilling, —Principles of Communication Systems , Tata McGraw-Hill, 3rd Edition, 2009.
4. George Kennedy and Bernard Davis, —Electronics & Communication System , TMH, 2004

Course Code	Course Title				Course Type	
	Computer Architecture (CA)				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives - The purpose of the course is to learn the basic structure and operations of a computer and the basic architectural concepts. It begins with basic organization, design, and programming of a simple digital computer and introduces simple register transfer language to specify various computer operations. - To learn fixed-point and floating point Arithmetic. - To learn the designing of micro programmed control unit. - To understand the memory hierarchies, cache memories and virtual memories. - To learn the different ways of communication with I/O devices.						
Course Outcomes - To Know basic principles of computer's working and Analyze the performance of computers. - Use fixed point multiplication (Booth's) algorithm. - To understand the development of a control unit . - To understand the memory organization. - Explain the concept of various I/O operations.						

Unit I: Basic Structure of Computers:

Functional units, software, performance issues software, machine instructions and programs, Types of instructions, Instruction sets: Instruction formats, addressing mode, Assembly language, Stacks, Queues, Subroutines.

Unit II: Information representation:

Number formats. Multiplication & division, ALU design, Floating Point arithmetic, IEEE 754 floating point formats.

Unit III: Control Design:

Instruction sequencing, Interpretation, Hard wired control - Design methods, and CPU control unit. Microprogrammed Control - Basic concepts, minimizing microinstruction size.

Unit IV: Memory organization:

Device characteristics, RAM, ROM, Memory management, Concept of Cache & associative memories, Virtual memory.

Unit V: I/O organization:

Input - Output systems, Interrupt, DMA, Standard I/O interfaces Concept of parallel processing, Pipelining, Forms of parallel processing.

Text Books

1. A.S.Tanenbum, "Structured Computer Organisation", PHI, Third edition
2. M.M.Mano, "Computer System Architecture", Edition

REFERENCES

1. V.CarlHammacher, "Computer Organisation", Fifth Edition.
2. Y.Chu, "Computer Organization and Microprogramming", II, Englewood Chiffs, N.J., Prentice Hall Edition.

MOOCs

1. <https://www.youtube.com/watch?v=4TzMyXmzL8M&list=PL59E5B57A04EAE09C>
2. <https://www.youtube.com/watch?v=msqxxkEKfg8I&list=PLgHucKw979AvcnTpPNZMZyORdL5HvTr9m>

Course Code	Course Title					Course Type
	Digital Signal Processing (DSP)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To introduce the basic concepts of discrete-time signals and systems. • To teach students how to analyze discrete signals and systems in frequency domains. • To explain the effects of finite word length in digital filter design. • To introduce students to different digital filter structures and their realizations. • To teach students the design methodologies of the Digital filter design						
Course Outcomes 1. Understand the fundamentals of Signals and systems. 2. Learn the analysis of discrete signals and systems in frequency domain. 3. Learn characteristics of Linear phase systems and understand the effects of finite word length in digital filter design. 4. Understand the different digital filter structures and their realizations. 5. To learn the design methodologies for Digital FIR and IIR filters.						

UNIT I

Discrete time signals and Systems:

Introduction to DSP, Applications of DSP, Sequences; representation of signals on orthogonal basis; Sampling and reconstruction of signals;

Discrete systems attributes, Representation of system with Difference equations and Impulse response calculation, LSI systems, Circular Convolution with examples

UNIT II

Frequency domain analysis using DTFT, DFT, Z transform

Review of DTFT, Discrete Fourier Transform (DFT) with Properties, Computation of Linear and circular convolution using DFT, Fast Fourier Transform Algorithm,

Z transform, ROC, Properties, System description in the frequency domain.

UNIT III

Digital filters and finite word length effects

Linear Phase filters, Analysis of simple digital filters, Comb filters, all-pass functions, Procedure for stability criteria of discrete systems

Effect of finite register length in FIR filter design.

Unit IV

Digital Filter Structures:

Direct, parallel, cascade, ladder and lattice for Infinite Impulse Response (IIR) filters

Possible realizations for FIR or Finite Impulse Response filters, including poly phase.

UNIT V

Design of IIR and FIR Digital filters::

Design of IIR Digital Filters: Butterworth, Chebyshev and Elliptic Approximations; Lowpass, Bandpass, Bandstop and High pass filters, invariant and bilinear transformations

Design of FIR Digital filters: Window method, and frequency response sampling techniques

Text Books

1. S.K.Mitra, Digital Signal Processing: A computer based approach. TMH
2. A.V. Oppenheim and Schafer, Discrete Time Signal Processing, Prentice Hall, 1989.

References

1. John G. Proakis and D.G. Manolakis, Digital Signal Processing: Principles, Algorithms And Applications, Prentice Hall, 1997.
2. L.R. Rabiner and B. Gold, Theory and Application of Digital Signal Processing, Prentice Hall, 1992.
3. J.R. Johnson, Introduction to Digital Signal Processing, Prentice Hall, 1992
4. D.J.DeFatta, J. G. Lucas and W.S.Hodgkiss, Digital Signal Processing, John Wiley & Sons, 1988.

Course Code	Course Title					Course Type
	RF AND MICROWAVE ENGINEERING (RFMW)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To prepare students to understand basic principle of microwave and its applications. • To prepare students to understand different microwave components and analyzing different type of junctions used in microwave engineering. • To teach the students about various microwave solid state devices and their characteristics. • To understand and gain complete knowledge about RF basic concepts, RF filter design. • To understand and gain complete knowledge about RF amplifier design.						
Course Outcomes 1. Able to calculate cut off frequency, identify possible modes and obtain mode characteristics of Reflex Klystron and Gunn oscillator. 2. understand the principles of operation of waveguide, gyrator, isolator attenuator etc. and obtain scattering matrix for various junctions like E-plane, H plane, Circulator, Direction Coupler. 3. Analyze and design basic microwave amplifiers, particularly klystrons, magnetron, and RF filters, basic RF oscillator and mixer models. 4. Become proficient with microwave measurement of power, frequency and VSWR, impedance for the analysis and design of circuits. 5. Analyze T-R Module, microwave systems and microwave antennas..						

UNIT-I: Introduction to Microwaves

History of Microwaves, Microwave Frequency bands; Applications of Microwaves: Civil and Military, Medical, EMI/ EMC.

Mathematical Model of Microwave Transmission: Concept of Mode, Features of TEM, TE and TM Modes, Losses associated with microwave transmission, Concept of Impedance in Microwave transmission.

UNIT-II: Analysis of RF and Microwave Transmission Lines

Coaxial line, Rectangular waveguide, Circular waveguide, Resonator, Strip line, Micro strip line.

Microwave Network Analysis: Equivalent voltages and currents for non-TEM lines, Network parameters for microwave circuits, Scattering Parameters.

UNIT-III: Passive and Active Microwave Devices:

Microwave passive components: Directional Coupler, Power Divider, Magic Tee, Attenuator.

Microwave active components: Diodes, Transistors, Oscillators, Mixers. Microwave Semiconductor Devices: Gunn Diodes, IMPATT diodes, Schottky Barrier diodes, PIN diodes. Microwave Tubes: Klystron, TWT, Magnetron.

UNIT-IV: Microwave Design Principles:

Impedance transformation, Impedance Matching, Microwave Filter Design, RF and Microwave Amplifier Design, Microwave Power Amplifier Design, Low Noise Amplifier Design, Microwave Mixer Design, Microwave Oscillator Design.

Microwave Antennas: Antenna parameters, Antenna for ground based systems, Antennas for airborne and satellite borne systems, Planar Antennas.

UNIT-V: Microwave Measurements:

Power, Frequency and impedance measurement at microwave frequency, Network Analyser and measurement of scattering parameters, Spectrum Analyser and measurement of spectrum of a microwave signal, Noise at microwave frequency and measurement of noise figure. Measurement of Microwave antenna parameters.

Microwave Systems: Radar, Terrestrial and Satellite Communication, Radio Aidsto Navigation, RFID, GPS. Modern Trends in Microwaves Engineering- Effect of Microwaves on human body, Medical and Civil applications of microwaves, Electromagnetic interference and Electromagnetic Compatibility (EMI & EMC), Monolithic Microwave ICs, RFMEMS for microwave components, Microwave Imaging.

Text Books:

1. Microwave devices and circuits-Samuel Y. Liao, Pearson, 3rd Edition, 2003.
2. Microwave principles-Herbert J.Reich, J.G.Skalknik, P.F.OrdungandH.L.Krauss, CBS publishers and distributors, New Delhi, 2004.

References:

1. Foundations for microwave engineering-R.E.Collin, IEEE press, John Wiley, 2nd edition, 2002.
2. Microwave circuits and passive devices-M.L.Sisodia and G.S.Raghuvanshi,Wiley Eastern Ltd., New age International publishers Ltd., 1995.
3. Microwave engineering passive circuits-Peter A.Rizzi, PHI, 1999.
4. Electronic and Radio Engineering-F.E.Terman, McGraw-Hill, 4th Edition, 1995

Course Code	Course Title				Course Type	
	VLSI Engineering				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To study the MOS transistors with their characteristics succeeded by the fabrication process. • Making aware of VLSI design flow and gaining knowledge on its basic micron constraints for a panoramic view of transistors • Understanding various subsystem design concepts and its internal schematics. • Comparison of various programmable logic devices in terms of applications • To understand the need for testing a VLSI chip by applying the Engineering skills to meet the challenges in semiconductor industries.						
Course Outcomes 6. Students will be Analyzed with various processing steps involved in IC on monolithic devices followed by understanding MOSFETS electrical properties. 7. Applying the knowledge of layout, stick diagrams, static and switching characteristics of inverters by CMOS technology for designing a sequential circuit. 8. Students will be good at Realizing CMOS as a switch and its technology for designing a combinational circuit by implementing it using transmission gate/PLD's. 9. Students will be knowing the ability to identify, formulate, and analyze by creating an ability to use the techniques, skills and modern EDA tools necessary for design and test of VLSI circuits by keeping aware of contemporary issues. 10. Students will be good at designing VLSI systems by keeping a view on the design for testability concepts.						

UNIT I: INTRODUCTION

Introduction to IC Technology–MOS, PMOS, NMOS, CMOS & BiCMOS technologies- Oxidation, Lithography, Diffusion, Ion implantation, Metallization Basic Electrical Properties of MOS and BiCMOS Circuits: Ids-Vds relationships, MOS transistor threshold Voltage, gm, gds, figure of merit o; Pass transistor, NMOS Inverter, Various pull ups, CMOS Inverter analysis and design, Bi-CMOS Inverters.

UNIT II: VLSI CIRCUIT DESIGN PROCESSES

VLSI Design Flow, MOS Layers, Stick Diagrams, Design Rules and Layout, 2 μ m CMOS Design rules for wires, Contacts and Transistors Layout Diagrams for NMOS and CMOS Inverters and Gates, Scaling of MOS circuits, Limitations of Scaling.

UNIT III: GATE LEVEL DESIGN

Logic Gates and Other complex gates, Switch logic, Alternate gate circuits, Basic circuit concepts, Sheet Resistance R_S and its concept to MOS, Area Capacitance Units, Calculations –Delays, Driving large Capacitive Loads, Wiring Capacitances, Fan-in and fan-out, Choice of layers Data path subsystem : Subsystem Design, Shifters, Adders, ALUs, Multipliers, Parity generators, Comparators, Zero/One Detectors, Counters,.

UNIT IV: SEMICONDUCTOR INTEGRATED CIRCUIT DESIGN

Logic Families: characteristics of digital circuit (Fan-in, Fan-out, power dissipation, propagation delay, noise margin, Figure of Merit etc., Saturated logic families: DCTL, RTL, DTL, HTL, TTL, I²L etc., non-saturated logic families: STTL, ECL. PMOS, NMOS, CMOS. Array Sub Systems: SRAM, DRAM.

UNIT V: CMOS TESTING

CMOS Testing, Need for testing, Test Principles, Design Strategies for test, Chip level Test Techniques, System-level Test Techniques, Layout Design for improved Testability.

TEXTBOOKS

1. Essentials of VLSI circuits and systems – Kamran Eshraghian, Eshraghian Douglas and A. Pucknell, PHI, 2005 Edition.
2. Principles of CMOS VLSI Design – Weste and Eshraghian, Pearson Education, 1999.

REFERENCES

1. Chip Design for Submicron VLSI: CMOS Layout & Simulation, – John P. Uyemura, Thomson Learning.
2. Introduction to VLSI Circuits and Systems – John .P. Uyemura, John Wiley, 2003.
3. Digital Integrated Circuits – John M. Rabaey, PHI, EEE, 1997.
4. Modern VLSI Design – Wayne Wolf, Pearson Education, 3rd Edition, 1997.
5. VLSI Technology – S.M. SZE, 2nd Edition, TMH, 2003.

Externals: 60Marks**Internals: 40Marks****L-T-P-C****0-0-2-1****List of Experiments:**

1. Amplitude Modulation and Demodulation
2. DSB-SC Modulation and Demodulation
3. Frequency Modulation
4. Frequency division multiplexing
5. PAM, PWM, PPM
6. Automatic Gain Control circuit
7. Carrier recovery circuit
8. Mixer circuit
9. Verification of Sampling theorem
10. Quantizer design
11. PCM implementation
12. ASK, PSK, FSK, QPSK modulation demodulation
13. Decoding of corrupted repetition code
14. Time division multiplexing
15. Using MATLAB, plot the constellation of BPSK, QPSK, without noise and with AWGN (under different SNR values) and draw the decision boundaries. Observe the symbol errors, bit errors.
16. Using MATLAB monte-carlo simulations, to find the BER versus SNR curves for ASK, BPSK, FSK, QPSK, 16 PSK, 16-QAM with AWGN channel.
17. Using MATLAB program, find the Huffman code for given set of samples.

Course Outcomes:

At the end of this lab course, students will able to learn the following:

- ☐ Basic level circuit design for AM, DSBSC, FM, modulator and demodulators
- ☐ Basic level receiver circuit design for analog receivers
- ☐ Digitization of analog signals
- ☐ Various modulation techniques used for digital signal transmission and BER performance
- ☐ Demonstration of Frequency division multiplexing and Time division multiplexing
- ☐ Source coding and Channel coding demonstration

Externals: 60Marks**Internals: 40Marks****L-T-P-C****0-0-2-1****Course Objectives:**

- ☐ To implement Linear and Circular Convolution
- ☐ FFT algorithm using MATLAB.
- ☐ To implement FIR and IIR filters
- ☐ Design of digital filters using MATLAB.
- ☐ To study the architecture of DSP processor
- ☐ Implementation of digital filters on DSP Processor.

List of Experiments**(A) Experiments on signal processing using MATLAB.**

1. Basic matrix operations and Generation of test signals.
2. Even and odd parts of the given sequence
3. Linear Convolution, circular convolution.
4. Interpolation and Decimation
5. Discrete Fourier Transform(DFT) and Fast Fourier Transform(FFT)
6. Filter Analysis and Implementation.
7. Analog Filter Design
8. IIR filter design: Butter worth, Chebyshev type 1 and 2: LPF, HPF, BPF & BSF filter.
9. FIR filter design using different windows
10. Adaptive equalizer, LS, MMSE
11. Bayesian, ML estimators

(B) Experiments on DSK and CCS

1. Study of procedure to work in real- time
2. Linear Convolution
3. Decimation and Interpolation
4. Implementation of IIR filters
5. Implementation of FIR filter

Course Outcomes:

Students will be able to:

- ☐ Design and analyze the digital filters using MATLAB.
- ☐ Implement FFT algorithms for linear filtering using MATLAB.
- ☐ implement FIR and IIR filters using MATLAB.
- ☐ Design and Implement the digital filters on DSP processor.

References:

1. Digital Signal Processing – A Computer Based Approach By Sanjay K. Mitra, Tata McGraw Hill
2. Vinay K. Ingle and John G. Proakis, “Digital Signal Processing using MATLAB”, 4/e, Cengage learning, 2011.
3. B. Venkataramani and M. Bhaskar,”Digital Signal Processor architecture, programming and application”, 6/e, TMH, 2006.

Course Code	Course Title				Course Type	
	NUMERICAL METHODS				BSC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives •						
Course Outcomes						

Unit 1: Error in numerical calculations and Solutions of equations one variable

Sources of errors, significant digits and numerical instability. Solutions of nonlinear equations: Bisection method, Method of false position, Newton-Raphson method, Fixed-point iteration, Rates of convergence of these methods. Iteration based on second degree equation: Muller method.

Unit-II Solution of system of linear algebraic equations

Direct methods: Gauss and Gauss– Jordan methods. Crout’s triangularization method.

Iterative methods: Gauss-Jacobi and Gauss-Seidel methods, Relaxation method.

Unit –III Interpolation and Numerical Integrations

Interpolation: Polynomial Interpolation: Lagrange’s interpolation, Newton’s divided difference interpolation polynomial, Gregory-Newton Forward and Back ward difference interpolation formulae, Piecewise and Spline interpolation.

Numerical integration: Trapezoidal and Simpson rules.

Unit-IV Numerical solution of ODE

Differentiation formulas in the case of equally spaced points. Numerical solution of ordinary differential equations: Single step methods: Taylor series method, Picard’s Method, Euler and Modified Euler methods, Runge – Kutta methods of 2nd and 4th order. Multi-step methods: Milne’s Predictor-Corrector formulas, Adam-Bashforth and Adam-Moulton formulas.

Unit-V: Finite difference methods

Solution of Linear difference equations with constant coefficients, Solutions of boundary value problems in ordinary differential equations, Approximate solution of eigen value problems, Finite difference methods for solving two-dimensional Laplace’s equation for a

rectangular region, Finite difference method of solving heat equation and wave equation with given initial and boundary conditions.

Textbooks

1. Froberg C. E., Introduction to Numerical Analysis 2ndedition, Addison Wesley, 1970.
2. Gerald C. F., Wheatley P.O., Applied Numerical Analysis, 6thedition, Pearson Asia, 2002

References

1. Jain M.K., Iyengar S.R.K., Numerical methods for Scientific and Engineering Computation, 3rdedition, New Age International (P) Ltd, 1996.
2. Phillips G.M., Taylor P.J., Theory and Applications of Numerical Analysis, 2ndedition Academic Press,

Externals: 60Marks**Internals: 40Marks****Course Objectives:****L-T-P-C****0-0-2-1**

- ☐ The goal of this course is to introduce students to the concepts and principles of the advanced microwave engineering.
- ☐ To study the characteristics of RKO and Gunn oscillator.
- ☐ Measurement of frequency and wavelengths would be learnt by the student.
- ☐ VSWR various TEES, MHD and Circulator would be understood by the student.
- ☐ Radiation pattern would be learnt by the student for horn antenna.
- ☐ To study the usage of hand held Vector Network Analyzer, Spectrum Analyzer, Advanced Microwave Integrated Circuits.

List of Experiments:**SECTION-A**

1. Study of standing wave pattern.
2. Measurement of guide wavelength and frequency.
 - (i) By using Frequency meter i.e. Direct Method
 - (ii) By using Slotted line method i.e. Indirect Method
3. Repeller mode characteristics of Reflex klystron.
4. I-V characteristics of Gunn Diode.
5. Measurement of VSWR.
 - (i) By using Slotted line method ($S < 10$)
 - (ii) By using Double minimum method ($S > 10$)
6. Calibration of Crystal detector.
7. Calibration of Attenuator (Fixed attenuation i.e. Power Ratio method).
8. Measurement of attenuator (Variable attenuation i.e. RF substitution method)
9. Measurement of unknown impedance.
 - (i) By using Load impedance formula
 - (ii) By using Smith Chart
10. Radiation pattern of horn antenna and parabolic dish antenna.

SECTION-B**Resonant Microwave components**

1. Introduction regarding S-parameters (Study Experiment).
2. Characteristics of Magic-Tee with the help of S-matrix and observe the phase difference with the help of CRO.
3. Characteristics of Directional Coupler.
 - (i) Directivity
 - (ii) Isolation
 - (iii) Insertion loss
 - (iv) Coupling Factor

(v) S-matrix

(vi) And Prove $P^2 + q^2 = 1$.

Non-Resonant Microwave components

4. Characteristics of Circulator (3-port).
 - (i) Find S-matrix
 - (ii) Find VSWR
5. 4-port Circulator by using two magic tees and one gyrator.
6. Characteristics of Isolator (By using Y-circulator)
 - (i) Find S-matrix
 - (ii) Find VSWR

Course Outcomes:

1. Gain knowledge and understanding of microwave analysis methods.
2. Be able to apply analysis methods to determine circuit properties of passive/active microwave devices.
3. Analyze the characteristics of RKO and Gunn oscillator.
4. Measure the frequency and guided wavelength.
5. Estimate the VSWR for various loads and S-Matrix for various microwave devices.
6. Obtain the horn antenna radiation pattern.

Text Books:

1. Microwave devices and circuits-Samuel Y. Liao, Pearson, 3rd Edition, 2003.
2. Microwave principles-Herbert J.Reich, J.G.Skalnik, P.F.Ordung and H.L.Krauss, CBS publishers and distributors, New Delhi, 2004.
3. Microwave engineering- David M. Pozar, fourth edition, John Wiley & Sons Inc. publications.

References:

1. Foundations for microwave engineering- R.E.Collin, IEEE press, John Wiley, 2nd edition, 2002.
2. Microwave circuits and passive devices- M.L.Sisodia and G.S.Raghuvanshi, Wiley Eastern Ltd., New age International publishers Ltd., 1995.
3. Microwave engineering passive circuits-Peter A.Rizzi, PHI, 1999.
4. Electronic and Radio Engineering-F.E.Terman, McGraw-Hill, 4th Edition, 1995.
5. Microwave and Radar engineering- Dr. M. Kulakarni, Umesh publications, fifth edition, 2015

Course Code	Course Title					Course Type
	Dital System Design (DSD)					PCC
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To learn the Hardware Description language (HDL) to design the digital systems • To learn the various concepts of timing parameters and its effects of digital systems. • To learn the controller and datapath designs of digital systems. • To understand the structure of FPGAs.						
Course Outcomes 1. Students will be able to understand the basic concepts of Verilog Hardware Description Language. 2. Students will be able to understand the concepts of timing constraints of digital systems. 3. Students will understand the steps to follow to design the digital systems. 4. Students will understand the concepts to design the controller and data path of digital systems. 5. Students will understand the structure of programmable logic devices like FPGAs.						

UNIT-I: Introduction to Verilog

Evolution of CAD tools, Overview of Design Flow, Modeling Concepts, Modules and Ports, Different Abstractions- Gate level, Dataflow, Behavioral, Tasks and Functions, Useful Modeling Techniques

UNIT-II: Advanced Verilog

Timings and Delays, clocking of Flip Flops- effect of Propagation delay by considering timing constraints, clock skew, Global setup and Hold time.

UNIT-III: Basic systems design

Review of FSM- Mealy, Moore Machines, State graphs, State tables, Design of pattern identification, Hardware realizations- Sequential logic, combinational logic and Verilog Modeling, One hot controller-vending machine, Hardware realizations and Verilog Modeling.

UNIT-IV: Sophisticated designs

ASM-components, Meelay, Moore ASM, Bus Arbiter, Traffic Light Controller, Dice Game,

Micro programming techniques-SQDA,SQSA for Dice Game

UNIT-V: FPGA architectures –Programmable logic devices –Spartan 6 FPGA Block Diagram and Structure.

Text Books

1. Verilog HDL: A Guide to Digital Design and Synthesis - Samir Palnitkar, Prentice Hall, Second Edition, 2003.
2. Digital Design: Principles and Practices - Jon F Wakerly, Pearson, Fourth Edition, 2007.

References

1. Digital Design and Verilog HDL Fundamentals – Joseph Cavanagh,CRC Press, 2008.

Online Resources

1. <https://youtu.be/Y8FvvzcocT4> , Digital VLSI System Design ,Lecture Series on VLSI
2. Design by Prof S.Srinivasan, Dept of Electrical Engineering, IIT Madras

5.

Course Code	Course Title				Course Type	
	MICRO-CONTROLLERS AND INTERFACING				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3
Course Objectives • To illustrate various types of processors available in the market with their features. • To illustrate the architecture of 8086 microprocessor and microcontroller 8051. • To introduce the programming and interfacing techniques to 8086 microprocessor. • To analyse the basic concepts and programming of 8051 microcontroller. • To understand the interfacing circuits for various applications of 8051 microcontroller.						
Course Outcomes 1. Describe the architecture of 8085, 8051 and 8086. 2. Illustrate the organization of registers and memory in microprocessors. 3. CO3: Identify the addressing mode of an instruction. 4. Develop programming skills in assembly language. 5. Explain the need for different interfacing devices						

Unit 1: Architecture of Microprocessors:

General definitions of mini computers, microprocessors, micro controllers and digital signal processors. CISC Vs RISC and ARM processors, Overview of 8085 microprocessors.

Unit 2: Architecture and Assembly language of 8086:

Architecture, memory segmentation, signals and pins of 8086 microprocessors. Assembly directives, Addressing modes, Description of Instructions and Assembly software programs with algorithms.

Unit 3: Interfacing with 8086:

Interfacing with peripheral ICs like 8255-PPI, 8237-DMA controller, 8259- Programmable Interrupt Controller, Interfacing with key boards, LEDs, LCDs, ADCs, and DACs etc.

Unit 4: Micro-controller 8051:

Overview of the architecture of 8051 microcontroller, Description of Instructions. Assembly directives. Assembly software programs with Algorithms.

Unit 5: Interfacing with 8051:

Interfacing with keyboards, LEDs, 7 segment LEDs, LCDs, Interfacing with ADCs and DACs, stepper motor etc.

Text Books:

1. Douglas Hall, Microprocessors Interfacing, Tata McGraw Hill, 1991.
2. Kenneth J. Ayala, The 8051 Microcontroller, Penram International Publishing, 1996.

References

1. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996
2. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.

Course Objectives:

- To introduce students to the Pattern Recognition and its applications.
- To study Bayes decision theory and parameter estimation methods.
- To study the classical techniques of classification.
- To study clustering algorithms.
- To study some concepts of Artificial Neural Network

UNIT-1

Introduction to Pattern Recognition:

Types of human learning, Supervised learning, Unsupervised learning, Reinforcement learning. Some applications of Artificial Intelligence. Training data, test data, cross validation, data collection (Unbiased data), data cleaning, feature extraction, Properties of best features for classification. Linear Regression. Under-fitting, Over-fitting problems, Bias-Variance trade-off. Basics of Probability, Linear Algebra and Optimization:

Conditional and joint probability, Bayes' theorem; Expectation, Autocorrelation, Cross-Correlation, Jointly Gaussian Random variables and special cases, Binomial distribution. Linearly independent vectors, orthogonal vectors, Inner product, Various types of Norms, projections, Over-determined system and Least squares solution. Eigen Values, Eigen Vectors, SVD. Local minima, global minima, convex functions, gradient descent algorithm.

UNIT-2

Bayes Decision Theory:

Bayesian risk, Two category classification. Minimum-error-rate classification, Multi class Classifiers, Discriminant functions, Decision surfaces. Discriminant functions for Normal densities (with two special cases). Missing and Noisy features, Bayesian belief networks. Parameter Estimation Methods: Maximum-Likelihood estimation: Gaussian case; Maximum A posteriori estimation; Bayesian estimation: Gaussian case. Expectation-Maximization, Hidden Markov Models.

UNIT-3

Some other classification techniques:

Naive Bayes classifier, Parzen-window method for density estimation, K-Nearest neighbor, Perceptron learning algorithm, Feature normalization, Decision Trees, Support Vector Machines. Ensemble learning techniques: Bagging, Boosting, Stacking. Some variants of Gradient Descent algorithms: Batch, mini-batch, Stochastic Gradient descent. Momentum based gradient descent. Over-fitting and L1, L2 regularization techniques.

UNIT-4

Component analysis and clustering:

Problems of dimensionality, Principle component analysis, Independent component analysis, Linear discriminant analysis. Similarity measures, Mixture densities and identifiability, Gaussian Mixture Models. k-Means clustering, hierarchical clustering.

UNIT-5

Artificial Neural Networks:

Logistic Regression, Introduction to Neural networks, back propagation algorithm. Activation functions: Sigmoid, tanh, ReLU, SoftMax. Problems in Deep Neural Networks, dropout regularization. CNN model, data augmentation. Performance metrics for imbalanced classes

Course Outcomes:

At the end of the course, students will learn the various algorithms related to Supervised Learning,
Unsupervised learning techniques.

Text Books:

1. Pattern Classification, Richard O.Duda et.al. 2 nd edition.
2. Pattern Recognition and Machine Learning, Christopher Bishop, Springer

Further Reading:

1. Deep Learning, Ian Goodfellow et.al.
2. Reinforcement Learning, Richard S. Sutton and Andrew G. Barto.

Course Outcome:

At the end of this course, students will get the ability to
Do preprocessing task and applying suitable classification algorithm for a given classification task.

Apply clustering algorithm for a given data.

Apply regression analysis technique for prediction.

Work with association algorithms.

Represent the data with less number of features.

Reference books:

R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2002.

C.M.Bishop, Neural Networks and Pattern Recognition, Oxford University Press (Indian Edition), 2003.

Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Introduction to Data Mining, Pearson, 2016.

B.Yegnanarayana, Artificial Neural Networks, PHI, 1999.

CS3203

OPERATING SYSTEMS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

OBJECTIVES:

To learn the fundamentals of Operating Systems.

1. To learn the mechanisms of OS to handle processes and threads and their communication
2. To learn the mechanisms involved in memory management in contemporary OS
3. To gain knowledge on distributed operating system concepts that includes architecture, Mutual exclusion algorithms, deadlock detection algorithms and agreement protocols
4. To know the components and management aspects of concurrency management

Unit 1:

Introduction: Concept of Operating Systems, Generations of Operating systems, Types of Operating Systems, OS Services, System Calls, Structure of an OS – Layered, Monolithic, Microkernel Operating Systems, Concept of Virtual Machine.

Unit 2:

Processes: Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switching

Thread: Definition, Various states, Benefits of threads, Types of threads, Concept of multithreads,

Process Scheduling: Foundation and Scheduling objectives, Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time; Scheduling algorithms: Pre-emptive and Non pre-emptive, FCFS, SJF, RR

Unit 3:

Inter-process Communication: Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer\ Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing,

Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem etc.

Deadlocks: Definition, Necessary and sufficient conditions for Deadlock, Deadlock

Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.

Unit 4: Memory Management: Basic concept, Logical and Physical address map, Memory

allocation: Contiguous Memory allocation – Fixed and variable partition–

Internal and External fragmentation and Compaction; Paging: Principle of

operation – Page allocation – Hardware support for paging, Protection and sharing,

Disadvantages of paging.

Virtual Memory: Basics of Virtual Memory – Hardware and control structures – Locality

of reference, Page fault, Working Set, Dirty page/Dirty bit – Demand paging, Page

Replacement algorithms: Optimal, First in First Out (FIFO), Second Chance (SC), Not

recently used (NRU) and Least Recently used (LRU).

Unit 5: I/O Hardware: I/O devices, Device controllers, Direct memory access Principles of I/O

Software: Goals of Interrupt handlers, Device drivers, Device independent I/O software,

Secondary-Storage Structure: Disk structure, Disk scheduling algorithms

File Management: Concept of File, Access methods, File types, File operation, Directory

structure, File System structure, Allocation methods (contiguous, linked, indexed),

Free-space management (bit vector, linked list, grouping), directory implementation

(linear list, hash table), efficiency and performance.

Disk Management: Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN, Disk

reliability, Disk formatting, Boot-block, Bad blocks

Suggested books:

1. Operating System Concepts Essentials, 9th Edition by AviSilberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition.

2. Operating Systems: Internals and Design Principles, 5th Edition, William Stallings, Prentice Hall of India.

Suggested reference books:

1. Operating System: A Design-oriented Approach, 1st Edition by Charles Crowley, Irwin Publishing
2. Operating Systems: A Modern Perspective, 2nd Edition by Gary J. Nutt, Addison-Wesley
3. Design of the Unix Operating Systems, 8th Edition by Maurice Bach, Prentice-Hall of India
4. Understanding the Linux Kernel, 3rd Edition, Daniel P. Bovet, Marco Cesati, O'Reilly and Associates

Outcomes:

1. Create processes and threads.
2. Develop algorithms for process scheduling for a given specification of CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time.
3. For a given specification of memory organization develop the techniques for optimally allocating memory to processes by increasing memory utilization and for improving the access time.
4. Design and implement file management system.
5. For a given I/O devices and OS (specify) develop the I/O management functions in OS as part of a uniform device abstraction by performing operations for synchronization between CPU and I/O controllers.

Externals: 60Marks**Internals: 40Marks****L-T-P-C****0-0-2-1****Course Objective:**

- Familiarize with VLSI CAD tools like Xilinx14.4 and Mentor Graphics tool.
- Gives Basic concepts of Verilog HDL code to write a code for digital circuits.
- To have hands on experience to design digital circuits, simulate and synthesis the design with Xilinx 14.4 VLSI CAD tool with timing diagrams and RTL diagrams.
- To have hands on experience for transistor level design and simulate it with transient and dc analysis using mentor Graphics tool.
- FPGA implantation of the Verilog code written in the VLSI CAD tool.

LIST OF EXPERIMENTS:

1. Familiarization with Xilinx14.4 tool.
2. Simulate and Synthesis of all basis gates.
3. Simulate and synthesis of multiplexers, decoders and code converters.
4. Simulate and synthesis of all flipflops.
5. Simulate and synthesis of Universal shift register.
6. Simulate and synthesis of the binary counter, MOD counters.
7. FPGA implementation of basic gates and binary counter.
8. Familiarization of the mentor Graphic tool for transistor level design.
9. Design and synthesis of a CMOS amplifier
10. Transient and DC analysis of CMOS inverter.
11. Transient, DC and power analysis of the NAND and NOR gates using CMOS implementation.
12. Transient, DC and power analysis of the XOR gates using NAND gates cells.
13. Transient, DC and power analysis of the 2x1 MUX using NAND gates cell

Course Outcomes:

- Able to write a Verilog HDL code for the digital systems.
- Able to use the VLSI CAD tools to design digital systems and get synthesis the design to get RTL level diagram.
- Able to simulate the digital system to check the functionality with the timing diagrams.
- Able to do transient and dc analysis of the CMOS Inverter, Logic gates and analog circuits.
- Able to do FPGA Implementation of the combinational and sequential circuits.

EC3801

MICRO – CONTROLLERS LAB

Externals: 60Marks

Internals: 40Marks

L-T-P-C

0-0-2-1

Couse Objectives:

1. To Expose the students to hardware kits and dumping tools
2. to prepare the students for interfacing procedure of various I/O devices
3. To expose the students to various actuation motors and their modes of operation.
4. To Prepare the students to design for day to day applications such as TLC and Elevator systems.
5. To understand the practical usage of ADCs and DACs.

List of Experiment:

Familiarization with TITAN II Kit's hardware and usage of Triton IDE along with Flash Magic for dumping the code to the controller by blinking on board LEDs.

1. Interface simple seven segment LED display with controller.
2. To Display "DEPT OF ECE" on LCD in 8-bit as well as 4-bit mode
3. Interface Keyboard and LCD with controller.
4. Interface Stepper Motor by controlling its direction and make it spin faster or slower.
5. Interface DC motor and control its speed using PWM technique.
6. Design, program and implement Traffic Light system using microcontroller.
(prefer the design on breadboard)
7. Design, program and implement Elevatorsystem using microcontroller.
(prefer the design on breadboard)
8. Interfacing ADC to Microcontroller.
9. Interface DAC with Microcontroller and generate multiple waveforms.
10. Interface Temperature Sensor to ADC and measure it on LCD with microcontroller.

Course Outcomes:

Upon completion of the course the students will have

- ☐ Ability to understand the hardware kits and way of dumping the program in IC.

- Gain the knowledge of various input and display output devices and their interfacing to μ c-8051.
- Understanding of various motors and ability to interface with microcontrollers in various modes.
- Ability to design and demonstrate various applications such as traffic light controller and elevator control.
- Understanding the interfacing of ADCs and DACs, and various analog sensors.

Text Books:

3. Douglas Hall, Microprocessors Interfacing, Tata McGraw Hill, 1991.
4. Kenneth J. Ayala, The 8051 Microcontroller, Penram International Publishing, 1996.

Reference Textbooks:

1. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996
2. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.

BM0010 Professional Law and Ethics**Externals: 60Marks****Internals: 40Marks****L-T-P-C****3-0-0-3**

Course Objective: To understand the Legal and Regulatory Framework for doing business in India.

Course Outcome: Students will be able to understand a) Business Laws related to contracts b) Importance of Ethics in Business c) IPR and Legal Aspects.

UNIT - I : Business Ethics Definition - Importance of Ethics in Business - Distinction between Values and Ethics -Characteristics of Ethical Organization - Morality and Professional Ethics - Ethical Dilemmas- How to create an ethical working environment-Ethical Decision making in Business- Role of corporate Governance in ensuring ethics in workplace - Indian Ethical Traditions.

UNIT - II: ETHICS IN FUNCTIONAL AREAS OF BUSINESS- Ethics in Marketing: Ethical practices in product packaging and labeling - Pricing - Advertising -Direct marketing – Green marketing - Ethical vs. Unethical marketing behavior. Ethics in HRM: Ethical implications of Privacy – Harassment – Discrimination – Whistle blowing. Ethics in Finance: Accountability – Window dressing and disclosure practices – Insider trading.

UNIT –III: Law of Contract: Nature of Contract and Essential elements of valid contract, Offer and Acceptance, Consideration, Capacity to contract and Free Consent, Legality of Object, Performance and discharge of Contracts, Remedies for breach of contract.

Unit-IV

Contracts-II: Indemnity and guarantee, Contract of Agency, Sale of goods Act -1930: General Principles, Conditions & Warranties, Performance of Contract of Sale.

Unit- V: Law relating to Intellectual Property: Introduction – meaning of intellectual property, main forms of IP, Copyright, Trademarks, Patents and Designs, Secrets;

Suggested readings:

1. Maheshwari & Maheswari - A Manual of Business Laws, Himalaya Publishing House.
2. D. Chandra Bose - Business Law PHI-Private Limited, New Delhi.
3. A.C. Fernando - Business Ethics An Indian Perspective Pearson Education
4. Manuel G. Velasquez - Business Ethics Concepts and Cases Prentice-Hall of India Pvt .Ltd, 2008.
5. S.S. Gulshan - Business Laws Excel Books, New Delhi

CS4101**COMPUTER NETWORKS**

Externals: 60Marks

Internals: 40Marks

Objectives:

L-T-P-C

3-0-0-3

1. To Introduce The Fundamental Various Types Of Computer Networks.
2. To Demonstrate The TCP/IP And OSI Models With Merits And Demerits.
3. To Introduce UDP And TCP Models.

Detailed Contents:

UNIT - I:

Introduction- Hardware And Software, Data Communication, Networking, Protocol, Layering Scenario, TCP/IP Protocol Suite: The OSI Model, Internet History Standards And Administration; Comparison Of The OSI And TCP/IP Reference Model, Digital And Analog Data And Signals.

Physical Layer: Guided Transmission Media, Wireless Transmission Media.

Data Link Layer: Design Issues, CRC Codes, Elementary Data Link Layer Protocols, Sliding Window Protocol, Flow Control. Error Detection And Error Control. HDLC And other Data Link Protocols.

UNIT - II:

Band Width Utilization: Multiplexing – Frequency-Division, Synchronous Time-Division, And Statistical Time-Division Multiplexing.

Multi Access Protocols: ALOHA, CSMA, Collision Free Protocols, Ethernet- Physical Layer, Ethernet Mac Sub Layer, Data Link Layer Switching & Use Of Bridges, Learning Bridges, Spanning Tree Bridges, Repeaters, Hubs, Bridges, Switches, Routers And Gateways.

UNIT-III:

Network Layer: Network Layer Design Issues, Store And Forward Packet Switching Connection Less And Connection Oriented Networks-Routing Algorithms-Optimality Principle, Shortest Path, Flooding, Distance Vector Routing, Control To Infinity Problem, Hierarchical Routing, Congestion Control Algorithms, Admission Control.

UNIT-IV: Internetworking: Tunneling, Internetwork Routing, Packet Fragmentation, Ipv4, Ipv6 Protocol, IP Addresses, CIDR, IMCP, BOOTP, ARP, RARP, DHCP, Network Address Translation (NAT) Internetworking

Transport Layer: TCP Introduction, Reliable/Un- Reliable Transport ,Connection Establishment, Connection Release, Crash Recovery, Intra-Domain Routing: Distance-Vector,

Intra-Domain Routing: Link- State, Wireless Networks: 802.11 MAC, Efficiency Considerations

UNIT-V:The Internet Transport Protocols: UDP-RPC, Real Time Transport Protocols, The Internet Transport Protocols- Introduction To TCP, The TCP Service Model, The TCP Segment Header, The Connection Establishment, The TCP Connection Release, The TCP Connection Management Modeling, The TCP Sliding Window, The TCP Congestion Control, The Future Of TCP.

Application Layer: Introduction, Providing Services, Applications Layer Paradigms, Client Server Model, Standard Client-Server Application-HTTP, FTP, Electronic Mail, TELNET, DNS, SSH,SNMP,WWW.

Text Books:

- 1.Computer Networks, by Andrew s Tanenbaum,PHI(2010)
- 2.Data and Computer Communications,by William Stallings,PHI(2002)

References Books:

1. Data Communications and Networking - Behrouz A. Forouzan, Fifth Edition TMH, 2013.
2. Computer Networks - Andrew S Tanenbaum, 4th Edition, Pearson Education.
3. An Engineering Approach to Computer Networks - S. Keshav, 2nd Edition, Pearson Education.
4. Understanding communications and Networks, 3rd Edition, W. A. Shay, Cengage Learning.
5. Introduction to Computer Networks and Cyber Security, Chwan-Hwa (John) Wu, J. David Irwin, CRC Press.
6. Computer Networks, L. L. Peterson and B. S. Davie, 4th edition, ELSEVIER.
7. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose, K. W. Ross, 3rd Edition, Pearson Education.

Course Outcomes:

- Students should understand and explore the basics of Computer Networks and Various Protocols. He/She will be in a position to understand the World Wide Web concepts.
- Students will be in a position to administrate a network and flow of information further he/she can understand easily the concepts of network security, Mobile and ad hoc networks.

CURRICULUM FOR PROFESSIONAL ELECTIVE SUBJECTS

Credit Distribution for Professional Elective Subjects									
Stream-I									
Sl. No.	Course Code	Course Title	Semester	Course Category	Hours per week			Total contact hours	Credits
					L	T	P		
1	ECPE1 –	Digital Image Processing	VII	PEC	3	0	0	3	3
2	ECPE1 –	Adaptive Signal Processing	VII	PEC	3	0	0	3	3
3	ECPE1 –	Biomedical Signal Processing	VII	PEC	3	0	0	3	3
4	ECPE1 –	Pattern Recognition	VII	PEC	3	0	0	3	3
5	ECPE1 –	Digital Image and Video Processing	VII	PEC	3	0	0	3	3
6	ECPE1 –	Speech and Audio Processing	VII	PEC	3	0	0	3	3
7	ECPE1 –	Detection and Estimation Theory	VII	PEC	3	0	0	3	3
8	ECPE1 –	Wavelets	VII	PEC	3	0	0	3	3
9	ECPE1	Machine Learning Techniques	VII	PEC	3	0	0	3	3
10	ECPE1	Deep Learning	VII	PEC	3	0	0	3	3
11	ECPE1	Multimedia Communication	VII	PEC	3	0	0	3	3
12	ECPE1	DIGITAL SIGNAL PROCESSORS AND ARCHITECTURES	VII	PEC	3	0	0	3	3

13	ECPE1	Data Mining	VII	PEC	3	0	0	3	3
Stream-II									
1	ECPE2 –	Wireless Communications	VII	PEC	3	0	0	3	3
2	ECPE2 –	Satellite Communication	VII	PEC	3	0	0	3	3
3	ECPE2 –	Wireless Sensor Networks	VII	PEC	3	0	0	3	3
4	ECPE2 –	Large MIMO Systems	VII	PEC	3	0	0	3	3
5	ECPE2 –	Applied Optimization for Wireless, Signal Processing, Machine Learning	VII	PEC	3	0	0	3	3
6	ECPE2 –	Principles of Signal Estimation for MIMO – OFDM Communications	VII	PEC	3	0	0	3	3
7	ECPE2 –	Error Correcting Codes	VII	PEC	3	0	0	3	3
8	ECPE2 –	Information Theory and Coding	VII	PEC	3	0	0	3	3
9	ECPE2 –	Radar Systems	VII	PEC	3	0	0	3	3
10	ECPE2 –	Adhoc Wireless Sensor Networks	VII	PEC	3	0	0	3	3
Stream-III									

1	ECPE3 –	IoT and Applications + (Lab for 1 credit)	VII	PEC	3 0	0 0	0 2	3 2	3 1
2	ECPE3 –	Embedded Systems	VII	PEC	3	0	0	3	3
3	ECPE3 –	Nano Electronics	VII	PEC	3	0	0	3	3
4	ECPE3 –	Introduction to MEMS	VII	PEC	3	0	0	3	3
5	ECPE3 –	Mixed Signal Design	VII	PEC	3	0	0	3	3
6	ECPE3 –	CMOS Design	VII	PEC	3	0	0	3	3
7	ECPE3 –	Power Electronics	VII	PEC	3	0	0	3	3
8	ECPE3 –	High Speed Electronics	VII	PEC	3	0	0	3	3
9	ECPE3 –	Embedded System Design	VII	PEC	3	0	0	3	3
10	ECPE3 –	Analog VLSI and Mixed Signal Design	VII	PEC	3	0	0	3	3

Course Objectives:

1. Learn and understand the representation of Two Dimensional Linear shift invariant Systems using Matrices
2. To understand the acquisition of digital images.
3. To learn and implement orthonormal and separable image transforms
4. Learn point processing, neighborhood processing and frequency domain methods for image enhancement
5. Formulate and solve the optimization problems to achieve image restoration from degraded images

Course Outcomes:

1. To Introduce the applications of the Digital image processing in different research fields, and learn the Mathematical preliminaries required for analyzing two dimensional systems.
2. Learn the acquisition process of a Digital images.
3. Demonstrated understanding of Image transforms such as Discrete Fourier Transform, Cosine Transform, Hadamard Transform, and KLT.
4. Demonstrated understanding of image enhancement techniques
5. Understanding of formulation and solution of image restoration techniques

Unit-I: INTRODUCTION:

Mathematical Preliminaries and Two dimensional Systems Introduction to image Processing and applications of image processing in different fields, Fundamentals of Linear algebra, and Probability, one dimensional and two dimensional Linear shift invariant systems and their representation using matrices, one dimensional and Two Dimensional Convolution, Separable operations using matrices, Two dimensional Discrete time Fourier transform, Two dimensional Z transform and Properties.

Unit-II: Image sampling and Quantization

Sampling of One dimensional signals, Sampling of Two dimensional signals, Anti-aliasing filter, Quantization: Liyod Max quantizer, Uniform quantizer, Signal to quantization noise ratio.

Unit-III: Image Transforms

Unitary transforms and properties, 1D & 2D Discrete Fourier transform, 1D & 2D Discrete cosine transform, 1D & 2D Discrete sine transform, 1D & 2D Discrete Walsh transform, 1D & 2D Discrete Hadamard transform, 1D & 2D Discrete Haar transform, 1D & 2D Discrete KLT transform, Application of KLT for Face recognition.

Unit-IV: Image Enhancement

Point operations: contrast stretching, digital negative, Power law correction, dynamic range compression, intensity level slicing, Thresholding, Bit plane extraction; Histogram equalization and histogram specification; spatial operations: Linear and non linear filtering in spatial domain using spatial masks, Unsharp masking; Transform Operations: Filtering in transform domain; Pseudo coloring

Unit – V: Image Restoration

Classification of restoration methods, Characteristic metrics for Image restoration, Linear and nonlinear degradation models, Inverse filtering, Pseudo inverse filtering, Wiener filtering: Least squares approach.

REFERENCES:

A.K Jain, Fundamentals of Digital Image Processing, Prentice Hall.

R. C. Gonzalez, R.E. Woods, Digital Image Processing, Pearson.

ECPE1_

ADAPTIVE SIGNAL PROCESSING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

General concept of adaptive filtering and estimation, applications and motivation, Review of probability, random variables and stationary random processes, Correlation structures, properties of correlation matrices.

Unit II:

Optimal FIR (Wiener) filter, Method of steepest descent, extension to complex valued The LMS algorithm (real, complex), convergence analysis, weight error correlation matrix, excess mean square error and mis-adjustment

Unit III:

Variants of the LMS algorithm: the sign LMS family, normalized LMS algorithm, block LMS and FFT based realization, frequency domain adaptive filters, Sub-band adaptive filtering. Signal space concepts - introduction to finite dimensional vector space theory, subspace, basis, dimension, linear operators, rank and nullity, inner product space, orthogonality, Gram Schmidt orthogonalization, concepts of orthogonal projection, orthogonal decomposition of vector spaces.

Unit IV:

Vector space of random variables, correlation as inner product, forward and backward projections, Stochastic lattice filters, recursive updating of forward and backward prediction errors, relationship with AR modeling, joint process estimator, gradient adaptive lattice.

Unit V:

Introduction to recursive least squares (RLS), vector space formulation of RLS estimation, pseudo-inverse of a matrix, time updating of inner products, development of RLS lattice filters, RLS transversal adaptive filters. Advanced topics: affine projection and subspace based adaptive filters, partial update algorithms, QR decomposition and systolic array.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand the non-linear control and the need and significance of changing the control parameters w.r.t. real-time situation.

Mathematically represent the ‘adaptability requirement’.

Understand the mathematical treatment for the modeling and design of the signal processing systems.

Text/Reference Books:

S. Haykin, Adaptive filter theory, Prentice Hall, 1986.

C.Widrow and S.D. Stearns, Adaptive signal processing, Prentice Hall, 1984.

EC4435

BIOMEDICAL SIGNAL PROCESSING

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Outcomes:

1. To understand the origin, acquisition and processing of Bio signals and their application for diagnosis, and different Imaging modalities
2. To understand the cardiac and brain signal processing for various applications
3. To learn fundamentals of digital images, and 2 D systems, and various image transforms
4. To learn image enhancement and image restoration methods
5. To learn different classification techniques

Chapter 1: Introduction to bio Signals and Bio images

The Nature of Biomedical Signals Examples of Biomedical Signals, Origin of bio potentials, The action potential of a cardiac myocyte, The action potential of a neuron, The electroencephalogram (ENG), The electromyogram (EMG), The electrocardiogram (ECG). The electroencephalogram (EEG), Event-related potentials (ERPs), The electrogastrogram (EGG), The phonocardiogram (PCG), The carotid pulse, Signals from catheter-tip sensors, The speech signal, The vibromyogram (VMG), The vibroarthrogram (VAG), Otoacoustic emission (OAE) signals, Bioacoustic signals. Biomedical images: Different imaging modalities, Computer aided Tomography

Chapter 2: Cardiac and Brain signal Processing

Acquisition of ECG: Standard 12 lead system, ambulatory ECG signal, Fundamental Problems of Cardiac signal Processing, Pre processing techniques, QRS detection: Categorisation, Pan Tompkins technique, Automatic heart beat classification, AAMI standard, Review of methods of heart beat classification,

Chapter 3:

Digital image acquisition, sampling and quantization, Response of 1D LSI system for non-periodic and periodic inputs (Using matrix method), Response of Two dimensional LSI systems for non periodic and periodic inputs using row ordered vector form of images, Image

transforms: Unitary transforms and properties, Discrete Fourier transform, Discrete Cosine transform, Discrete sine transform, Walsh transform, Hadmard transform, Haar transform, KL transform

Chapter 4:

Image enhancement: Point processing techniques: Contrast stretching, dynamic range compression, Power law correction, Bit plane slicing, Gray level slicing, thresholding, Histogram equalization, Histogram matching, Neighbour hood processing: LSI filtering, Low pass filters, High pass filters: First order and second order derivative filters, High boost filtering, Frequency domain processing

Image Restoration: Difference between Image restoration and Image enhancement, Performance metrics, Simplified model for degradation using LSI approximation, methods of estimation of degradation function, Models for different degradations, Inverse filtering, effect of noise on inverse filtering: Least squares filtering, Constrained Least squares filtering, Wiener filtering

Chapter 5:

Data preprocessing, Feature extraction methods, Regression, Classification: Supervised and Unsupervised learning, Nearest neighbour classification, Bayesian classifier, K Nearest neighbour classification, Support vector machine, Artificial neural networks, K means clustering,

Reference Books:

1. "BIOMEDICAL SIGNAL ANALYSIS", RANGARAJ M. RANGAYYAN, IEEE press, Wiley
2. "Fundamentals of Digital image processing" by A.K Jain, PHI
3. "Digital image processing", R. Gonzalez and Woods

ECPE1_ DIGITAL IMAGE AND VIDEO PROCESSING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures.

Unit II:

Image Enhancements and Filtering-Gray level transformations, histogram equalization and specifications, pixel-domain smoothing filters – linear and order-statistics, pixel-domain sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass.

Unit III:

Color Image Processing-Color models–RGB, YUV, HSI; Color transformations–formulation, color complements, color slicing, tone and color corrections; Color image smoothing and sharpening; Color Segmentation.

Image Segmentation- Detection of discontinuities, edge linking and boundary detection, thresholding – global and adaptive, region-based segmentation.

Unit IV:

Wavelets and Multi-resolution image processing- Uncertainty principles of Fourier Transform, Time-frequency localization, continuous wavelet transforms, wavelet bases and multi-resolution analysis, wavelets and Subband filter banks, wavelet packets.

Unit V:

Image Compression-Redundancy–inter-pixel and psycho-visual; Lossless compression – predictive, entropy; Lossy compression- predictive and transform coding; Discrete Cosine Transform; Still image compression standards – JPEG and JPEG-2000.

Fundamentals of Video Coding- Inter-frame redundancy, motion estimation techniques – full search, fast search strategies, forward and backward motion prediction, frame classification – I, P and B; Video sequence hierarchy – Group of pictures, frames, slices, macro-blocks and blocks; Elements of a video encoder and decoder; Video coding standards – MPEG and H.26X.

Video Segmentation- Temporal segmentation–shot boundary detection, hard-cuts and soft-cuts; spatial segmentation – motion-based; Video object detection and tracking.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Mathematically represent the various types of images and analyze them.

Process these images for the enhancement of certain properties or for optimized use of the resources.

Develop algorithms for image compression and coding

Text/Reference Books:

R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008

Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India. 2nd edition 2004

Murat Tekalp, Digital Video Processing" Prentice Hall, 2nd edition 2015

ECPE1_

SPEECH AND AUDIO PROCESSING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Introduction- Speech production and modeling - Human Auditory System; General structure of speech coders; Classification of speech coding techniques – parametric, waveform and hybrid ; Requirements of speech codecs –quality, coding delays, robustness. Speech Signal Processing- Pitch-period estimation, all-pole and all-zero filters, convolution; Power spectral density, periodogram, autoregressive model, autocorrelation estimation.

Unit II:

Linear Prediction of Speech- Basic concepts of linear prediction; Linear Prediction Analysis of non-stationary signals –prediction gain, examples; Levinson-Durbin algorithm; Long term and short-term linear prediction models; Moving average prediction.

Unit III:

Speech Quantization- Scalar quantization–uniform quantizer, optimum quantizer, logarithmic quantizer, adaptive quantizer, differential quantizers; Vector quantization – distortion measures, codebook design, codebook types.

Scalar Quantization of LPC- Spectral distortion measures, Quantization based on reflection coefficient and log area ratio, bit allocation; Line spectral frequency – LPC to LSF conversions, quantization based on LSF.

Unit IV:

Linear Prediction Coding- LPC model of speech production; Structures of LPC encoders and decoders; Voicing detection; Limitations of the LPC model.

Unit V:

Code Excited Linear Prediction-CELP speech production model; Analysis-by-synthesis; Generic CELP encoders and decoders; Excitation codebook search – state-save method, zero-input zerostate method; CELP based on adaptive codebook, Adaptive Codebook search; Low Delay CELP and algebraic CELP.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Mathematically model the speech signal

Analyze the quality and properties of speech signal.

Modify and enhance the speech and audio signals.

Text/Reference Books:

“Digital Speech” by A.M.Kondoz, Second Edition (Wiley Students Edition), 2004.

“Speech Coding Algorithms: Foundation and Evolution of Standardized Coders”, W.C. Chu, WileyInter science, 2003.

ECPE1_ DETECTION AND ESTIMATION THEORY

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Course Objectives:

To enable the students to acquire the fundamental concepts of Signal detection and estimation
To expose the conceptual basics of Hypotheses.

To introduce the methods of Detection and estimation of signals in white and non-white Gaussian noise.

To familiarize with the detection of random signals.

To enable the students to understand the time varying waveform detection and its estimation.

UNIT –I: Random Processes

Discrete Linear Models, Markov Sequences and Processes, Point Processes, and Gaussian Processes.

UNIT –II: Detection Theory

Basic Detection Problem, Maximum A posteriori Decision Rule, Minimum Probability of Error Classifier, Bayes Decision Rule, Multiple-Class Problem (Bayes)-minimum probability error with and without equal a priori probabilities, Neyman-Pearson Classifier, General Calculation of Probability of Error, General Gaussian Problem, Composite Hypotheses.

UNIT –III: Linear Minimum Mean-Square Error Filtering

Linear Minimum Mean Squared Error estimators, Nonlinear Minimum Mean Squared Error Estimators. Innovations, Digital Wiener Filters with tored Data, Real-time Digital Wiener Filters, Kalman Filters.

UNIT –IV: Statistics

Measurements, Nonparametric Estimators of Probability Distribution and Density Functions, Point Estimators of Parameters, Measures of the Quality of Estimators, Introduction to Interval estimates, Distribution of Estimators, Tests of Hypotheses, Simple Linear Regression, Multiple Linear Regression.

UNIT –V: Estimating the Parameters of Random Processes from Data

Tests for Stationarity and Ergodicity, Model-free Estimation, Model-based Estimation of Autocorrelation Functions, Power Spectral Density Functions.

TEXT BOOKS:

Random Signals: Detection, Estimation and Data Analysis - K. Sam Shanmugan & A.M. Breipohl, Wiley India Pvt. Ltd, 2011.

Random Processes: Filtering, Estimation and Detection - Lonnie C. Ludeman, Wiley India Pvt. Ltd., 2010.

REFERENCES:

Fundamentals of Statistical Signal Processing: Volume I Estimation Theory—Steven M. Kay, Prentice Hall, USA, 1998.

Fundamentals of Statistical Signal Processing: Volume I Detection Theory—Steven M. Kay, Prentice Hall, USA, 1998.

Introduction to Statistical Signal Processing with Applications - Srinath, Rajasekaran, Viswanathan, 2003, PHI.

ECPE1

WAVELETS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Introduction to time frequency analysis; the how, what and why about wavelets, Short-time Fourier transform, Wigner-Ville transform.; Continuous time wavelet transform, Discrete wavelet transform, tiling of the time-frequency plane and wave packet analysis, Construction of wavelets. Multiresolution analysis. Introduction to frames and biorthogonal wavelets, Multirate signal processing and filter bank theory, Application of wavelet theory to signal denoising, image and video compression, multi-tone digital communication, transient detection.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand time-frequency nature of the signals.

Apply the concept of wavelets to practical problems.

Mathematically analyze the systems or process the signals using appropriate wavelet functions.

Text/Reference Books:

Y.T. Chan, Wavelet Basics, Kluwer Publishers, Boston, 1993.

I. Daubechies, Ten Lectures on Wavelets, Society for Industrial and Applied Mathematics, Philadelphia, PA, 1992.

C. K. Chui, An Introduction to Wavelets, Academic Press Inc., New York, 1992.

Gerald Kaiser, A Friendly Guide to Wavelets, Birkhauser, New York, 1995.

P. P. Vaidyanathan, Multirate Systems and Filter Banks, Prentice Hall, New Jersey, 1993.

A.N. Akansu and R.A. Haddad, Multiresolution signal Decomposition: Transforms, Subbands and Wavelets, Academic Press, Oranld, Florida, 1992.

B.Boashash, Time-Frequency signal analysis, In S.Haykin, (editor), Advanced Spectral Analysis, pages 418--517. Prentice Hall, New Jersey, 1991.

Course Objectives:

- To introduce students to the basic concepts and classical techniques of Machine Learning.
- To study optimization algorithms used in Machine learning.
- To study various classification techniques, regression, clustering, ANN.
- To study Deep learning concepts: CNN, RNN and hyper parameter tuning, some case studies.

UNIT-1(Introduction):

Introduction to Machine Learning: Supervised learning, Unsupervised learning, Reinforcement learning. Machine Learning applied to AI examples. Structured data, Unstructured data, training data, test data, cross validation, data collection (Unbiased data), data cleaning, feature extraction, Properties of best features for classification. Linear Regression, Logistic Regression. Over fitting problem. Bias-Variance tradeoff.

UNIT-2(Classical Techniques of ML):

Error metrics, error metrics for skewed classes, Gradient Descent algorithms: Batch, mini-batch, Stochastic Gradient descent.

Classification Techniques: Bayes classifier, Naïve Bayes classifier, K-Nearest neighbor, Perceptron learning algorithm, Multi-layer Perceptron, Regularization, Support Vector Machines, Decision tree algorithm.

Clustering: K-means clustering

Dimensionality Reduction, Anomaly detection.

UNIT-3(Artificial Neural Networks):

Introduction to Neural networks, back propagation algorithm. Activation functions: Sigmoid, tanh, ReLU, SoftMax. Regularization techniques.

UNIT-4(CNN)

Deep learning, vanishing gradients problem, Hyper parameters.

CNN and applications to Computer Vision.

Hyper parameters tuning.

Case studies: LeNet-5, AlexNet, VGG-16, ResNets

Self-driving car application.

Generative Adversarial Networks (GAN)

UNIT-5(RNN)RNN and application to sequence modeling. Hyper parameters tuning.

Some Case studies.

Course Outcomes:

At the end of the course, students will learn the various algorithms related to Supervised Learning, Unsupervised learning, Deep learning concepts.

Text Books:

1. Deep Learning, Goodfellow et al, MIT Press
2. and Machine Learning, Christopher Bishop, Springer

Reference MOOCs:

1. “Machine Learning” by Prof.Andrew NG, Coursera (Stanford)
2. “Deep Learning” by Prof.Andrew NG, Coursera (Stanford)

EC4432

MULTIMEDIA COMMUNICATION

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives

- The course is designed
- To provide an introduction to the fundamental principles and techniques in Multimedia Signal coding and compression.
- To give an overview of current multimedia standards and technologies.
- To provide techniques related to computer and multimedia networks.
- To provide knowledge related to Multimedia Network Communications and Applications.

Course Outcomes

- Upon completing the course, the student will be able to:
- Understand the fundamentals behind multimedia signal processing.
- Understand the fundamentals behind multimedia compression.
- Understand the basic principles behind existing multimedia compression and communication standards.
- Understand future multimedia technologies.
- Apply the acquired knowledge to specific multimedia related problems and projects at work.
- Take advanced courses in this area.

UNIT -I

Introduction to Multimedia: Multimedia, World Wide Web, Overview of Multimedia Tools, Multimedia Authoring, Graphics/Image Data Types, and File Formats.

Color in Image and Video: Color Science — Image Formation, Camera Systems, Gamma Correction, Color Matching Functions, CIE Chromaticity Diagram, Color Monitor Specifications, Out-of-Gamut Colors, White Point Correction, XYZ to RGB Transform, Transform with Gamma Correction, L*A*B* Color Model. Color Models in Images — RGB Color Model for CRT Displays, Subtractive Color: CMV Color Model, Transformation from RGB to CMV, Under Color Removal: CMYK System, Printer Gamuts, Color Models in Video — Video Color Transforms, YUV Color Model, YIQ Color Model, Ycber Color Model.

UNIT -II

Video Concepts: Types of Video Signals, Analog Video, Digital Video.

Audio Concepts: Digitization of Sound, Quantization and Transmission of Audio.

UNIT -III

Compression Algorithms

Lossless Compression Algorithms: Run Length Coding, Variable Length Coding, Arithmetic Coding, Lossless JPEG, Image Compression.

Lossy Image Compression Algorithms: Transform Coding: KLT And DCT Coding, Wavelet Based Coding.

Image Compression Standards: JPEG and JPEG2000.

UNIT – IV

Video Compression Techniques: Introduction to Video Compression, Video Compression Based on Motion Compensation, Search for Motion Vectors, H.261-Intra-Frame and Inter-Frame Coding, Quantization, Encoder and Decoder, Overview of MPEG 1 and MPEG2.

UNIT -V

Audio Compression Techniques: ADPCM in Speech Coding, G.726 ADPCM, Vocoder Phase Insensitivity, Channel Vocoder, Formant Vocoder, Linear Predictive Coding, CELP, Hybrid Excitation Vocoder, MPEG Audio — MPEG Layers, MPEG Audio Strategy, MPEG Audio Compression Algorithms, MPEG-2 AAC, MPEG-4 Audio.

TEXT BOOKS

Fundamentals of Multimedia — Ze- Nian Li, Mark S. Drew, PHI, 2010.

Multimedia Signals & Systems — Mrinal Kr. Mandal Springer International Edition 1st Edition, 2009

REFERENCE BOOKS

Multimedia Communication Systems — Techniques, Standards & Networks KR. Rao, Zorans. Bojkoric, Dragorad A. Mijovanj 1st Edition, 2002.

Fundamentals of Multimedia Ze- Man Li, Mark S. Drew, Pearson Education (LPE), 1st Edition, 2009.

Multimedia Systems John F. Koegel Buford Pearson Education (LPE), 1st Edition, 2003.

Digital Video Processing — A. Murat Tekalp, PHI, 1996.

Video Processing and Communications — Yaowang, Jorn Ostermann, Ya-Qin Zhang, Pearson, 2002

EC4433 DIGITAL SIGNAL PROCESSORS AND ARCHITECTURES

Externals: 60 Marks

(L-T)-P-C

Internals: 40 Marks

4-0-3

Course Objectives:

- To introduce architectural features of programmable DSP Processors of TI and Analog Devices.
- To recall digital transform techniques.
- To give practical examples of DSP Processor architectures for better understanding.
- To develop the programming knowledge using Instruction set of DSP Processors.
- To understand interfacing techniques to memory and I/O devices.

Course Outcomes:

- Student will be able to:
- To distinguish between the architectural features of general purpose processors and DSP processors
- Understand the architectures of TMS 320C54XX and ADSP2100 DSP devices
- Able to write assembly language programs using instruction set of TMS320C54XX
- Can interface various devices to DSP Processors

UNIT-I: Introduction to Digital Signal Processing:

Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences. Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.

UNIT-II: Architectures for Programmable DSP Devices:

Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation UNIT, Programmability and Program Execution, Speed Issues, Features for External interfacing.

UNIT-III: Programmable Digital Signal Processors:

Commercial Digital signal-processing Devices, Data Addressing modes of TMS320C54XX DSPs, Data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX Processors, Program Control, TMS320C54XX instructions and Programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, Pipeline operation of TMS320C54XX Processors.

UNIT-IV: Analog Devices Family of DSP Devices:

Analog Devices Family of DSP Devices – ALU and MAC block diagram, ShifterInstruction, Base Architecture of ADSP 2100, ADSP-2181 high performance Processor. introduction to Blackfin Processor - The Blackfin Processor, Introduction to Micro Signal Architecture, Overview of Hardware Processing Units and Register files, Address Arithmetic Unit, Control Unit, Bus Architecture and Memory, Basic Peripherals.

UNIT-V: Interfacing Memory and I/O Peripherals to Programmable DSP Devices:

Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).

TEXT BOOKS:

Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.

A Practical Approach to Digital Signal Processing - K Padmanabhan, R. Vijayarajeswaran, Ananthi. S, New Age International, 2006/2009

Embedded Signal Processing with the Micro Signal Architecture Publisher: Woon-Seng Gan, Sen M. Kuo, Wiley-IEEE Press, 2007

REFERENCES:

Digital Signal Processors, Architecture, Programming and Applications – B. Venkataramani and M. Bhaskar, 2002, TMH.

Digital Signal Processing – Jonatham Stein, 2005, John Wiley.

DSP Processor Fundamentals, Architectures & Features – Lapsley et al. 2000, S. Chand & Co.

Digital Signal Processing Applications Using the ADSP-2100 Family by The Applications Engineering Staff of Analog Devices, DSP Division, Edited by Amy Mar, PHI

The Scientist and Engineer's Guide to Digital Signal Processing by Steven W. Smith, Ph.D., California Technical Publishing, ISBN 0-9660176-3-3, 1997 6. Embedded Media Processing by David J. Katz and Rick Gentile of Analog Devices, Newnes, ISBN 0750679123, 2005.

EC4502

DATA MINING

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Objectives:

1. To impart an introduction to Data Mining.
2. To develop basic knowledge of how data is transformed to Data Warehouses.

UNIT - I:

Introduction to Data Mining: What is data mining? Related technologies - Machine Learning, DBMS, OLAP and Statistics. Data Mining Goals, Stages of the Data Mining Process, Data Mining Techniques. Knowledge Representation Methods, Applications. Example: weather data
Data Warehouse and OLAP: Data Warehouse and DBMS, Multidimensional data model, OLAP operations, Example: loan dataset.

UNIT - II:

Data preprocessing: Data cleaning, Data transformation, Data reduction, Discretization and generating concept hierarchies.

Data mining knowledge representation: Task relevant data, Background knowledge, Interestingness measures, Representing input data and output knowledge

Attribute-oriented analysis: Attribute generalization, Attribute relevance, Class comparison, Statistical measures

UNIT - III:

Data mining algorithms: Association rules - Motivation and terminology, Example: mining weather data, Basic idea: item sets, Generating item sets and rules efficiently, Correlation analysis.

Data mining algorithms: Classification - Basic learning/mining tasks, Bayesian, Naïve Bayes, Decision trees, Covering rules, Random Forest.

UNIT - IV:

Data mining algorithms: Prediction - The prediction task, Statistical (Bayesian) classification, Bayesian networks, Instance-based methods (nearest neighbor), Linear models

Evaluating what's been learned: Basic issues, Training and testing, Estimating classifier accuracy (holdout, cross-validation, leave-one-out), Combining multiple models (bagging, boosting, stacking), Minimum Description Length Principle (MLD)

UNIT - V:

Clustering: Basic issues in clustering, First conceptual clustering system: Cluster/2 , Partitioning methods: k-means, expectation maximization (EM), Hierarchical methods: distance-based agglomerative and divisible clustering
Basics of ANN, Perceptron, MLP

Suggested References:

1. I. H. Witten and E. Frank. Data Mining: Practical Machine Learning Tools and Techniques. Morgan Kaufmann. 2000.
2. J. Han and M. Kamber. Data Mining: Concepts and Techniques, 2nd Ed. Morgan Kaufman. 2006.
3. M. H. Dunham. Data Mining: Introductory and Advanced Topics. Pearson Education. 2001.
4. D. Hand, H. Mannila and P. Smyth. Principles of Data Mining. Prentice-Hall. 2001.
5. Pang-Ning Tan, Michael Steinbach, Vipin Kumar. Introduction to Data Mining. Addison- Wesley Longman Publishing Co.

ECPE1_
Externals: 60Marks

WIRELESS COMMUNICATIONS

(L-T)-P-C

Internals: 40Marks

4-0-3

Course objectives:

- To provide the students with the fundamental theoretical and practical concepts of wireless communication
- To equip the students with various kinds of wireless networks and its operations.
- To prepare the students to understand the concept of frequency reuse and be able to apply it in the design of mobile cellular system
- To prepare the students to understand various multiple access techniques that are used in wireless communications
- To train the students to understand the basic concepts of OFDM and MIMO.
-

Course outcomes:

- On successful completion of this course, the students should be able to:
- Select appropriate value of C/I to design the Antenna system
- Analyze the characteristics of different setups for the wireless communication using new models for the coverage improvement.
- Select different technologies to solve numerical problems using multiple access technique.
- Demonstrate the technical aspects of diversity for wireless communication.
- Understand the basic PAPR problem in OFDM and understanding basic concepts in SIMO.

UNIT - I

Basic Cellular system and its operation: frequency reuse, channel assignment strategies, Handoff process, factors influencing handoffs, handoffs in different Generations, Interference and system capacity, Crosstalk, Enhancing capacity and cell coverage, Trunked radio system, grade of service as per Erlang's B system.

UNIT – II

Propagation models: Free space propagation model, three basic propagation mechanisms, practical link budget design using path loss models, outdoor propagation models: Ground reflection (2-ray) Model, log normal shadowing, Okumura model, Hata model and indoor propagation model. Rayleigh fading, BER in Rayleigh fading channel with BPSK

transmission, doppler spread, rms delay spread, Coherence time and Coherence bandwidth of wireless channel.

UNIT – III

Basic equalizers: LS, MMSE estimators for channel, Multiple Access Techniques: FDMA, TDMA, CDMA, RAKE receiver, SDMA.

UNIT – IV

SIMO: Diversity, SIMO (multiple receive antennas) model, maximal ratio combining (MRC) receiver, BER with MRC (high SNR approximation), diversity order.

UNIT – V

OFDM: Multicarrier basics, OFDM transmitter and receiver blocks, cyclic prefix in OFDM, PAPR problem in OFDM, SC-FDMA transmitter and receiver blocks.

Text Books:

Theodore.S. Rappaport, “Wireless Communications: Principles and Practice”, 2/e, Pearson Education, 2010

Aditya K Jagannatham, “Principles of Modern Wireless Communication Systems”, Mc-Graw Hill

Suggested Reading:

David Tse and Pramod Viswanath, “Fundamentals of Wireless Communications”, Cambridge University Press.

EC4403

SATELLITE COMMUNICATIONS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To prepare the student to excel in basic knowledge of satellite communication principles.
- To provide students with solid foundation in orbital mechanics and launches for the satellite communication
- To train the students with the basic knowledge of link design of satellite with a design examples.
- To provide the better understanding of multiple access systems and earth station technology.
- To prepare the students with knowledge in satellite navigation and GPS and satellite packet communication

Course Outcomes:

- At the end of this course Students will be able to
- Explain, basic concepts and frequency allocations for satellite communications.
- Describe the orbital mechanics, launch vehicles and launchers.
- Design satellite links for specified C/N.
- Visualize satellites sub systems like telemetry, tracking, command and monitoring power systems etc.,
- Explain the different multiple access systems and their need in satellite communications and GPS Receivers,.

UNIT-I: Communication Satellite Orbit and Description

A Brief history of satellite Communication, satellite Frequency Bands, Satellite Systems, Applications, Orbital Period and Velocity, effects of orbital Inclination, Azimuth and Elevation, Coverage angle and slant Range, Eclipse, Orbital Perturbations, Placement of a Satellite in a Geo-Stationary orbit.

UNIT-II: Satellite Sub-Systems

Attitude and Orbit Control system, TT&C subsystem, Attitude Control subsystem, Power systems, Communication subsystems, Satellite Antenna Equipment. Satellite Link: Basic Transmission Theory, System Noise Temperature and G/T ratio, Basic Link Analysis, Interference Analysis, Design of satellite Links for a specified C/N, (With and without frequency Re-use), Link Budget

UNIT-III: Propagation Effects

Introduction, Atmospheric Absorption, Cloud Attenuation, Tropospheric and Ionospheric Scintillation and Low angle fading, Rain induced attenuation, rain induced cross polarization interference. Multiple Access : Frequency Division Multiple Access (FDMA) -Intermodulation, Calculation of C/N, Time Division Multiple Access (TDMA) -Frame Structure, Burst Structure, Satellite switched TDMA, On-board Processing, Demand Assignment Multiple Access (DAMA) –Types of Demand Assignment, Characteristics, CDMA Spread Spectrum Transmission and Reception.

UNIT-IV: Earth Station Technology

Transmitters, Receivers, Antennas, Tracking Systems, Terrestrial Interface, Power Test Methods, Lower Orbit Considerations. Satellite Navigation and Global Positioning Systems: Radio and Satellite Navigation, GPS Position Location Principles, GPS Receivers, GPS C/A Code Accuracy, Differential GPS.

UNIT-V: Satellite Packet Communications

Message Transmission by FDMA: M/G/1 Queue, Message Transmission by TDMA, PURE ALOHA-Satellite Packet Switching, Slotted Aloha, Packet Reservation, Tree Algorithm.

TEXT BOOKS:

1. Satellite Communications –Timothy Pratt, Charles Bostian, Jeremy Allnutt, 2nd Edition, 2003, John Wiley & Sons.
2. Satellite Communications Engineering –Wilbur, L. Pritchard, Robert A. Nelson and Henri G. Suyderhoud, 2nd Ed., Pearson Publications.
3. Digital Satellite Communications-Tri. T. Ha, 2nd Edition, 1990, McGraw Hill.

REFERENCES:

1. Satellite Communications-Dennis Roddy, 2nd Edition, 1996, McGraw Hill.
2. Satellite Communications: Design Principles –M. Richcharia, 2nd Ed., BSP, 2003.

3. Digital Satellite Communications –Tri. T. Ha, 2ndEd., MGH, 1990.
4. Fundamentals of Satellite Communications –K. N. Raja Rao, PHI, 2004.

ECPE1_

WIRELESS SENSOR NETWORKS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Introduction to Sensor Networks, unique constraints and challenges, Advantage of Sensor Networks, Applications of Sensor Networks, Types of wireless sensor networks

Unit II:

Mobile Ad-hoc Networks (MANETs) and Wireless Sensor Networks, Enabling technologies for Wireless Sensor Networks. Issues and challenges in wireless sensor networks

Unit III:

Routing protocols, MAC protocols: Classification of MAC Protocols, S-MAC Protocol, B-MAC protocol, IEEE 802.15.4 standard and ZigBee,

Dissemination protocol for large sensor network. Data dissemination, data gathering, and data fusion; Quality of a sensor network; Real-time traffic support and security protocols.

Unit IV:

Design Principles for WSNs, Gateway Concepts Need for gateway, WSN to Internet Communication, and Internet to WSN Communication.

Unit V:

Single-node architecture, Hardware components & design constraints, Operating systems and execution environments, introduction to TinyOS and nesC.

Course Outcomes:

At the end of the course the students will be able to

Design wireless sensor networks for a given application

Understand emerging research areas in the field of sensor networks

Understand MAC protocols used for different communication standards used in WSN

Explore new protocols for WSN

Text/Reference Books:

WaltenegusDargie , Christian Poellabauer, “Fundamentals Of Wireless Sensor Networks Theory And Practice”, By John Wiley & Sons Publications ,2011

SabrieSoloman, “Sensors Handbook" by McGraw Hill publication. 2009

Feng Zhao, Leonidas Guibas, “Wireless Sensor Networks”, Elsevier Publications,2004

Kazem Sohrby, Daniel Minoli, “Wireless Sensor Networks”: Technology, Protocols and Applications, Wiley-Inter science

Philip Levis, And David Gay "TinyOS Programming" by Cambridge University Press 2009

ECPE1_

LARGE MIMO SYSTEMS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Course Objectives:

1. To study the various concepts of MIMO communication.
2. To study Large MIMO systems encoding and decoding techniques.
3. To study MIMO channel models and MIMO channel estimation.

Unit-I:

Introduction: Multi-antenna wireless channels, MIMO system model, MIMO communication with CSIR-only, Slow fading channels, Fast fading channels, MIMO communication with CSIT and CSIR, Increasing spectral efficiency: quadrature amplitude modulation (QAM) vs MIMO, Multiuser MIMO communication.

Large MIMO systems: Opportunities in large MIMO systems, Channel hardening in large dimensions, Technological challenges and solution approaches.

Unit-II:

MIMO encoding: Spatial multiplexing, Space-time coding, Space-time block codes, High-rate NO-STBCs, NO-STBCs from CDAs, Spatial modulation (SM).

MIMO detection: System model, Optimum detection, Linear detection, Interference cancellation, LR-aided linear detection, Sphere decoding

Unit-III:

Detection based on local search, Detection based on probabilistic data association (PDA), Detection/decoding based on message passing on graphical models, Detection based on MCMC techniques.

Unit-IV:

Channel estimation in large MIMO systems, MIMO capacity with imperfect CSI, Point-to-point MIMO training, Multi-user MIMO training, Large multi-user MIMO systems, Iterative channel estimation/detection in frequency-flat fading, Iterative channel

estimation/equalization in ISI channels, Equalization using initial channel estimates, Equalization using the MGS-MR algorithm.

Precoding in point-to-point MIMO, Precoding in a multiuser MIMO downlink, Precoding in large multiuser MISO systems, Multicell precoding

Unit-V:

MIMO channel models: Analytical channel models, Spatial correlation based models, Propagation based models, Effect of spatial correlation on large MIMO performance: an illustration, Pinhole effect, Effect of spatial correlation on LAS detector performance, Standardized channel models, Models in IEEE 802.11 WiFi, Models in 3GPP/LTE, Large MIMO channel measurement campaigns, Compact antenna arrays, PIFAs as elements in compact arrays, MIMO cubes

Text Books:

1. A.Chockalingam and B.SundarRajan, “Large MIMO Systems”, Cambridge University Press, 2014.

Reference Books:

1. D. Tse and P. Viswanath, Fundamentals of Wireless Communication. Cambridge, UK: Cambridge University, 2005.
2. H. Jafarkhani, Space-Time Coding: Theory and Practice. Cambridge, UK: Cambridge University Press, 2005.

ECPE1_ Applied Optimization for Wireless, Signal Processing, Machine Learning

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Course Objectives:

1. To study the concepts of local minima and global minima, and study of some Numerical optimization techniques.
2. To study the Convex optimization concepts.
3. To study the formulation and solving various optimization problems related to Wireless communications, signal processing and Machine learning.

Unit-I:

Introduction to properties of Vectors, Norms, Positive Semi-Definite matrices, Gaussian Random

Vectors, Local minimum, global minimum, Optimization Problem formulation examples: Bus terminus problem, Transportation problem, regression problem. Steepest Descent algorithm.

Unit-II:

Introduction to Convex Optimization – Convex sets, Hyperplanes/ Half-spaces etc. Application: Power constraints in Wireless Systems, Convex/ Concave Functions, Examples, Conditions for Convexity. Application: Beamforming in Wireless Systems, Multi-User Wireless, Cognitive Radio Systems, Convex Optimization problems, Linear Program, Application: Power allocation in Multi-cell cooperative OFDM

Unit-III:

SOCP Problems, Application: Channel shortening for Wireless Equalization, Robust Beamforming in Wireless Systems, Duality Principle and KKT Framework for Optimization. Application: Water-filling power allocation, Optimization for MIMO Systems, OFDM Systems and MIMO-OFDM systems

Unit-IV:

Optimization for signal estimation, LS, WLS, Regularization. Application: Wireless channel estimation, Image Reconstruction-Deblurring, Application: Convex optimization for Machine Learning, Principal Component Analysis (PCA), Support Vector Machines

Unit-V:

Application: Cooperative Communication, Optimal Power Allocation for cooperative Communication, Application: Compressive Sensing, Sparse Signal Processing, OMP (Orthogonal Matching Pursuit), LASSO (Least Absolute Shrinkage and Selection Operator) for signal estimation

Text Books:

1. Stephen Boyd, Convex Optimization, Cambridge University Press.

Reference Books:

1. Convex Optimization Algorithms, by Dimitri P. Bertsekas, 2015, Athena scientific

Reference NPTEL MOOCs:

Applied Optimization for Wireless, Machine Learning, Big-Data by Prof. Aditya K. Jagannatham

ECPE1_ Principles of Signal Estimation for MIMO/OFDM Wireless Communications

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Course Objectives:

1. To study the ML estimation concept.
2. To study the MMSE estimation concept.
3. To study the estimation problems related to wireless sensor network, wireless fading channel, channel equalizer, OFDM.

Unit-I:

Basics of Estimation, Maximum Likelihood (ML), Application: Wireless Sensor Network, Reliability of Estimation, Wireless Fading Channel Estimation, Cramer-Rao Bound for Estimation

Unit-II:

Vector Parameter Estimation, Properties of Estimate; Applications: Multi-antenna Wireless Channel Estimation, MIMO Wireless Channel Estimation, Error Covariance of Estimation, Equalization for Frequency Selective Channels

Unit-III:

OFDM Estimation, Sequential Estimation, Minimum Mean-Squared Error (MMSE) Estimate, Gaussian Parameter, Application: Wireless Sensor Network, Wireless Fading Channel Estimation

Unit-IV:

MMSE Estimation for Multi-Antenna Channel, MMSE for MIMO Channel Estimation, Properties of Estimate

Unit-V:

MMSE for Equalization of Wireless Channel, MMSE for OFDM Channel Estimation

Reference NPTEL MOOCs:

1. Principles of Signal Estimation for MIMO/OFDM Wireless Communications by Prof. Aditya K. Jagannatham

EC4402

ERROR CORRECTING CODES

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To study the importance of channel coding techniques in digital communications.
- To learn the mathematical structure of various codes
- To learn the algorithms for various codes
- To study the various application of codes.
- To acquire the knowledge of measurement of information
-

Course Outcomes:

- At the end of this course Students will be able to
- Understand principles of channel Coding techniques.
- Analyze the performance of different codes.
- Design various codes like block codes, cyclic codes, convolution codes, turbo codes etc.
- Generate different codes.
- Estimate the information content and errors

UNIT-I

Coding for Reliable Digital Transmission and Storage: Introduction, Types of codes, Types of errors, Channels models, Modulation and coding, channel coding Theorem, Channel coding gain.

UNIT-II

Linear Block codes: Introduction, encoding, syndrome decoding, error-detecting and correcting

capabilities, Maximum likelihood decoding. Cyclic codes: Description, encoding and syndrome decoding.

UNIT- III

Galois Fields: Groups, Fields, Binary arithmetic, Construction of Galois Fields $GF(2^m)$, Basic properties of Galois Fields. RS codes: Introduction, encoding and decoding (Berlekamp-Massey algorithm).

UNIT- IV

Convolution codes: Introduction, Encoding, State diagram, Trellis diagram, Decoding -Maximum-Likelihood decoding, soft decision and hard decision decoding, Viterbi algorithm.

UNIT- V

Turbo codes: Concatenation, Types of Concatenation, interleaving, types of interleavers, Turbo codes: Introduction, encoding and decoding (BCJR Algorithm).

Text books:

Shulin and Daniel J. Costello, Jr. "Error Control Coding," 2/e, Pearson, 2011.

L.H.Charles LEE "Error control block codes for Communication Engineers", Artech, 2000.

Suggested readings:

Simon Haykin, "Communication Systems", 4/e, Wiley, 2000.

K Sam Shanmugum, "Digital and Analog Communication Systems", Wiley, 2005.

ECPE1_

INFORMATION THEORY AND CODING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Basics of information theory, entropy for discrete ensembles; Shannon's noiseless coding theorem; Encoding of discrete sources.

Markov sources; Shannon's noisy coding theorem and converse for discrete channels; Calculation of channel capacity and bounds for discrete channels; Application to continuous channels.

Techniques of coding and decoding; Huffman codes and uniquely detectable codes; Cyclic codes, convolutional arithmetic codes.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand the concept of information and entropy

Understand Shannon's theorem for coding

Calculation of channel capacity

Apply coding techniques

Text/Reference Books:

N. Abramson, Information and Coding, McGraw Hill, 1963.

M. Mansurpur, Introduction to Information Theory, McGraw Hill, 1987.

R.B. Ash, Information Theory, Prentice Hall, 1970.

Shu Lin and D.J. Costello Jr., Error Control Coding, Prentice Hall, 1983.

EC4405

RADAR SYSTEMS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To learn working principle of Radar Operating frequencies and derive Radar Range Equation,
- To understand the basic concepts of different types of Radars for surveillance & Tracking.
- To learn functioning of MTI radar and its performance limitations.
- To get acquainted with the working principles of CW radar, FM-CW radar.
- To understand concept of a Matched Filter in Radar Receivers gain knowledge of different receiver blocks and understand receiver functioning

Course Outcomes:

- At the end of this course Students will learn:
- Distinguish between the functioning of CW FM-CW and MTI radars,
- Apply Doppler principle to radars and hence detect moving targets.
- Distinguish between Sequential Lobing, Conical Scan, Monopulse type Of Tracking Radars, specify their requirements and compare their characteristic features.
- Derive the matched filter response characteristics for radar applications and account for correlation receivers; to distinguish between different radar displays and duplexers.
- Account for the electronic scanning principle and implement the same through phased array antennas, knowing their requirements and utilities.

UNIT-I: Basics of Radar

Introduction, Maximum Unambiguous Range, Radar Waveforms, Simple form of Radar Equation, Radar Block Diagram and Operation, Radar Frequencies and Applications, Prediction of Range Performance, Minimum Detectable Signal, Receiver Noise, Modified Radar Range Equation, Related Problems. Radar Equation: SNR, Envelope Detector-False Alarm Time and Probability, Integration of Radar Pulses, Radar Cross Section of Targets (simple targets - sphere, cone-sphere), Transmitter Power, PRF and Range Ambiguities, System Losses (qualitative treatment), Related Problems.

UNIT-II: CW and Frequency Modulated Radar Doppler Effect

CW Radar – Block Diagram, Isolation between Transmitter and Receiver, Non-zero IF Receiver, Receiver Bandwidth Requirements, Applications of CW radar, Related Problems. FM-CW Radar: FM-CW Radar, Range and Doppler Measurement, Block Diagram and Characteristics (Approaching/ Receding Targets), FM-CW altimeter, Multiple Frequency CW Radar.

UNIT-III: MTI and Pulse Doppler Radar

Introduction, Principle, MTI Radar with - Power Amplifier Transmitter and Power Oscillator Transmitter, Delay Line Cancellers – Filter Characteristics, Blind Speeds, Double Cancellation, Staggered PRFs. Range Gated Doppler Filters. MTI Radar Parameters, Limitations to MTI Performance, MTI versus Pulse Doppler Radar. Tracking Radar: Tracking with Radar, Sequential Lobing, Conical Scan, Monopulse Tracking Radar –Amplitude Comparison Monopulse (one- and two- coordinates), Angular Accuracy, Tracking in Range, Acquisition and Scanning Patterns, Comparison of Trackers.

UNIT-IV: Detection of Radar Signals in Noise

Introduction, Matched Filter Receiver – Response Characteristics and Derivation, Correlation Function and Cross-correlation Receiver, Efficiency of Non- matched Filters, Matched Filter with Non-white Noise.

UNIT-V: Radar Receivers

Noise Figure and Noise Temperature, Displays – types, Introduction to Phased Array Antennas –Basic Concepts, Radiation Pattern, Beam Steering and Beam Width changes, Applications, Advantages and Limitations. Electronic Warfare : Introduction to ESM, ECM and ECCM systems.

TEXT BOOKS:

1. Introduction to Radar Systems – Merrill I. Skolnik, TMH Special Indian Edition, 2 nd Edition, Tata McGraw-Hill, 2007.

REFERENCES:

1. Introduction to Radar Systems – Merrill I. Skolnik, 3 rd Edition Tata McGraw-Hill, 2001.
2. Radar: Principles, Technology, Applications-Byron Edde, Pearson Education, 2004.
3. Principles of Modern Radar: Basic Principles-Mark A. Richards, James A. Scheer, William A. Holm, Yesdee, 2013.

4. 'Radar Hand Book ' Ed. By M.I Skolnik, 2 nd Edition, Tata McGraw Hill.
5. 'Understanding Radar Systems' by Simon Kinsley and Shaun Quegan, Scitech Publishing, McGraw-Hill.

EC4425 ADHOC WIRELESS SENSOR NETWORKS

Externals: 60Marks **(L-T)-P-C**

Internals: 40Marks **4-0-3**

Course Objectives:

- To introduction of fundamentals of Wireless LANS and PANS and its design issues.
- To understand the MAC protocols for Ad Hoc Wireless Networks and its designingissues.
- To introduction of different kinds of Routing algorithms for effective design of Ad HocWireless Networks.
- To introduction and designing issues in Transport Layer Protocol for Ad Hoc WirelessNetworks.
- Introduction of Wireless Sensor Networks and its Architecture.

Course Outcomes:

- At the end of this course Students will learn
- Students will be good at fundamentals of Wireless LANS and PANS and its designissues.
- Students will know the MAC protocols for Ad Hoc Wireless Networks and its designing issues.
- Student knows the different kinds of Routing algorithms for effective design of Ad Hoc Wireless Networks.
- Student will be ability to overcome the issues in Transport Layer Protocol for Ad Hoc Wireless Networks.
- Finally, student will be good at design and architecture of Wireless Sensor Networks.

UNIT -I:

Wireless LANS and PANS: Introduction, Fundamentals of WLANS, IEEE 802.11 Standards,HIPERLAN Standard, Bluetooth, Home RF.AD HOC Wireless Networks: Introduction, Issues in Ad Hoc Wireless Networks.

UNIT -II:

MAC Protocols: Introduction, Issues in Designing a MAC protocol for Ad Hoc Wireless Networks, Design goals of a MAC Protocol for Ad Hoc Wireless Networks, Classifications of MAC Protocols, Contention - Based Protocols, Contention - Based Protocols with reservation

Mechanisms, Contention – Based MAC Protocols with Scheduling Mechanisms, MAC Protocols that use Directional Antennas, Other MAC Protocols.

UNIT -III:

Routing Protocols: Introduction, Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks, Classification of Routing Protocols, Table –Driven Routing Protocols, On – Demand Routing Protocols, Hybrid Routing Protocols, Routing Protocols with Efficient Flooding Mechanisms, Hierarchical Routing Protocols, Power – Aware Routing Protocols.

UNIT –IV:

Transport Layer Protocols: Introduction, Issues in Designing a Transport Layer Protocol for Ad Hoc Wireless Networks, Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks, Classification of Transport Layer Solutions, TCP Over Ad Hoc Wireless Networks, Other Transport Layer Protocol for Ad Hoc Wireless Networks.

UNIT –V:

Wireless Sensor Networks: Introduction, Sensor Network Architecture, Data Dissemination, Data Gathering, MAC Protocols for Sensor Networks, Location Discovery, Quality of a Sensor Network, Evolving Standards, Other Issues.

TEXT BOOKS:

Ad Hoc Wireless Networks: Architectures and Protocols - C. Siva Ram Murthy and B.S. Manoj, 2004, PHI.

Wireless Ad- hoc and Sensor Networks: Protocols, Performance and Control – Jagannathan Sarangapani, CRC Press.

REFERENCES:

Ad- Hoc Mobile Wireless Networks: Protocols & Systems, C.K. Toh , 1st Ed. Pearson Education.

Wireless Sensor Networks - C. S. Raghavendra, Krishna M. Sivalingam, 2004, Springer.

IoT and Applications

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Course Objective.

1. The course aims to introduce students to the concepts underlying the Internet of Things (IoT) through a series of lectures on the various topics that are important to understand the state-of-the-art as well as the trends for IoT.
2. The students will be introduced to the history and evolution of IoT, as well as case studies from various industry domains. In addition, students will be able to work in teams to design, build, evaluate and test an innovative IoT system for a specific industry domain, such as sports and agriculture.
3. Students will also be required to present their innovations to their peers in class as well as to the public (at the end of the course), and will also be required to document their findings in the form of a conference-style research paper.
4. Students will also be exposed to real-world sports technologies, to witness these technologies in action behind-the-scenes, and to interact with experts from the industry.**
5. The lectures will be focused around industry domains (the verticals where IoT is applicable, or has been applied), platforms (the hardware or software platforms that are applicable for IoT), protocols (the communication protocols that are applicable to IoT) and services (the types of services that can layer over IoT).

UNIT-I: Introduction & Concepts

Introduction & Concepts: Introduction to Internet of Things- Definitions & Characteristics of

IoT, Physical Design of IOT-Things in IoT, IoT Protocols, Logical Design of IOT-IoT Functional Blocks, IoT Communication Models, IoT Communication APIs, IOT Enabling Technologies- Wireless Sensor Networks, Cloud Computing, Big Data Analytics, Communication Protocols, Embedded Systems, IOT Levels & Deployment Templates.

UNIT-II: Domain Specific IoTs and M2M

IoT applications for Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, health and Lifestyle

IoT and M2M – Introduction to M2M, Similarities and Differences between IoT and M2M.

UNIT-III: : IoT Platforms Design Methodology

Introduction, IoT Design Methodology Steps-Purpose and Requirements Specification, Process

Specification, Domain Model Specification, Information Model Specification, Service

Specifications, IoT Level Specification, Functional View Specification, Operational View

Specification, Device and Component Integration, Application Development, Case Study on IoT System for Weather Monitoring.

UNIT-IV: Python Packages for IoT

Motivation for using Python for designing IoT systems, Language features of Python, Data

types- Numbers, Strings, Lists, Tuples, Dictionaries, Type Conversions, Data Structures: Control

of flow-if, for, while, range, break/continue, pass, functions, modules, packaging, file handling,

data/time operations, classes, Exception handling, Python packages of Interest for IoT - JSON,

XML, HTTPLib, URLLib, SMTPLib

UNIT-V: IoT Physical Devices and End Points

Basic building blocks of an IoT device, Raspberry Pi-About the Raspberry Pi board, Raspberry Pi interfaces-Serial, SPI, I2C, Interfacing an LED and switch with RPi and controlling.

Other IoT Devices- pcDuino, BeagleBone Black, Cubieboard IoT

Course Outcome: The Student is able to

1. Demonstrate the ability to capture the trends of IoT in the Market.
2. Demonstrate the ability to find the various domain of IoT applications.
3. Demonstrate the ability to systemically design with the learned steps.
4. Demonstrate the ability to use the various IoT packages available in Python.
5. The Student is expected to design and develop an IoT real-world application in a specific domain arming with knowledge of Python and choosing hardware for specific application.

Textbook:

1. “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014. by Vijay Madiseti and ArshdeepBahga.
2. “Introduction to IoT “ 1st Edition, 2015, Sudip Mishra and Arijit Roy

Referencens:

Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013

Various research papers and articles recommended during course work.

EC4413

EMBEDDED SYSTEMS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- Understand the basics of an embedded system.
- Program an embedded system.
- To learn the design process of embedded system applications.
- To understand the RTOS and inter-process communication.
- To understand different communication interfaces.

Course Outcomes:

- At the end of this course Students will learn:
- Understand and design the embedded systems
- Learn the basics of OS and RTOS.
- Understand types of memory and interfacing to external world.
- Understand embedded firmware design approaches.
- Understand the interfacing of communication devices.

UNIT-I: INTRODUCTION TO EMBEDDED SYSTEMS

Complex systems and microprocessors-embedding computers, characteristics of embedded computing applications, challenges in embedded computing system design, performance in embedded computing; The embedded system design process-requirements, specification, architecture design, designing hardware and software, components, system integration, design example.

UNIT-II: TYPICAL EMBEDDED SYSTEM

Core of the embedded system-general purpose and domain specific processors, ASICs, PLDs, COTs; Memory-ROM, RAM, memory according to the type of interface,

memory shadowing, memory selection for embedded systems; Sensors, actuators and other components-sensors, actuators, seven segment LED, relay, piezo buzzer, push button switch, reset circuit, brownout protection circuit, oscillator circuit real time clock, watch dog timer.

UNIT-III: EMBEDDED FIRMWARE DESIGN AND DEVELOPMENT

Embedded firmware design approaches-super loop based approach, operating system based approach; Embedded firmware development languages-assembly language based development, high level language based development; Programming in embedded C.

UNIT-IV: RTOS BASED EMBEDDED SYSTEM DESIGN

Operating system basics, types of operating systems, tasks, process and threads, multiprocessing and multitasking, task scheduling: non-preemptive and pre-emptive scheduling; task communication-shared memory, message passing.

UNIT-V: COMMUNICATION INTERFACE

Onboard communication interfaces-I2C, SPI, UART, 1 wire interface, parallel interface; External communication interfaces-RS232 and RS485,USB, infrared, Bluetooth, wi-Fi, zigbee, GPRS; Automotive networks and sensor networks.

TEXT BOOKS:

Computers as Components –Wayne Wolf, Morgan Kaufmann (second edition).

Introduction to Embedded Systems - shibu k v, Mc Graw Hill Education.

REFERENCES:

Embedded System Design -frank vahid, tony grivargis, john Wiley.

Embedded Systems- An integrated approach - Lyla b das, Pearson education 2012.

Embedded Systems – Raj kamal, TMH. 4. An embedded Software Primer, David e Simon, Pearson education

ECPE1_

NANO ELECTRONICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Introduction to nanotechnology, meso structures, Basics of Quantum Mechanics: Schrodinger equation, Density of States. Particle in a box Concepts, Degeneracy. Band Theory of Solids. Kronig-Penny Model. Brillouin Zones.

Shrink-down approaches: Introduction, CMOS Scaling, The nanoscale MOSFET, Finfets,

Vertical MOSFETs, limits to scaling, system integration limits (interconnect issues etc.),

Resonant Tunneling Diode, Coulomb dots, Quantum blockade, Single electron transistors, Carbon nanotube electronics, Bandstructure and transport, devices, applications, 2D semiconductors and electronic devices, Graphene, atomistic simulation

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand various aspects of nano-technology and the processes involved in making nano components and material.

Leverage advantages of the nano-materials and appropriate use in solving practical problems.

Understand various aspects of nano-technology and the processes involved in making nano components and material.

Leverage advantages of the nano-materials and appropriate use in solving practical problems.

Text/ Reference Books:

G.W. Hanson, Fundamentals of Nanoelectronics, Pearson, 2009.

W. Ranier, Nanoelectronics and Information Technology (Advanced Electronic Material and Novel Devices), Wiley-VCH, 2003.

K.E. Drexler, Nanosystems, Wiley, 1992.

J.H. Davies, The Physics of Low-Dimensional Semiconductors, Cambridge University Press, 1998.

C.P. Poole, F. J. Owens, Introduction to Nanotechnology, Wiley, 2003

ECPE1_

INTRODUCTION TO MEMS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Introduction and Historical Background, Scaling Effects. Micro/Nano Sensors, Actuators and Systems overview: Case studies. Review of Basic MEMS fabrication modules: Oxidation, Deposition Techniques, Lithography (LIGA), and Etching. Micromachining: Surface Micromachining, sacrificial layer processes, Stiction; Bulk

Micromachining, Isotropic Etching and Anisotropic Etching, Wafer Bonding. Mechanics of solids in MEMS/NEMS: Stresses, Strain, Hooke's law, Poisson effect, Linear

Thermal Expansion, Bending; Energy methods, Overview of Finite Element Method, Modeling of Coupled Electromechanical Systems.

Course Outcomes:

At the end of the course the students will be able to

Appreciate the underlying working principles of MEMS and NEMS devices.

Design and model MEM devices.

Text/Reference Book:

G. K. Ananthasuresh, K. J. Vinoy, S. Gopalkrishnan K. N. Bhat, V. K. Aatre, Micro and Smart Systems, Wiley India, 2012.

S. E. Lyshevski, Nano- and Micro-Electromechanical systems: Fundamentals of Nano- and Microengineering (Vol. 8). CRC press, (2005).

S. D. Senturia, Microsystem Design, Kluwer Academic Publishers, 2001.

M. Madou, Fundamentals of Microfabrication, CRC Press, 1997.

G. Kovacs, Micromachined Transducers Sourcebook, McGraw-Hill, Boston, 1998.

M.H. Bao, Micromechanical Transducers: Pressure sensors, accelerometers, and Gyroscopes, Elsevier, New York, 2000.

ECPE1_ MIXED SIGNAL DESIGN

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Analog and discrete-time signal processing, introduction to sampling theory; Analog continuous-time filters: passive and active filters; Basics of analog discrete-time filters and Z-transform.

Unit II:

Switched-capacitor filters- Nonidealities in switched-capacitor filters; Switched-capacitor filter architectures; Switched-capacitor filter applications.

Unit III:

Basics of data converters; Successive approximation ADCs, Dual slope ADCs, Flash ADCs, Pipeline ADCs, Hybrid ADC structures, High-resolution ADCs, DACs.

Unit IV:

Mixed-signal layout, Interconnects and data transmission; Voltage-mode signaling and data transmission; Current-mode signaling and data transmission.

Unit V:

Introduction to frequency synthesizers and synchronization; Basics of PLL, Analog PLLs; Digital PLLs; DLLs.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand the practical situations where mixed signal analysis is required.

Analyze and handle the inter-conversions between signals.

Design systems involving mixed signals

Text/Reference Books:

R. Jacob Baker, CMOS mixed-signal circuit design, Wiley India, IEEE press, reprint 2008.

Behzad Razavi , Design of analog CMOS integrated circuits, McGraw-Hill, 2003.

R. Jacob Baker, CMOS circuit design, layout and simulation, Revised second edition, IEEE press, 2008.

Rudy V. dePlassche, CMOS Integrated ADCs and DACs, Springer, Indian edition, 2005.

Arthur B. Williams, Electronic Filter Design Handbook, McGraw-Hill, 1981.

R. Schauman, Design of analog filters by, Prentice-Hall 1990 (or newer additions).

M. Burns et al., An introduction to mixed-signal IC test and measurement by, Oxford university press, first Indian edition, 2008.

ECPE1_

CMOS DESIGN

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Review of MOS transistor models, Non-ideal behavior of the MOS Transistor. Transistor as a switch. Inverter characteristics, Integrated Circuit Layout: Design Rules, Parasitics. Delay: RC Delay model, linear delay model, logical path efforts. Power, interconnect and Robustness in CMOS circuit layout. Combinational Circuit Design: CMOS logic families including static, dynamic and dual rail logic. Sequential Circuit Design: Static circuits. Design of latches and Flip-flops.

Course Outcomes:

At the end of the course the students will be able to

Design different CMOS circuits using various logic families along with their circuit layout.

Use tools for VLSI IC design.

Text/Reference Books:

N.H.E. Weste and D.M. Harris, CMOS VLSI design: A Circuits and Systems

Perspective, 4th Edition, Pearson Education India, 2011.

C.Mead and L. Conway, Introduction to VLSI Systems, Addison Wesley, 1979.

J. Rabaey, Digital Integrated Circuits: A Design Perspective, Prentice Hall India, 1997.

P. Douglas, VHDL: programming by example, McGraw Hill, 2013.

L. Glaser and D. Dobberpuhl, The Design and Analysis of VLSI Circuits, Addison Wesley, 1985.

ECPE1_

POWER ELECTRONICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Characteristics of Semiconductor Power Devices: Thyristor, power MOSFET and IGBT- Treatment should consist of structure, Characteristics, operation, ratings, protections and thermal considerations. Brief introduction to power devices viz. TRIAC, MOS controlled thyristor (MCT), Power Integrated Circuit (PIC) (Smart Power), Triggering/Driver, commutation and snubber circuits for thyristor, power MOSFETs and IGBTs (discrete and IC based). Concept of fast recovery and schottky diodes as freewheeling and feedback diode.

Unit II:

Controlled Rectifiers: Single phase: Study of semi and full bridge converters for R, RL, RLE and level loads. Analysis of load voltage and input current- Derivations of load form factor and ripple factor, Effect of source impedance, Input current Fourier series analysis of input current to derive input supply power factor, displacement factor and harmonic factor.

Unit III:

Choppers: Quadrant operations of Type A, Type B, Type C, Type D and type E choppers, Control techniques for choppers – TRC and CLC, Detailed analysis of Type A chopper. Step up chopper. Multiphase Chopper

Unit IV:

Single-phase inverters: Principle of operation of full bridge square wave, quasi-square wave, PWM inverters and comparison of their performance. Driver circuits for above inverters and mathematical analysis of output (Fourier series) voltage and harmonic control at output of

inverter (Fourier analysis of output voltage). Filters at the output of inverters, Single phase current source inverter

Unit V:

Switching Power Supplies: Analysis of fly back, forward converters for SMPS, Resonant converters - need, concept of soft switching, switching trajectory and SOAR, Load resonant converter - series loaded half bridge DC-DC converter.

Applications: Power line disturbances, EMI/EMC, power conditioners. Block diagram and configuration of UPS, salient features of UPS, selection of battery and charger ratings, sizing of UPS. Separately excited DC motor drive. P M Stepper motor Drive.

Course Outcomes:

At the end of this course students will demonstrate the ability to

Build and test circuits using power devices such as SCR

Analyze and design controlled rectifier, DC to DC converters, DC to AC inverters,

Learn how to analyze these inverters and some basic applications.

Design SMPS.

Text /Reference Books:

Muhammad H. Rashid, “Power electronics” Prentice Hall of India.

Ned Mohan, Robbins, “Power electronics”, edition III, John Wiley and sons.

P.C. Sen., “Modern Power Electronics”, edition II, Chand& Co.

V.R.Moorthi, “Power Electronics”, Oxford University Press.

Cyril W., Lander,” Power Electronics”, edition III, McGraw Hill.

G K Dubey, S R Doradla, “Thyristorised Power Controllers”, New Age International Publishers. SCR manual from GE, USA.

ECPE1_

HIGH SPEED ELECTRONICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Transmission line theory (basics) crosstalk and nonideal effects; signal integrity: impact of packages, vias, traces, connectors; non-ideal return current paths, high frequency power delivery, methodologies for design of high speed buses; radiated emissions and minimizing system noise; Noise Analysis: Sources, Noise Figure, Gain compression, Harmonic distortion, Intermodulation, Cross-modulation, Dynamic range

Unit II:

Devices: Passive and active, Lumped passive devices (models), Active (models, low vs high frequency)

Unit III:

RF Amplifier Design, Stability, Low Noise Amplifiers, Broadband Amplifiers (and Distributed) Power Amplifiers, Class A, B, AB and C, D E Integrated circuit realizations, Cross-over distortion Efficiency RF power output stages

Unit IV:

Mixers – Upconversion Downconversion, Conversion gain and spurious response. Oscillators Principles. PLL Transceiver architectures

Unit V:

Printed Circuit Board Anatomy, CAD tools for PCB design, Standard fabrication, Microvia Boards. Board Assembly: Surface Mount Technology, Through Hole Technology, Process Control and Design challenges.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand significance and the areas of application of high-speed electronics circuits.

Understand the properties of various components used in high speed electronics

Design High-speed electronic system using appropriate components.

Text/Reference Books:

Stephen H. Hall, Garrett W. Hall, James A. McCall “High-Speed Digital System Design: A Handbook of Interconnect Theory and Design Practices”, August 2000, Wiley-IEEE Press

Thomas H. Lee, “The Design of CMOS Radio-Frequency Integrated Circuits”, Cambridge University Press, 2004, ISBN 0521835399.

Behzad Razavi, “RF Microelectronics”, Prentice-Hall 1998, ISBN 0-13-887571-5.

Guillermo Gonzalez, “Microwave Transistor Amplifiers”, 2nd Edition, Prentice Hall.

Kai Chang, “RF and Microwave Wireless systems”, Wiley.

R.G. Kaduskar and V.B. Baru, Electronic Product design, Wiley India, 2011

ECPE1_

EMBEDDED SYSTEMS DESIGN

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Prerequisite: Microprocessor and Microcontrollers Course

Objectives:

1. To provide an overview of Design Principles of Embedded System.
2. To provide clear understanding about the role of firmware , operating systems in correlation with hardware systems.

Course Outcomes:

1. Expected to understand the selection procedure of Processors in the Embedded domain.
2. Design Procedure for Embedded Firmware.
3. Expected to visualize the role of Real time Operating Systems in Embedded Systems
4. Expected to evaluate the Correlation between task synchronization and latency issues

UNIT -I:

Introduction to Embedded Systems Definition of Embedded System, Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems.

UNIT -II:

Typical Embedded System: Core of the Embedded System: General Purpose and Domain Specific Processors, ASICs, PLDs, Commercial Off-The-Shelf Components (COTS), Memory: ROM, RAM, Memory according to the type of Interface, Memory Shadowing, Memory selection for Embedded Systems, Sensors and Actuators, Communication Interface: Onboard and External Communication Interfaces.

UNIT -III:

Embedded Firmware: Reset Circuit, Brown-out Protection Circuit, Oscillator Unit, Real Time Clock, Watchdog Timer, Embedded Firmware Design Approaches and Development Languages.

UNIT -IV:

RTOS Based Embedded System Design: Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling.

UNIT -V:

Task Communication: Shared Memory, Message Passing, Remote Procedure Call and Sockets, Task Synchronization: Task Communication/Synchronization Issues, Task Synchronization Techniques, Device Drivers, How to Choose an RTOS.

TEXT BOOKS:

Introduction to Embedded Systems - Shibu K.V, Mc Graw Hill.

REFERENCE BOOKS:

Embedded Systems - Raj Kamal, TMH.

Embedded System Design - Frank Vahid, Tony Givargis, John Wiley.

Embedded Systems – Lyla, Pearson, 2013

An Embedded Software Primer - David E. Simon, Pearson Education.

EC4421

ANALOG VLSI AND MIXED SIGNAL DESIGN

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objective:

- To understand the types of active filters and its operation.
- To understand the types of continuous time filters and digital filters and its operation.
- To understand various ADC and DAC converters and its importance in the Electronic systems.
- Gives Knowledge on VHDL Programming language for Mixed Signal Circuit Design.
- Extension the Verilog concepts for analog VLSI circuits.

Course Outcomes:

- Able to design filters with active devices only.
- Able to design the first and second order digital filters.
- The ability to use and design DAC and ADC techniques for data conversions.
- The ability to program, Mixed Signal VLSI Circuits.
- Verilog program for analog VLSI circuits.

UNIT I: Introduction to Active Filters (PLL) & Switched capacitor filters

Active RC Filters for monolithic filter design: First & Second order filter realizations - universal active filter (KHN)- self tuned filter - programmable filters - Switched capacitor filters: Switched capacitor resistors - amplifiers –comparators - sample & hold circuits – Integrator- Biquad.

UNIT II: Continuous Time filters & Digital Filters

Introduction to Gm - C filters - bipolar transconductors - CMOS Transconductors using Triode transistors, active transistors - BiCMOS transconductors – MOSFET C Filters -

Tuning Circuitry - Dynamic range performance -Digital Filters: Sampling – decimation – interpolation - implementation of FIR and IIR filters.

UNIT III: Digital to Analog & Analog to Digital Converters

Non-idealities in the DAC - Types of DAC's: Current switched, Resistive, Charge redistribution (capacitive), Hybrid, segmented DAC's - Techniques for improving linearity - Analog to Digital Converters: quantization errors -non-idealities - types of ADC's: Flash, two step, pipelined, successive approximation, folding ADC's.Sigma Delta Converters: Over sampled converters - over sampling without noise & with noise – implementation imperfections - first order modulator - decimation filters- second order modulator - sigma delta DAC & ADC's

UNIT IV: Analog and Mixed Signal Extensions to VHDL

Introduction - Language design objectives - Theory of differential algebraic equations - the 1076 .1 Language - Tolerance groups - Conservative systems - Time and the simulation cycle -A/D and D/A Interaction - Quiescent Point - Frequency domain modeling and examples.

UNIT V: Analog Extensions to Verilog

Introduction –data types –Expressions-Signals-Analog Behavior-Hierarchical structures-Mixed Signal Interaction. Introduction - Equation construction - solution - waveform Filter functions - simulator - Control Analysis - Multi -disciplinary model.

TEXT BOOKS:

David A. Johns, Ken Martin, “Analog Integrated Circuit Design” John Wiley & Sons, 2002.

Rudy van de Plassche “Integrated Analog-to-Digital and Digital-to-Analog Converters “,Kluwer 1999.

Antoniou, “Digital Filters Analysis and Design” Tata McGraw Hill, 1998.

REFERENCES:

Phillip Allen and Douglas Holmberg "CMOS Analog Circuit Design" Oxford University. Press, 2000.

BenhardRazavi, “Data Converters”, Kluwer Publishers, 1999.

Jacob Baker, Harry W LI, and David E Boyce “CMOS, Circuit Design Layout and Simulation”, Wiley- IEEE Press, 1 st Edition, 1997.

Tsividis Y P, “Mixed Analog and Digital VLSI Devices and Technology”, Mc-Graw Hill,1996.

ECPE1_ CMOS ANALOG VLSI DESIGN

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

COURSE OBJECTIVES

1. To understand CMOS analog circuits design
2. To simulate Analog circuits using H SPICE.
3. To learn noise modeling of CMOS analog circuits

UNIT I - ANALOG CMOS SUB-CIRCUITS

Introduction to analog design, Passive and active current mirrors, band-gap references, Switched Capacitor circuits - basic principles, sampling switches, switched capacitor integrator, switched capacitor amplifier, simulation of CMOS sub circuits using SPICE.

UNIT II - CMOS SINGLE STAGE AMPLIFIERS

Common-Source stage (with resistive load, diode connected load, current-source load, triode load, source degeneration), source follower, common-gate stage, cascode stage, folded cascode stage. Frequency responses of CS stage, CD stage, CG stage, cascode stage, simulation of CMOS amplifiers using SPICE.

UNIT III - DIFFERENTIAL AMPLIFIER & OPERATIONAL AMPLIFIERS

Single-ended and differential operation, basic differential pair – qualitative and quantitative analyses, common-mode response, differential pair with MOS loads, Performance parameters of op-amp, one stage op-amp, two-stage CMOS op-amp, Gain boosting, slew rate, power supply rejection, Simulation of differential amplifiers using SPICE.

UNIT IV - OSCILLATORS

General considerations, Ring oscillators, LC oscillators – cross-coupled oscillators, Colpitts oscillator, One-port oscillator, and voltage controlled oscillators. Simulation of oscillators using SPICE.

UNIT V - NOISE CHARACTERISTICS

Statistical characteristics of noise, Types of noise - thermal noise, flicker noise, Representation of noise in circuits, noise in single-stage amplifiers (CS, CD and CG stages), noise bandwidth.

TEXT BOOK

Razavi, "*Design of analog CMOS integrated circuits*", McGraw Hill, Edition 2002.

REFERENCES

1. Gray, Meyer, Lewis, Hurst, "*Analysis and design of Analog Integrated Circuits*", Willey International, 4th Edition, 2002.
2. Allen, Holberg, "*CMOS analog circuit design*", Oxford University Press, 2nd Edition, 2012.

COURSE OUTCOMES

1. Analog circuits are essential in interfacing and building amplifiers and low pass filters.
2. Able to do the design of analog sublevel blocks.
3. Learn some design methods for CMOS analog circuit.

ECPE1

CMOS DIGITAL VLSI DESIGN

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Course Objectives:

1. This is an introductory course which covers basic theories and techniques of digital VLSI Circuit design in CMOS technology.
2. In this course, we will study the fundamental concepts and structures of MOS Transistor and designing digital VLSI Circuits, static and dynamic Power Dissipation, interconnect analysis, Propagation delays, MOS Inverters and their Characteristics.
3. The course is designed to give the student an understanding of the different Combinational circuit design, Sequential MOS Logic Gates and CMOS Dynamic Logic Circuits including their Transient Analysis, design steps and behavior.

Unit I MOS Transistor-First Glance at the MOS device, MOS Transistor under static conditions, threshold voltage, Resistive operation, saturation region, channel length modulation, velocity saturation, Hot carrier effect-drain current Vs voltage charts, sub threshold conduction, equivalent resistance, MOS structure capacitance & CMOS logics.

Unit II MOS Inverter, Switching characteristics & Interconnect Effects- Delay Time, Interconnect Parasitic Capacitances, Resistance, RC Delays, Inductances, Gate Delays, Stage Ratio, Power Dissipation, CMOS Logic Gate Design, Transmission Gate & BiCMOS.

Unit III Combinational Circuit Design: NAND Gate, NOR Gate, Transient Analysis of NAND & NOR Gate. Sequential MOS Logic Gates: Behavior of Bistable element, CMOS latches & Clocked Flip-Flops, Clock Skew & Clocking Strategies.

Unit IV CMOS Dynamic Logic Circuits: Pass Transistor Logic-0 and 1 transfer, Charge Storage & Leakage. Voltage Bootstrapping. High Performance Dynamic CMOS Circuits: Domino CMOS Logic. NORA CMOS Logic, Zipper CMOS Circuits, TSPC Dynamic CMOS.

Unit V Semiconductor Memories: ROM, DRAM, SRAM, PLA, Cell, Leakage Circuit and Input/output Circuit.

Text books:

1. Jan.M.Rabaey,, AnithaChandrakasanBorivoje Nikolic, "Digital Integrated Circuits", SecondEdition.
2. Sung-Mo Kang, Yusuf Leblebici, "CMOS Digital IC- Analysis and Design", 3rd Edition, Tata McGraw Hill.

Reference books:

1. Neil H.E Weste and Kamran Eshraghian, "Principles of CMOS VLSI Design", 2nd Edition, Addition Wesley, 1998.

COURSE OUTCOMES

1. Able to design digital circuits in circuit-level.
2. Able to have a digital circuit design with optimizing speed and area.
3. Semiconductor memories cell can be enhanced with the existed memory cells.

EC4411

REAL TIME OPERATING SYSTEMS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To understand the need of real time operating system.
- To learn the basic concepts of inter process communication (IPC).
- To analyze various scheduling algorithms related to RTOS.
- To introduce the elementary concepts of Vx works.
- To study the basic concepts of UNIX operating system.

Course Outcomes:

- At the end of this course Students will learn:
- 1.Understand Real-time operating system requirements and applications.
- Categorize different scheduling approaches for real time scheduler.
- Compare different real time systems.
- Analyze a module and understand design issues.
- Develop a real time embedded system module.

UNIT-I:

Introduction to Real Time Systems Structures of Operating System (Monolithic, Microkernel, Layered, Exo-kernel and Hybrid kernel structures), Operating system objectives and functions, Virtual Computers, Interaction of OS and Hardware architecture, Evolution of operating systems, Batch, multi programming. Multitasking, Multiuser, parallel, distributed and real-time OS.

UNIT-II:

Process Management of OS/RTOS Hard versus Soft Real-Time System: Jobs and Processors, release time, deadlines, and timing constraints, hard and soft timing constraints, hard real-time systems. Uniprocessor Scheduling: Types of scheduling, scheduling algorithms: FCFS, SJF, Priority, Round Robin, UNIX Multi-level feedback queue scheduling, Thread scheduling, Multiprocessor scheduling concept, Real Time scheduling concept.

UNIT-III:

Real Time Operating System Concepts Foreground and Background Systems, Shared Resource, Critical section of a Code, Multi-Tasking, Task, Context switch, Kernel, Scheduler, Preemptive and non-preemptive kernel, Inter Task Communication: Message Mailboxes, Message queues or pipes and Event flags, Semaphores, Interrupts

UNIT-IV:

Introduction to Vx works/UNIX OS Elementary Concepts of Vx Works: Multitasking, Task State Transition, Task Control- Task Creation and Activation, Task Stack, Task Names and IDs, Task Options, Task Information, Task Deletion and Deletion Safety. Fundamental Concepts of UNIX Operating Systems UNIX Kernel – File system, Concepts of – Process, Concurrent Execution & Interrupts. Process Management – forks & execution. Basic level Programming with system calls, Shell programming and filters.

UNIT-V:

Linux development process Types of Host /Target Development and debug setup, Generic Architecture of an Embedded Linux System, System start up, Types of Boot configurations, System Memory Layout, Development Tools: Project Workspace, IDE, GNCC cross platform, selecting and configuring kernel, setting up boot loader.

TEXT BOOKS:

Tanenbaum, “Modern Operating Systems,” 4/e, Pearson Edition, 2014.

Jane W.S.Liu, Real Time Systems, Pearson Education, Asia, 2001. REFERENCES: 1. Jean J Labrosse, “Embedded Systems Building Blocks Complete and Ready-to-use Modules in C” ,2/e, CRC Press ,1999. 2. Karim Yaghmour, Jon Masters, Gilad Ben-Yesset, Philippe Gerum, “Building Embedded Linux Systems”, O’Reilly Media, 2008.

Wind River Systems, “VxWorks Programmers Guide 5.5”, Wind River Systems Inc.2002.

EC4423

CAD FOR VLSI CIRCUITS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To make understand the VLSI CAD tools flow with Algorithmic Graphs.
- Gives knowledge on layout design rules and various algorithms for placement and partitioning of circuits.
- Understands the Floor planning and routing process in the VLSI CAD tools.
- Knows the various concepts of simulation and synthesis process in VLSI CAD tools.
- Gives Knowledge on modelling concepts of synthesis process.

Course Outcomes:

- Understands the complete VLSI CAD tool flow.
- Able to follow the design rules and debug it while creating layouts for circuits in VLSI CAD tool and also understands the placement and partitioning of the digital systems.
- Able to debug the floor planning and routing problems in the VLSI CAD tools for the digital systems.
- Understands the simulation and synthesis process and make advantage to debug the errors when developing and using the VLSI CAD tools.
- Able to develop a synthesis process in the VLSI CAD tools

UNIT I: VLSI DESIGN METHODOLOGIES

Introduction to VLSI Design methodologies - Review of Data structures and algorithms - Review of VLSI Design automation tools - Algorithmic Graph Theory and Computational Complexity - Tractable and Intractable problems - general purpose methods for combinatorial optimization.

UNIT II:DESIGN RULES

Layout Compaction - Design rules - problem formulation - algorithms for constraint graph compaction - placement and partitioning - Circuit representation - Placement algorithms - partitioning

UNIT III:FLOOR PLANNING

Floor planning concepts - shape functions and floorplan sizing - Types of local routing problems - Area routing - channel routing - global routing - algorithms for global routing.

UNIT IV:SIMULATION

Simulation - Gate-level modeling and simulation - Switch-level modeling and simulation - Combinational Logic Synthesis - Binary Decision Diagrams - Two Level Logic Synthesis.

UNIT V:MODELLING AND SYNTHESIS

High level Synthesis - Hardware models - Internal representation - Allocation assignment and scheduling - Simple scheduling algorithm - Assignment problem - High level transformations.

TEXTBOOKS:

S.H. Gerez, "Algorithms for VLSI Design Automation", John Wiley & Sons,2002.

N.A. Sherwani, "Algorithms for VLSI Physical Design Automation", Kluwer Academic Publishers, 2002.

EC4424

RF INTEGRATED CIRCUITS

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

- To educate students fundamental RF circuit and system design skills.
- To introduce students, the basic transmission line theory, single and multiport networks, RF component modelling.
- To offer students experience on designing matching and biasing networks & RF transistor amplifier design.

Course Outcomes:

- At the end of this course Students will learn:
- Understand the design bottlenecks specific to RF IC design, linearity related issues, ISI.
- Identify noise sources, develop noise models for the devices and systems.
- Specify noise and interference performance metrics like noise figure, IIP3 and different matching criteria.
- Comprehend different multiple access techniques, wireless standards and various transceiver architectures
- Design various constituents' blocks of RF receiver front end.

Unit I: INTRODUCTION TO RF AND WIRELESS TECHNOLOGY:

Complexity comparison, Design bottle necks, Applications, Analog and digital systems, Choice of Technology. BASIC CONCEPTS IN RF DESIGN: Nonlinearity and time variance, ISI, Random process and noise, sensitivity and dynamic range, passive impedance transformation.

Unit II: MULTIPLE ACCESS:

Techniques and wireless standards, mobile RF communication, FDMA, TDMA, CDMA, Wireless standards.

Unit III: TRANSCEIVER ARCHITECTURES:

General considerations, receiver architecture, Transmitter Architecture, transceiver performance tests, case studies.

Unit IV: AMPLIFIERS, MIXERS AND OSCILLATORS:

LNAs, down conversion mixers, Cascaded Stages, oscillators, Frequency synthesizers.

Unit V: POWER AMPLIFIERS:

General considerations, linear and nonlinear Pas, classification, High Frequency power amplifier, large signal impedance matching, linearization techniques.

Text Books:

Behzad Razavi, RF Microelectronics Prentice Hall of India, 2001

Thomas H. Lee, the Design of CMOS Radio Integrated Circuits, Cambridge University Press.

ECPE1_

FIBER OPTIC COMMUNICATION

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Introduction to vector nature of light, propagation of light, propagation of light in a cylindrical dielectric rod, Ray model, wave model.

Unit II:

Different types of optical fibers, Modal analysis of a step index fiber.

Signal degradation on optical fiber due to dispersion and attenuation. Fabrication of fibers and measurement techniques like OTDR.

Unit III:

Optical sources - LEDs and Lasers, Photo-detectors - pin-diodes, APDs, detector responsivity, noise, optical receivers. Optical link design - BER calculation, quantum limit, power penalties.

Unit IV:

Optical switches - coupled mode analysis of directional couplers, electro-optic switches.

Optical amplifiers - EDFA, Raman amplifier.

Unit V:

WDM and DWDM systems. Principles of WDM networks.

Nonlinear effects in fiber optic links. Concept of self-phase modulation, group velocity dispersion and soliton based communication.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand the principles fiber-optic communication, the components and the bandwidth advantages.

Understand the properties of the optical fibers and optical components.

Understand operation of lasers, LEDs, and detectors

Analyze system performance of optical communication systems

Design optical networks and understand non-linear effects in optical fibers.

EC4434 ANTENNAS AND WAVE PROPAGATION

Externals: 60Marks

L-T-P-C

Internals: 40Marks

4-0-3

Course Objectives:

1. Understand basic terminology and concepts of Antennas.
2. To attain knowledge on the basic parameters those are considered in the antenna design process and the analysis while designing that.
3. Analyze the electric and magnetic field emission from various basic antennas and
4. mathematical Formulation of the analysis.
5. To have knowledge on antenna operation and types as well as their usage in real time filed.
6. Aware of the wave spectrum and respective band based antenna usage and also to know the propagation of the waves at different frequencies through different layers in the existing layered free space environment structure.

UNIT-I: Antenna Basics:

Introduction, Basic Antenna Parameters – Patterns, Beam Area, Radiation Intensity, Beam Efficiency, Directivity-Gain-Resolution, Antenna Apertures, Illustrative Problems. Fields from Oscillating Dipole, Field Zones, Front - to-back Ratio, Antenna Theorems, Radiation, Retarded Potentials – Helmholtz Theorem. Thin Linear Wire Antennas – Radiation from Small Electric Dipole, Quarter Wave Monopole and Half Wave Dipole – Current Distributions, Field Components, Radiated Power, Radiation Resistance, Beam Width, Directivity, Effective Area, Effective Height, Natural Current Distributions, Far Fields and Patterns of Thin Linear Centre-fed Antennas of Different Lengths, Illustrative Problems. Loop Antennas - Introduction, Small Loop, Comparison of Far Fields of Small Loop and Short Dipole, Radiation Resistances.

UNIT-II: Antenna Arrays:

Point Sources – Definition, Patterns, arrays of 2 Isotropic Sources - Different Cases, Principle of Pattern Multiplication, Uniform Linear Arrays – Broadside Arrays, Endfire Arrays, EFA with Increased Directivity, Derivation of their Characteristics and Comparison, BSAs with Non-uniform Amplitude Distributions –General Considerations and Binomial Arrays, Illustrative Problems.

Antenna Measurements: Introduction, Concepts - Reciprocity, Near and Far Fields, Coordinate System Patterns to be Measured, Pattern Measurement Arrangement, Directivity Measurement, Gain Measurements (by Comparison, Absolute and 3- Antenna Methods)

UNIT-III: VHF, UHF and Microwave Antennas-I:

Arrays with Parasitic Elements, Yagi-Uda Array, Folded Dipoles and their Characteristics, Helical Antennas – Helical Geometry, Helix Modes, Practical Design Considerations for Monofilar Helical Antenna in Axial and Normal Modes, Horn Antennas – Types, Optimum Horns, Design Considerations of Pyramidal Horns, Illustrative Problems.

VHF, UHF and Microwave Antennas - II:

Microstrip Antennas – Introduction, Features, Advantages and Limitations, Rectangular Patch Antennas– Geometry and Parameters, Characteristics of Microstrip Antennas. Impact of Different Parameters on Characteristics, 103 Reflector Antennas – Introduction, Flat Sheet and Corner Reflectors, Paraboloidal Reflectors – Geometry, Pattern Characteristics, Feed Methods, Reflector Types – Related Features, Illustrative Problems. Lens Antennas – Introduction, Geometry of Non-metallic Dielectric Lenses, Fermat's Principle, Zoning, Applications.

UNIT-IV: Wave Propagation – I:

Introduction, Definitions, Categorizations and General Classifications, Different Modes of Wave Propagation, Ray/Mode Concepts, Ground Wave Propagation (Qualitative Treatment) Introduction, Plane Earth Reflections, Space and Surface Waves, Wave Tilt, Curved Earth Reflections. Space Wave Propagation – Introduction, Field Strength Variation with Distance and Height, Effect of Earth's Curvature, Absorption, Super Refraction, M-Curves and Duct Propagation, Scattering Phenomena, Tropospheric Propagation.

UNIT-V: Wave Propagation – II:

Sky Wave Propagation – Introduction, Structure of Ionosphere, Refraction and Reflection of Sky Waves by Ionosphere, Ray Path, Critical Frequency, MUF, LUF, OF, Virtual Height and Skip Distance, Relation between MUF and Skip Distance, Multihop Propagation.

Course Outcomes:

Student will be able to:

1. Aware of antenna parameter considerations
2. Capable to analyze the designed antenna and field evaluation under various conditions and formulate the electric as well as magnetic fields equation set for far field and near field conditions
3. Understand the array system of different antennas and field analysis under application of different currents to the individual antenna elements
4. Understand the design issues, operation of fundamental antennas and their operation methodology in practice.
5. Design a lens structure and also the bench set up for antenna parameter measurement of testing for their effectiveness
6. Knowledge about the means of propagation of electromagnetic waves

TEXT BOOKS:

1. Antennas and Wave Propagation – J.D. Kraus, R.J. Marhefka and Ahmad S. Khan, TMH, New Delhi, 4th ed., (Special Indian Edition), 2010.
2. Antenna Theory - C.A. Balanis, John Wiley & Sons, 3rd Ed., 2005.

REFERENCE BOOKS :

1. A.Harish, M.Sachidanada,” Antennas and Wave Propagation”, Oxford University Press, 2007
2. Electromagnetic Waves and Radiating Systems – E.C. Jordan and K.G. Balmain, PHI, 2nd ed. 2000.
3. Antennas and Wave Propagation – K.D. Prasad, Satya Prakashan, Tech India Publications, New Delhi, 2001.
4. Transmission and Propagation – E.V.D. Glazier and H.R.L. Lamont, The Services Text Book of Radio, vol. 5, Standard Publishers Distributors, Delhi.
5. Antennas – John D. Kraus, McGraw-Hill (International Edition), 2nd Ed. 1988.
6. Electronic and Radio Engineering – F.E. Terman, McGraw-Hill, 4th edition, 1955.

MOOCS :

1. https://youtu.be/wx_tlvaajAI
2. <https://youtube.com/playlist?list=PLgwJf8NK-2e7tzLIDL4aXUbtRFY3ykmkT>

ECPE1_ MICROWAVE THEORY AND TECHNIQUES

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Unit I:

Introduction to Microwaves-History of Microwaves, Microwave Frequency bands; Applications of Microwaves: Civil and Military, Medical, EMI/ EMC. Mathematical Model of Microwave Transmission-Concept of Mode, Features of TEM, TE and TM Modes, Losses associated with microwave transmission, Concept of Impedance in Microwave transmission.

Unit II:

Analysis of RF and Microwave Transmission Lines- Coaxial line, Rectangular waveguide, Circular waveguide, Strip line, Micro strip line. Microwave Network Analysis- Equivalent voltages and currents for non-TEM lines, Network parameters for microwave circuits, Scattering Parameters.

Unit III:

Passive and Active Microwave Devices- Microwave passive components: Directional Coupler, Power Divider, Magic Tee, Attenuator, Resonator. Microwave active components: Diodes, Transistors, Oscillators, Mixers. Microwave Semiconductor Devices: Gunn Diodes, IMPATT diodes, Schottky Barrier diodes, PIN diodes. Microwave Tubes: Klystron, TWT, Magnetron.

Unit IV:

Microwave Design Principles- Impedance transformation, Impedance Matching, Microwave Filter Design, RF and Microwave Amplifier Design, Microwave Power Amplifier Design, Low Noise Amplifier Design, Microwave Mixer Design, Microwave Oscillator Design. Microwave Antennas- Antenna parameters, Antenna for ground based systems, Antennas for airborne and satellite borne systems, Planar Antennas. Microwave Measurements- Power, Frequency and impedance measurement at microwave frequency, Network Analyzer and measurement of scattering parameters, Spectrum Analyzer and measurement of spectrum of a

microwave signal, Noise at microwave frequency and measurement of noise figure. Measurement of Microwave antenna parameters.

Unit V:

Microwave Systems- Radar, Terrestrial and Satellite Communication, Radio Aidsto Navigation, RFID, GPS. Modern Trends in Microwaves Engineering- Effect of Microwaves on human body, Medical and Civil applications of microwaves, Electromagnetic interference and Electromagnetic Compatibility (EMI & EMC), Monolithic Microwave ICs, RFMEMS for microwave components, Microwave Imaging.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

Understand various microwave system components their properties.

Appreciate that during analysis/ synthesis of microwave systems, the different mathematical treatment is required compared to general circuit analysis.

Design microwave systems for different practical application.

Text/Reference Books:

R.E. Collins, Microwave Circuits, McGraw Hill

K.C. Gupta and I.J. Bahl, Microwave Circuits, Artech house

EC4426 ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

Externals: 60Marks

(L-T)-P-C

Internals: 40Marks

4-0-3

Course Objectives:

This course provides

- An introduction to measurement techniques and instrumentation design and operation
- The basic concepts of units measurement error and accuracy, the construction and design of measuring devices and circuits measuring instruments and their proper applications
- To use different measuring techniques and measurement of different physical parameters using different transducers

Course Outcomes:

- Upon a successful completion of this course, the student will be able to:
- Describe the fundamental concepts and principles of instrumentation.
- Explain the operations of the various instruments required in measurements.
- Apply the measurement techniques for different types of tests.
- To select specific instrument for specific measurement function.
- Understand principle of operation, working of different electronic instruments like digital multi meter, vector voltmeter.
- Learners will apply knowledge of different oscilloscopes like CRO, DSO.
- Students will understand functioning, specification, and applications of signal analyzing instruments.

UNIT - I:

Block Schematics of Measuring Systems: Performance characteristics, Static characteristics, Accuracy, Precision, Resolution, Types of Errors, Gaussian Error, Root Sum Squares formula, Dynamic Characteristics, Repeatability, Reproducibility, Fidelity, Lag; Measuring Instruments: DC Voltmeters, D' Arsonval Movement, DC Current Meters, AC Voltmeters and

Current Meters, Ohmmeters, Multimeters, Meter Protection, Extension of Range, True RMS Responding Voltmeters, Specifications of Instruments.

UNIT - II:

Signal Analyzers: AF, HF Wave Analyzers, Harmonic Distortion, Heterodyne wave Analyzers, Spectrum Analyzers, Power Analyzers, Capacitance-Voltage Meters, Oscillators. Signal Generators: AF, RF Signal Generators, Sweep Frequency Generators, Pulse and Square wave Generators, Function Generators, Arbitrary waveform Generator, Video Signal Generators, and Specifications.

UNIT - III:

Oscilloscopes: CRT, Block Schematic of CRO, Time Base Circuits, Lissajous Figures, CRO Probes, High Frequency CRO Considerations, Delay lines, Applications: Measurement of Time, Period and Frequency Specifications.

Special Purpose Oscilloscopes: Dual Trace, Dual Beam CROs, Sampling Oscilloscopes, Storage Oscilloscopes, Digital Storage CROs.

UNIT - IV:

Transducers: Classification, Strain Gauges, Bounded, unbounded; Force and Displacement Transducers, Resistance Thermometers, Hotwire Anemometers, LVDT, Thermocouples, Synchros, Special Resistance Thermometers, Digital Temperature sensing system, Piezoelectric Transducers, Variable Capacitance Transducers, Magneto Strictive Transducers.

UNIT - V:

Bridges: Wheat Stone Bridge, Kelvin Bridge, and Maxwell Bridge.

Measurement of Physical Parameters: Flow Measurement, Displacement Meters, Liquid level Measurement, Measurement of Humidity and Moisture, Velocity, Force, Pressure - High Pressure, Vacuum level, Temperature - Measurements, Data Acquisition Systems.

TEXTBOOKS:

Electronic instrumentation: H.S.Kalsi - TMH, 2nd Edition 2004.

Modern Electronic Instrumentation and Measurement Techniques: A.D. Helbins, W.D. Cooper: PHI, 5th Edition, 2003.

REFERENCES:

Electronic Instrumentation and Measurements - David A. Bell, Oxford Univ. Press, 1997.

Electronic Measurements and Instrumentation: B. M. Oliver, J. M. Cage TMH Reprint.

Measurement Systems - Ernest O. Doebelin and Dhanesh N Manik, 6th Ed., TMH.

Electronic Measurements and Instrumentations by K. Lal Kishore, Pearson Education - 2010.

Industrial Instrumentation: T. R. PadmanabhamSpiriger 2009.

SCHEME OF EVALUATION

Special Evaluation Pattern for the following courses:

S.NO.	Type of Course	Internals	Externals
1	HS1101: Communication Skills-I, HS1201: Communication Skills-II (Theory course)	Activities and Assignments = 40Marks	Written exam: 60Marks
2	HS1601: English Language Lab	Orals (50Marks)	Written exam: 50Marks
3	CE1001 (Engineering Graphics): (Course Type: Lab)	40	60

SCHEME OF EVALUATION		
Assessment Method	Description	Weightage
Assignment & Continuous monitoring	Assignments & Evaluation of drawings	25%
Mid Term	3 mid examinations will be conducted and best of 2 will be considered.	25%
End Term	Students will be evaluated based on the understanding of the principles, skills and practices of the course	50%

Evaluation Pattern for General Courses:

S.NO	Type of Course	Internals	Externals
1	Theory	Best two out of Three MTs (2x15) + Assignments (10)= 40Marks	60Marks
2	Lab	(Report+Record) (20)+ Attendance(10) + Internal Viva (10)= 40 Marks	Lab Experiment (40) + External Viva (20) = 60 Marks
3	19EC2901,19EC2902 , 19EC3901 (1 credit Mini project, 1 credit Technical seminar)	NA	100Marks
4	19ECP01 (2 credits Mini project)	Mid-Sem review (40 Marks)	End-Sem review (60 Marks)
5	19ECP02 (1 credit Summer Internship)	Industry Expert (30 Marks) + RGUKT panel (70 Marks)	
6	19ECP03(3 credits Project),19ECP04(8 credits project)	Three Internal Reviews (3X20) = 60Marks	External Review=40Marks

7	19ECCV-I,19ECCV-I I (0 credit Comprehensive Viva)	NA	100 Marks
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